1.2 Lengths and Dot Products

The first section backed off from multiplying vectors. Now we go forward to define the *"dot product"* of v and w. This multiplication involves the separate products v_1w_1 and v_2w_2, but it doesn't stop there. Those two numbers are added to produce the single number $v \cdot w$. *This is the geometry section (lengths and angles).*

DEFINITION The ***dot product*** or ***inner product*** of $v = (v_1, v_2)$ and $w = (w_1, w_2)$ is the number $v \cdot w$:

$$v \cdot w = v_1w_1 + v_2w_2. \tag{1}$$

Example 1 The vectors $v = (4, 2)$ and $w = (-1, 2)$ have a *zero* dot product:

Dot product is zero
Perpendicular vectors

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix} = -4 + 4 = 0.$$

In mathematics, zero is always a special number. For dot products, it means that *these two vectors are perpendicular.* The angle between them is 90°. When we drew them in Figure 1.1, we saw a rectangle (not just any parallelogram). The clearest example of perpendicular vectors is $i = (1, 0)$ along the x axis and $j = (0, 1)$ up the y axis. Again the dot product is $i \cdot j = 0 + 0 = 0$. Those vectors i and j form a right angle.

The dot product of $v = (1, 2)$ and $w = (3, 1)$ is 5. Soon $v \cdot w$ will reveal the angle between v and w (not 90°). Please check that $w \cdot v$ is also 5.

The dot product $w \cdot v$ ***equals*** $v \cdot w$. The order of v and w makes no difference.

Example 2 Put a weight of 4 at the point $x = -1$ (left of zero) and a weight of 2 at the point $x = 2$ (right of zero). The x axis will balance on the center point (like a see-saw). The weights balance because the dot product is $(4)(-1) + (2)(2) = 0$.

This example is typical of engineering and science. The vector of weights is $(w_1, w_2) = (4, 2)$. The vector of distances from the center is $(v_1, v_2) = (-1, 2)$. The weights times the distances, w_1v_1 and w_2v_2, give the "moments". The equation for the see-saw to balance is $w_1v_1 + w_2v_2 = 0$.

Example 3 Dot products enter in economics and business. We have three goods to buy and sell. Their prices are (p_1, p_2, p_3) for each unit—this is the "price vector" p. The quantities we buy or sell are (q_1, q_2, q_3)—positive when we sell, negative when we buy. *Selling* q_1 *units at the price* p_1 *brings in* q_1p_1. The total income (quantities q times prices p) is ***the dot product*** $q \cdot p$ ***in three dimensions***:

$$\textit{Income} = (q_1, q_2, q_3) \cdot (p_1, p_2, p_3) = q_1p_1 + q_2p_2 + q_3p_3 = \textit{dot product.}$$

A zero dot product means that "the books balance". Total sales equal total purchases if $q \cdot p = 0$. Then p is perpendicular to q (in three-dimensional space). A supermarket with thousands of goods goes quickly into high dimensions.

Small note: Spreadsheets have become essential in management. They compute linear combinations and dot products. What you see on the screen is a matrix.

Main point To compute $v \cdot w$, multiply each v_i times w_i. Then add $\Sigma v_i w_i$.

Lengths and Unit Vectors

An important case is the dot product of a vector *with itself.* In this case v equals w. When the vector is $v = (1, 2, 3)$, the dot product with itself is $v \cdot v = \|v\|^2 = 14$:

Dot product $v \cdot v$
Length squared

$$\|v\|^2 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 + 4 + 9 = \mathbf{14}.$$

Instead of a 90° angle between vectors we have 0°. The answer is not zero because v is not perpendicular to itself. The dot product $v \cdot v$ gives the *length of v squared.*

DEFINITION The *length* $\|v\|$ of a vector v is the square root of $v \cdot v$:

Length = norm(v) $\qquad$ **length** $= \|v\| = \sqrt{v \cdot v}$.

In two dimensions the length is $\sqrt{v_1^2 + v_2^2}$. In three dimensions it is $\sqrt{v_1^2 + v_2^2 + v_3^2}$. By the calculation above, the length of $v = (1, 2, 3)$ is $\|v\| = \sqrt{14}$.

Here $\|v\| = \sqrt{v \cdot v}$ is just the ordinary length of the arrow that represents the vector. In two dimensions, the arrow is in a plane. If the components are 1 and 2, the arrow is the third side of a right triangle (Figure 1.6). The Pythagoras formula $a^2 + b^2 = c^2$, which connects the three sides, is $1^2 + 2^2 = \|v\|^2$.

For the length of $v = (1, 2, 3)$, we used the right triangle formula twice. The vector $(1, 2, 0)$ in the base has length $\sqrt{5}$. This base vector is perpendicular to $(0, 0, 3)$ that goes straight up. So the diagonal of the box has length $\|v\| = \sqrt{5 + 9} = \sqrt{14}$.

The length of a four-dimensional vector would be $\sqrt{v_1^2 + v_2^2 + v_3^2 + v_4^2}$. Thus the vector $(1, 1, 1, 1)$ has length $\sqrt{1^2 + 1^2 + 1^2 + 1^2} = 2$. This is the diagonal through a unit cube in four-dimensional space. The diagonal in n dimensions has length $\sqrt{n}$.

The word "**unit**" is always indicating that some measurement equals "one". The unit price is the price for one item. A unit cube has sides of length one. A unit circle is a circle with radius one. Now we define the idea of a "unit vector".

DEFINITION *A unit vector u is a vector whose length equals one.* Then $u \cdot u = 1$.

An example in four dimensions is $u = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Then $u \cdot u$ is $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 1$. We divided $v = (1, 1, 1, 1)$ by its length $\|v\| = 2$ to get this unit vector.

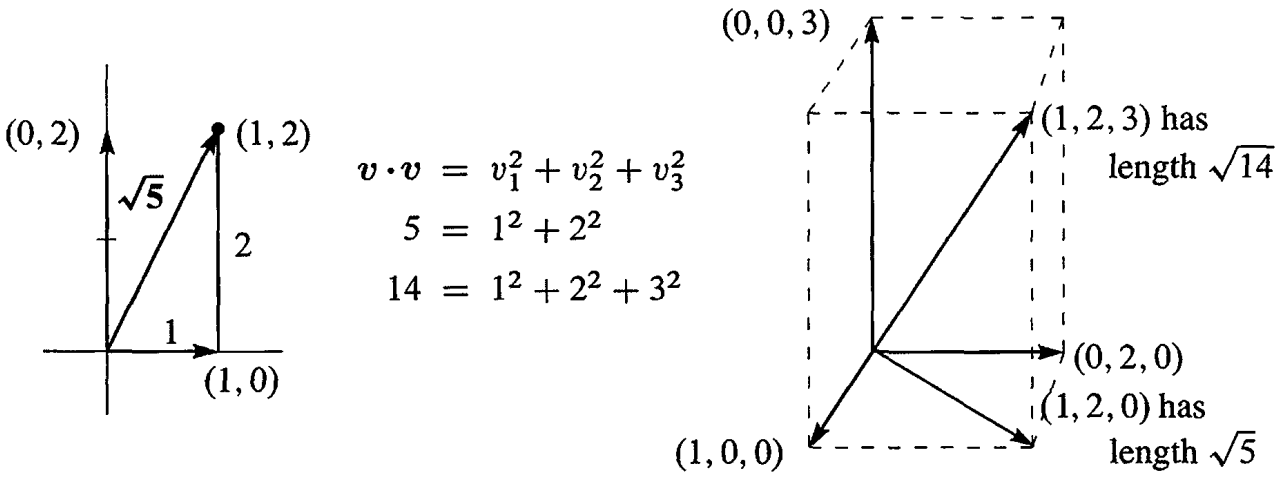


Figure 1.6: The length $\sqrt{v \cdot v}$ of two-dimensional and three-dimensional vectors.

Example 4 The standard unit vectors along the x and y axes are written $\boldsymbol{i}$ and $\boldsymbol{j}$. In the xy plane, the unit vector that makes an angle "theta" with the x axis is $(\cos\theta, \sin\theta)$:

$$\textbf{Unit vectors} \quad \boldsymbol{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \boldsymbol{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{and} \quad \boldsymbol{u} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}.$$

When $\theta = 0$, the horizontal vector $\boldsymbol{u}$ is $\boldsymbol{i}$. When $\theta = 90°$ (or $\frac{\pi}{2}$ radians), the vertical vector is $\boldsymbol{j}$. At any angle, the components $\cos\theta$ and $\sin\theta$ produce $\boldsymbol{u} \cdot \boldsymbol{u} = 1$ because $\cos^2\theta + \sin^2\theta = 1$. These vectors reach out to the unit circle in Figure 1.7. Thus $\cos\theta$ and $\sin\theta$ are simply the coordinates of that point at angle θ on the unit circle.

Since $(2, 2, 1)$ has length 3, the vector $(\frac{2}{3}, \frac{2}{3}, \frac{1}{3})$ has length 1. Check that $\boldsymbol{u} \cdot \boldsymbol{u} = \frac{4}{9} + \frac{4}{9} + \frac{1}{9} = 1$. For a unit vector, **divide any nonzero v by its length $\|v\|$.**

Unit vector $\boldsymbol{u} = v/\|v\|$ **is a unit vector in the same direction as v.**

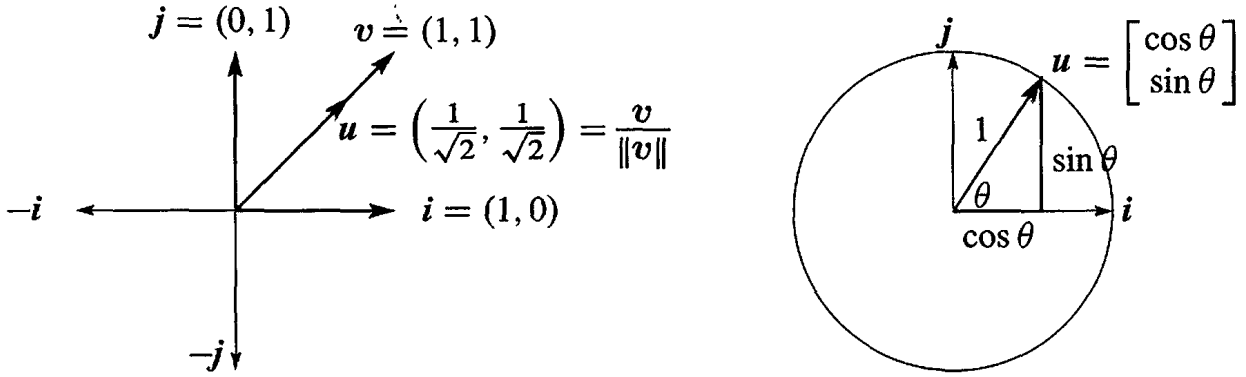


Figure 1.7: The coordinate vectors $\boldsymbol{i}$ and $\boldsymbol{j}$. The unit vector $\boldsymbol{u}$ at angle 45° (left) divides $v = (1, 1)$ by its length $\|v\| = \sqrt{2}$. The unit vector $\boldsymbol{u} = (\cos\theta, \sin\theta)$ is at angle θ.

The Angle Between Two Vectors

We stated that perpendicular vectors have $v \cdot w = 0$. The dot product is zero when the angle is 90°. To explain this, we have to connect angles to dot products. Then we show how $v \cdot w$ finds the angle between any two nonzero vectors v and w.

Right angles *The dot product is $v \cdot w = 0$ when v is perpendicular to w.*

Proof When v and w are perpendicular, they form two sides of a right triangle. The third side is $v - w$ (the hypotenuse going across in Figure 1.8). The *Pythagoras Law* for the sides of a right triangle is $a^2 + b^2 = c^2$:

$$\textbf{Perpendicular vectors} \quad \|v\|^2 + \|w\|^2 = \|v - w\|^2 \tag{2}$$

Writing out the formulas for those lengths in two dimensions, this equation is

$$\textbf{Pythagoras} \quad (v_1^2 + v_2^2) + (w_1^2 + w_2^2) = (v_1 - w_1)^2 + (v_2 - w_2)^2. \tag{3}$$

The right side begins with $v_1^2 - 2v_1w_1 + w_1^2$. Then v_1^2 and w_1^2 are on both sides of the equation and they cancel, leaving $-2v_1w_1$. Also v_2^2 and w_2^2 cancel, leaving $-2v_2w_2$. (In three dimensions there would be $-2v_3w_3$.) Now divide by -2:

$$0 = -2v_1w_1 - 2v_2w_2 \quad \text{which leads to} \quad \boldsymbol{v_1w_1 + v_2w_2 = 0}. \tag{4}$$

Conclusion Right angles produce $v \cdot w = 0$. The dot product is zero when the angle is $\theta = 90°$. Then $\cos\theta = 0$. The zero vector $v = \mathbf{0}$ is perpendicular to every vector w because $\mathbf{0} \cdot w$ is always zero.

Now suppose $v \cdot w$ is **not zero**. It may be positive, it may be negative. The sign of $v \cdot w$ immediately tells whether we are below or above a right angle. The angle is less than 90° when $v \cdot w$ is positive. The angle is above 90° when $v \cdot w$ is negative. The right side of Figure 1.8 shows a typical vector $v = (3, 1)$. The angle with $w = (1, 3)$ is less than 90° because $v \cdot w = 6$ is positive.

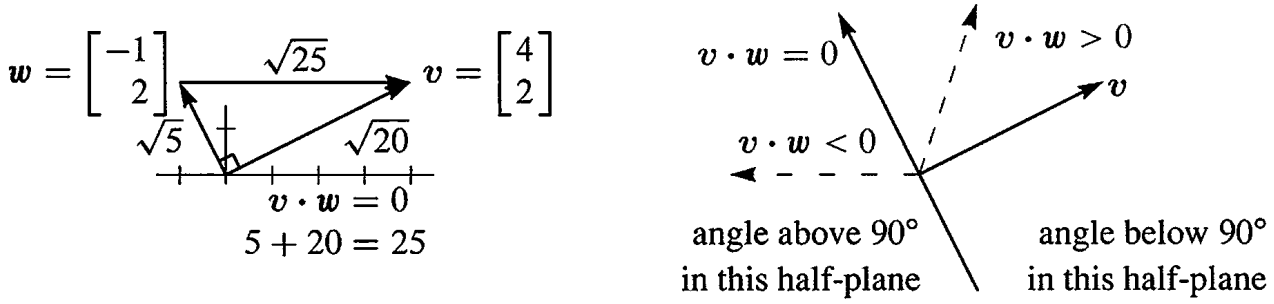


Figure 1.8: Perpendicular vectors have $v \cdot w = 0$. Then $\|v\|^2 + \|w\|^2 = \|v - w\|^2$.

The borderline is where vectors are perpendicular to v. On that dividing line between plus and minus, $(1, -3)$ is perpendicular to $(3, 1)$. The dot product is zero.

The dot product reveals the exact angle θ. This is not necessary for linear algebra—you could stop here! Once we have matrices, we won't come back to θ. But while we are on the subject of angles, this is the place for the formula.

Start with **unit vectors u and U**. The sign of $u \cdot U$ tells whether $\theta < 90°$ or $\theta > 90°$. Because the vectors have length 1, we learn more than that. ***The dot product $u \cdot U$ is the cosine of θ.*** This is true in any number of dimensions.

Unit vectors u and U at angle θ have $u \cdot U = \cos\theta$. Certainly $|u \cdot U| \leq 1$.

Remember that $\cos\theta$ is never greater than 1. It is never less than -1. ***The dot product of unit vectors is between -1 and 1.***

Figure 1.9 shows this clearly when the vectors are $u = (\cos\theta, \sin\theta)$ and $i = (1, 0)$. The dot product is $u \cdot i = \cos\theta$. That is the cosine of the angle between them.

After rotation through any angle α, these are still unit vectors. The vector $i = (1, 0)$ rotates to $(\cos\alpha, \sin\alpha)$. The vector u rotates to $(\cos\beta, \sin\beta)$ with $\beta = \alpha + \theta$. Their dot product is $\cos\alpha\cos\beta + \sin\alpha\sin\beta$. From trigonometry this is the same as $\cos(\beta - \alpha)$. But $\beta - \alpha$ is the angle θ, so the dot product is $\cos\theta$.

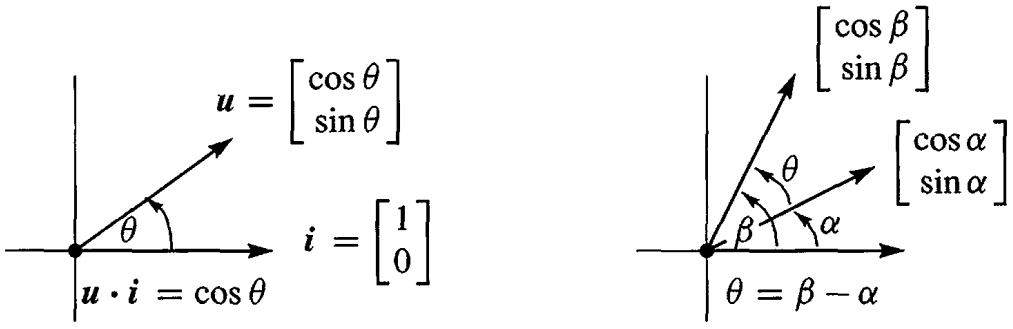


Figure 1.9: The dot product of unit vectors is the cosine of the angle θ.

Problem 24 proves $|u \cdot U| \leq 1$ directly, without mentioning angles. The inequality and the cosine formula $u \cdot U = \cos\theta$ are always true for unit vectors.

What if v and w are not unit vectors? Divide by their lengths to get $u = v/\|v\|$ and $U = w/\|w\|$. Then the dot product of those unit vectors u and U gives $\cos\theta$.

COSINE FORMULA If v and w are nonzero vectors then $\dfrac{v \cdot w}{\|v\|\,\|w\|} = \cos\theta$.

Whatever the angle, this dot product of $v/\|v\|$ with $w/\|w\|$ never exceeds one. That is the ***"Schwarz inequality"*** $|v \cdot w| \leq \|v\| \, \|w\|$ for dot products—or more correctly the Cauchy-Schwarz-Buniakowsky inequality. It was found in France and Germany and Russia (and maybe elsewhere—it is the most important inequality in mathematics).

Since $|\cos\theta|$ never exceeds 1, the cosine formula gives two great inequalities:

SCHWARZ INEQUALITY $\quad |v \cdot w| \leq \|v\| \, \|w\|$

TRIANGLE INEQUALITY $\quad \|v + w\| \leq \|v\| + \|w\|$

Example 5 Find $\cos\theta$ for $v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and check both inequalities.

Solution The dot product is $v \cdot w = 4$. Both v and w have length $\sqrt{5}$. The cosine is $4/5$.

$$\cos\theta = \frac{v \cdot w}{\|v\| \, \|w\|} = \frac{4}{\sqrt{5}\sqrt{5}} = \frac{4}{5}.$$

The angle is below 90° because $v \cdot w = 4$ is positive. By the Schwarz inequality, $v \cdot w = 4$ is less than $\|v\| \, \|w\| = 5$. Side $3 = \|v + w\|$ is less than side 1 + side 2, by the triangle inequality. For $v + w = (3, 3)$ that says $\sqrt{18} < \sqrt{5} + \sqrt{5}$. Square this to get $18 < 20$.

Example 6 The dot product of $v = (a, b)$ and $w = (b, a)$ is $2ab$. Both lengths are $\sqrt{a^2 + b^2}$. The Schwarz inequality in this case says that $2ab \leq a^2 + b^2$.

This is more famous if we write $x = a^2$ and $y = b^2$. The "geometric mean" $\sqrt{xy}$ is not larger than the "arithmetic mean" = average $\frac{1}{2}(x + y)$.

$$\begin{matrix} \textbf{Geometric} \\ \textbf{mean} \end{matrix} \leq \begin{matrix} \textbf{Arithmetic} \\ \textbf{mean} \end{matrix} \qquad ab \leq \frac{a^2 + b^2}{2} \quad \text{becomes} \quad \sqrt{xy} \leq \frac{x + y}{2}.$$

Example 5 had $a = 2$ and $b = 1$. So $x = 4$ and $y = 1$. The geometric mean $\sqrt{xy} = 2$ is below the arithmetic mean $\frac{1}{2}(1 + 4) = 2.5$.

Notes on Computing

Write the components of v as $v(1), \ldots, v(N)$ and similarly for w. In FORTRAN, the sum $v + w$ requires a loop to add components separately. The dot product also uses a loop to add the separate $v(j)w(j)$. Here are VPLUSW and VDOTW:

FORTRAN

```
DO 10 J = 1,N                          DO 10 J = 1,N
10 VPLUSW(J) = v(J) + w(J)             10 VDOTW = VDOTW + V(J) * W(J)
```

MATLAB *and also* PYTHON *work directly with whole vectors, not their components.* No loop is needed. When v and w have been defined, $v + w$ is immediately understood.

Input v and w as rows—the prime $'$ transposes them to columns. $2v + 3w$ uses $*$ for multiplication by 2 and 3. The result will be printed unless the line ends in a semicolon.

MATLAB $v = [2 \quad 3 \quad 4]' \;\; ; \;\; w = [1 \quad 1 \quad 1]' \;\; ; \;\; u = 2 * v + 3 * w$

The dot product $v \cdot w$ is usually seen as ***a row times a column (with no dot)***:

Instead of $\begin{bmatrix}1\\2\end{bmatrix} \cdot \begin{bmatrix}3\\4\end{bmatrix}$ we more often see $[1\ 2]\begin{bmatrix}3\\4\end{bmatrix}$ or $v' * w$

The length of v is known to MATLAB as norm (v). We could define it ourselves as sqrt $(v' * v)$, using the square root function—also known. The cosine we have to define ourselves! The angle (in radians) comes from the *arc cosine* (acos) function:

Cosine formula	cosine $= v' * w/(\text{norm}\,(v) * \text{norm}\,(w))$
Angle formula	angle $=$ acos (cosine)

An M-file would create a new function **cosine** (v, w) for future use. The M-files created especially for this book are listed at the end. R and PYTHON are open source software.

■ REVIEW OF THE KEY IDEAS ■

1. The dot product $v \cdot w$ multiplies each component v_i by w_i and adds all $v_i w_i$.
2. The length $\|v\|$ of a vector is the square root of $v \cdot v$.
3. $u = v/\|v\|$ is a ***unit vector.*** Its length is 1.
4. The dot product is $v \cdot w = 0$ when vectors v and w are perpendicular.
5. The cosine of θ (the angle between any nonzero v and w) never exceeds 1:

$$\cos\theta = \frac{v \cdot w}{\|v\|\|w\|} \qquad \textit{Schwarz inequality} \quad |v \cdot w| \le \|v\|\,\|w\|.$$

Problem 21 will produce the ***triangle inequality*** $\|v + w\| \le \|v\| + \|w\|$.

■ WORKED EXAMPLES ■

1.2 A For the vectors $v = (3, 4)$ and $w = (4, 3)$ test the Schwarz inequality on $v \cdot w$ and the triangle inequality on $\|v + w\|$. Find $\cos\theta$ for the angle between v and w. When will we have *equality* $|v \cdot w| = \|v\|\,\|w\|$ and $\|v + w\| = \|v\| + \|w\|$?

Solution The dot product is $v \cdot w = (3)(4) + (4)(3) = 24$. The length of v is $\|v\| = \sqrt{9+16} = 5$ and also $\|w\| = 5$. The sum $v + w = (7,7)$ has length $7\sqrt{2} < 10$.

Schwarz inequality	$\|v \cdot w\| \le \|v\| \, \|w\|$ is $24 < 25$.
Triangle inequality	$\|v + w\| \le \|v\| + \|w\|$ is $7\sqrt{2} < 5 + 5$.
Cosine of angle	$\cos\theta = \frac{24}{25}$ Thin angle from $v = (3,4)$ to $w = (4,3)$

Suppose one vector is a multiple of the other as in $w = cv$. Then the angle is $0°$ or $180°$. In this case $|\cos\theta| = 1$ and $|v \cdot w|$ *equals* $\|v\| \, \|w\|$. If the angle is $0°$, as in $w = 2v$, then $\|v + w\| = \|v\| + \|w\|$. The triangle is completely flat.

1.2 B Find a unit vector u in the direction of $v = (3,4)$. Find a unit vector U that is perpendicular to u. How many possibilities for U?

Solution For a unit vector u, divide v by its length $\|v\| = 5$. For a perpendicular vector V we can choose $(-4,3)$ since the dot product $v \cdot V$ is $(3)(-4) + (4)(3) = 0$. For a *unit* vector U, divide V by its length $\|V\|$:

$$u = \frac{v}{\|v\|} = \left(\frac{3}{5}, \frac{4}{5}\right) \qquad U = \frac{V}{\|V\|} = \left(-\frac{4}{5}, \frac{3}{5}\right) \qquad u \cdot U = 0$$

The only other perpendicular unit vector would be $-U = (\frac{4}{5}, -\frac{3}{5})$.

1.2 C Find a vector $x = (c,d)$ that has dot products $x \cdot r = 1$ and $x \cdot s = 0$ with the given vectors $r = (2,-1)$ and $s = (-1,2)$.

How is this question related to Example **1.1 C**, which solved $cv + dw = b = (1,0)$?

Solution Those two dot products give linear equations for c and d. Then $x = (c,d)$.

$x \cdot r = 1$	$2c - d = 1$	**The same equations as**
$x \cdot s = 0$	$-c + 2d = 0$	**in Worked Example 1.1 C**

The second equation makes x perpendicular to $s = (-1,2)$. So I can see the geometry: Go in the perpendicular direction $(2,1)$. When you reach $x = \frac{1}{3}(2,1)$, the dot product with $r = (2,-1)$ has the required value $x \cdot r = 1$.

Comment on n equations for $x = (x_1, \ldots, x_n)$ in n-dimensional space

Section 1.1 would start with column vectors $v_1, \ldots, v_n$. The goal is to combine them to produce a required vector $x_1 v_1 + \cdots + x_n v_n = b$. This section would start from vectors $r_1, \ldots, r_n$. Now the goal is to find x with the required dot products $x \cdot r_i = b_i$.

Soon the v's will be the columns of a matrix A, and the r's will be the rows of A. Then the (one and only) problem will be to solve $Ax = b$.

Problem Set 1.2

1 Calculate the dot products $u \cdot v$ and $u \cdot w$ and $u \cdot (v + w)$ and $w \cdot v$:

$$u = \begin{bmatrix} -.6 \\ .8 \end{bmatrix} \qquad v = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \qquad w = \begin{bmatrix} 8 \\ 6 \end{bmatrix}.$$

2 Compute the lengths $\|u\|$ and $\|v\|$ and $\|w\|$ of those vectors. Check the Schwarz inequalities $|u \cdot v| \leq \|u\| \, \|v\|$ and $|v \cdot w| \leq \|v\| \, \|w\|$.

3 Find unit vectors in the directions of v and w in Problem 1, and the cosine of the angle θ. Choose vectors a, b, c that make 0°, 90°, and 180° angles with w.

4 For any *unit* vectors v and w, find the dot products (actual numbers) of

(a) v and $-v$ (b) $v + w$ and $v - w$ (c) $v - 2w$ and $v + 2w$

5 Find unit vectors u_1 and u_2 in the directions of $v = (3, 1)$ and $w = (2, 1, 2)$. Find unit vectors U_1 and U_2 that are perpendicular to u_1 and u_2.

6 (a) Describe every vector $w = (w_1, w_2)$ that is perpendicular to $v = (2, -1)$.

(b) The vectors that are perpendicular to $V = (1, 1, 1)$ lie on a _____.

(c) The vectors that are perpendicular to $(1, 1, 1)$ and $(1, 2, 3)$ lie on a _____.

7 Find the angle θ (from its cosine) between these pairs of vectors:

(a) $v = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}$ and $w = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ (b) $v = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$ and $w = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$

(c) $v = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}$ and $w = \begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix}$ (d) $v = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$.

8 True or false (give a reason if true or a counterexample if false):

(a) If u is perpendicular (in three dimensions) to v and w, those vectors v and w are parallel.

(b) If u is perpendicular to v and w, then u is perpendicular to $v + 2w$.

(c) If u and v are perpendicular unit vectors then $\|u - v\| = \sqrt{2}$.

9 The slopes of the arrows from $(0, 0)$ to (v_1, v_2) and (w_1, w_2) are v_2/v_1 and w_2/w_1. **Suppose the product $v_2 w_2 / v_1 w_1$ of those slopes is -1.** Show that $v \cdot w = 0$ and the vectors are perpendicular.

10 Draw arrows from $(0, 0)$ to the points $v = (1, 2)$ and $w = (-2, 1)$. Multiply their slopes. That answer is a signal that $v \cdot w = 0$ and the arrows are _____.

11 If $v \cdot w$ is negative, what does this say about the angle between v and w? Draw a 3-dimensional vector v (an arrow), and show where to find all w's with $v \cdot w < 0$.

12 With $v = (1, 1)$ and $w = (1, 5)$ choose a number c so that $w - cv$ is perpendicular to v. Then find the formula that gives this number c for any nonzero v and w. (*Note*: cv is the "projection" of w onto v.)

13 Find two vectors v and w that are perpendicular to $(1, 0, 1)$ and to each other.

14 Find nonzero vectors u, v, w that are perpendicular to $(1, 1, 1, 1)$ and to each other.

15 The geometric mean of $x = 2$ and $y = 8$ is $\sqrt{xy} = 4$. The arithmetic mean is larger: $\frac{1}{2}(x + y) =$ _____. This would come in Example 6 from the Schwarz inequality for $v = (\sqrt{2}, \sqrt{8})$ and $w = (\sqrt{8}, \sqrt{2})$. Find $\cos\theta$ for this v and w.

16 **How long is the vector $v = (1, 1, \ldots, 1)$ in 9 dimensions?** Find a unit vector u in the same direction as v and a unit vector w that is perpendicular to v.

17 What are the cosines of the angles α, β, θ between the vector $(1, 0, -1)$ and the unit vectors i, j, k along the axes? Check the formula $\cos^2\alpha + \cos^2\beta + \cos^2\theta = 1$.

Problems 18–31 lead to the main facts about lengths and angles in triangles.

18 The parallelogram with sides $v = (4, 2)$ and $w = (-1, 2)$ is a rectangle. Check the Pythagoras formula $a^2 + b^2 = c^2$ which is for ***right triangles only***:

$$(\text{length of } v)^2 + (\text{length of } w)^2 = (\text{length of } v + w)^2.$$

19 (Rules for dot products) These equations are simple but useful:

(1) $v \cdot w = w \cdot v$ **(2)** $u \cdot (v + w) = u \cdot v + u \cdot w$ **(3)** $(cv) \cdot w = c(v \cdot w)$

Use **(2)** with $u = v + w$ to prove $\|v + w\|^2 = v \cdot v + 2v \cdot w + w \cdot w$.

20 The "Law of Cosines" comes from $(v - w) \cdot (v - w) = v \cdot v - 2v \cdot w + w \cdot w$:

Cosine Law $$\|v - w\|^2 = \|v\|^2 - 2\|v\|\,\|w\|\cos\theta + \|w\|^2.$$

If $\theta < 90°$ show that $\|v\|^2 + \|w\|^2$ is larger than $\|v - w\|^2$ (the third side).

21 The ***triangle inequality*** says: (length of $v + w$) $\leq$ (length of v) + (length of w).

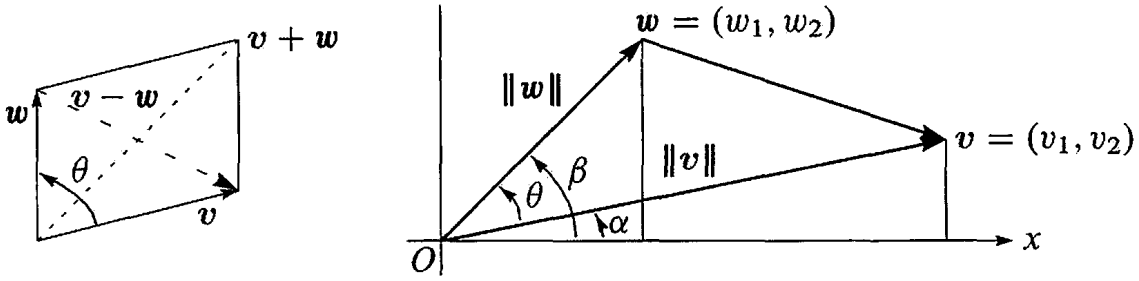
Problem 19 found $\|v + w\|^2 = \|v\|^2 + 2v \cdot w + \|w\|^2$. Use the Schwarz inequality $v \cdot w \leq \|v\|\,\|w\|$ to show that $\|\textbf{side 3}\|$ can not exceed $\|\textbf{side 1}\| + \|\textbf{side 2}\|$:

Triangle inequality $$\|v + w\|^2 \leq (\|v\| + \|w\|)^2 \quad \textbf{or} \quad \|v + w\| \leq \|v\| + \|w\|.$$

22 The Schwarz inequality $|v \cdot w| \leq \|v\|\,\|w\|$ by algebra instead of trigonometry:

(a) Multiply out both sides of $(v_1w_1 + v_2w_2)^2 \leq (v_1^2 + v_2^2)(w_1^2 + w_2^2)$.

(b) Show that the difference between those two sides equals $(v_1w_2 - v_2w_1)^2$. This cannot be negative since it is a square—so the inequality is true.



23 The figure shows that $\cos\alpha = v_1/\|v\|$ and $\sin\alpha = v_2/\|v\|$. Similarly $\cos\beta$ is _____ and $\sin\beta$ is _____. The angle θ is $\beta - \alpha$. Substitute into the trigonometry formula $\cos\beta\cos\alpha + \sin\beta\sin\alpha$ for $\cos(\beta - \alpha)$ to find $\cos\theta = v \cdot w/\|v\|\,\|w\|$.

24 One-line proof of the Schwarz inequality $|u \cdot U| \le 1$ for unit vectors:

$$|u \cdot U| \le |u_1|\,|U_1| + |u_2|\,|U_2| \le \frac{u_1^2 + U_1^2}{2} + \frac{u_2^2 + U_2^2}{2} = \frac{1+1}{2} = 1.$$

Put $(u_1, u_2) = (.6, .8)$ and $(U_1, U_2) = (.8, .6)$ in that whole line and find $\cos\theta$.

25 Why is $|\cos\theta|$ never greater than 1 in the first place?

26 If $v = (1, 2)$ draw all vectors $w = (x, y)$ in the xy plane with $v \cdot w = x + 2y = 5$. Which is the shortest w?

27 (*Recommended*) If $||v|| = 5$ and $||w|| = 3$, what are the smallest and largest values of $||v - w||$? What are the smallest and largest values of $v \cdot w$?

Challenge Problems

28 Can three vectors in the xy plane have $u \cdot v < 0$ and $v \cdot w < 0$ and $u \cdot w < 0$? I don't know how many vectors in xyz space can have all negative dot products. (Four of those vectors in the plane would certainly be impossible . . .).

29 Pick any numbers that add to $x + y + z = 0$. Find the angle between your vector $v = (x, y, z)$ and the vector $w = (z, x, y)$. Challenge question: Explain why $v \cdot w/\|v\|\,\|w\|$ is always $-\frac{1}{2}$.

30 How could you prove $\sqrt[3]{xyz} \le \frac{1}{3}(x+y+z)$ (geometric mean $\le$ arithmetic mean)?

31 Find four perpendicular unit vectors with all components equal to $\frac{1}{2}$ or $-\frac{1}{2}$.

32 Using $v = \text{randn}(3, 1)$ in MATLAB, create a random unit vector $u = v/\|v\|$. Using $V = \text{randn}(3, 30)$ create 30 more random unit vectors U_j. What is the average size of the dot products $|u \cdot U_j|$? In calculus, the average $\int_0^\pi |\cos\theta|\,d\theta/\pi = 2/\pi$.

Problem Set 1.3

1 Find the linear combination $2s_1 + 3s_2 + 4s_3 = b$. Then write b as a matrix-vector multiplication Sx. Compute the dot products (row of S)$\cdot x$:

$$s_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad s_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad s_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ go into the columns of } S.$$

2 Solve these equations $Sy = b$ with s_1, s_2, s_3 in the columns of S:

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ and } \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 9 \end{bmatrix}.$$

The sum of the first n odd numbers is _____.

3 Solve these three equations for y_1, y_2, y_3 in terms of B_1, B_2, B_3:

$$Sy = B \qquad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} B_1 \\ B_2 \\ B_3 \end{bmatrix}.$$

Write the solution y as a matrix $A = S^{-1}$ times the vector B. Are the columns of S independent or dependent?

4 Find a combination $x_1 w_1 + x_2 w_2 + x_3 w_3$ that gives the zero vector:

$$w_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad w_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \quad w_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}.$$

Those vectors are (independent) (dependent). The three vectors lie in a _____. The matrix W with those columns is *not invertible.*

5 The rows of that matrix W produce three vectors (I write them as columns):

$$r_1 = \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} \quad r_2 = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} \quad r_3 = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}.$$

Linear algebra says that these vectors must also lie in a plane. There must be many combinations with $y_1 r_1 + y_2 r_2 + y_3 r_3 = \mathbf{0}$. Find two sets of y's.

6 Which values of c give dependent columns (combination equals zero)?

$$\begin{bmatrix} 1 & 3 & 5 \\ 1 & 2 & 4 \\ 1 & 1 & c \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & c \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} c & c & c \\ 2 & 1 & 5 \\ 3 & 3 & 6 \end{bmatrix}$$

7 If the columns combine into $A\boldsymbol{x} = \mathbf{0}$ then each row has $\boldsymbol{r} \cdot \boldsymbol{x} = 0$:

$$\begin{bmatrix} & & \\ \boldsymbol{a}_1 & \boldsymbol{a}_2 & \boldsymbol{a}_3 \\ & & \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad \textbf{By rows} \begin{bmatrix} \boldsymbol{r}_1 \cdot \boldsymbol{x} \\ \boldsymbol{r}_2 \cdot \boldsymbol{x} \\ \boldsymbol{r}_3 \cdot \boldsymbol{x} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The three rows also lie in a plane. Why is that plane perpendicular to $\boldsymbol{x}$?

8 Moving to a 4 by 4 difference equation $A\boldsymbol{x} = \boldsymbol{b}$, find the four components x_1, x_2, x_3, x_4. Then write this solution as $\boldsymbol{x} = S\boldsymbol{b}$ to find the inverse matrix $S = A^{-1}$:

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \boldsymbol{b}.$$

9 What is the *cyclic* 4 by 4 difference matrix C? It will have 1 and -1 in each row. Find all solutions $\boldsymbol{x} = (x_1, x_2, x_3, x_4)$ to $C\boldsymbol{x} = \mathbf{0}$. The four columns of C lie in a "three-dimensional hyperplane" inside four-dimensional space.

10 A *forward* difference matrix Δ is *upper* triangular:

$$\Delta \boldsymbol{z} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} z_2 - z_1 \\ z_3 - z_2 \\ 0 - z_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \boldsymbol{b}.$$

Find z_1, z_2, z_3 from b_1, b_2, b_3. What is the inverse matrix in $\boldsymbol{z} = \Delta^{-1}\boldsymbol{b}$?

11 Show that the forward differences $(t+1)^2 - t^2$ are $2t+1$ = *odd numbers*. As in calculus, the difference $(t+1)^n - t^n$ will begin with the derivative of t^n, which is _____.

12 The last lines of the Worked Example say that the 4 by 4 centered difference matrix in (16) *is* invertible. Solve $C\boldsymbol{x} = (b_1, b_2, b_3, b_4)$ to find its inverse in $\boldsymbol{x} = C^{-1}\boldsymbol{b}$.

Challenge Problems

13 The very last words say that the 5 by 5 centered difference matrix *is not* invertible. Write down the 5 equations $C\boldsymbol{x} = \boldsymbol{b}$. Find a combination of left sides that gives zero. What combination of b_1, b_2, b_3, b_4, b_5 must be zero? (The 5 columns lie on a "4-dimensional hyperplane" in 5-dimensional space.)

14 If (a, b) is a multiple of (c, d) with $abcd \neq 0$, *show that* (a, c) *is a multiple of* (b, d). This is surprisingly important; two columns are falling on one line. You could use numbers first to see how a, b, c, d are related. The question will lead to:

The matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has dependent columns when it has dependent rows.

Chapter 4

Orthogonality

4.1 Orthogonality of the Four Subspaces

Two vectors are orthogonal when their dot product is zero: $v \cdot w = 0$ or $v^T w = 0$. This chapter moves to **orthogonal subspaces** and **orthogonal bases** and **orthogonal matrices**. The vectors in two subspaces, and the vectors in a basis, and the vectors in the columns, all pairs will be orthogonal. Think of $a^2 + b^2 = c^2$ for a *right triangle* with sides v and w.

Orthogonal vectors $\quad v^T w = 0 \quad$ and $\quad \|v\|^2 + \|w\|^2 = \|v + w\|^2.$

The right side is $(v + w)^T(v + w)$. This equals $v^T v + w^T w$ when $v^T w = w^T v = 0$.

Subspaces entered Chapter 3 to throw light on $Ax = b$. Right away we needed the column space (for b) and the nullspace (for x). Then the light turned onto A^T, uncovering two more subspaces. Those four fundamental subspaces reveal what a matrix really does.

A matrix multiplies a vector: *A times* x. At the first level this is only numbers. At the second level Ax is a combination of column vectors. The third level shows subspaces. But I don't think you have seen the whole picture until you study Figure 4.2. It fits the subspaces together, to show the hidden reality of A times x. The 90° angles between subspaces are new—and we have to say what those right angles mean.

The row space is perpendicular to the nullspace. Every row of A is perpendicular to every solution of $Ax = 0$. That gives the 90° angle on the left side of the figure. This perpendicularity of subspaces is Part 2 of the Fundamental Theorem of Linear Algebra.

The column space is perpendicular to the nullspace of A^T. When b is outside the column space—when we want to solve $Ax = b$ and can't do it—then this nullspace of A^T comes into its own. It contains the error $e = b - Ax$ in the "least-squares" solution. Least squares is the key application of linear algebra in this chapter.

Part 1 of the Fundamental Theorem gave the dimensions of the subspaces. The row and column spaces have the same dimension r (they are drawn the same size). The two nullspaces have the remaining dimensions $n - r$ and $m - r$. Now we will show that ***the row space and nullspace are orthogonal subspaces inside*** $\mathbf{R}^n$.

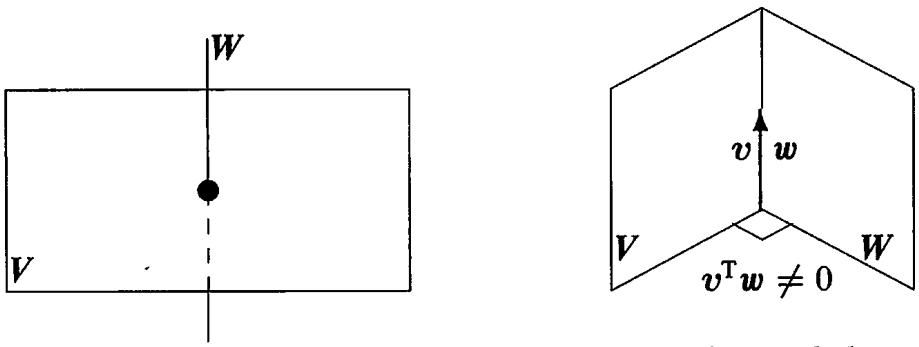
DEFINITION Two subspaces V and W of a vector space are ***orthogonal*** if every vector v in V is perpendicular to every vector w in W:

Orthogonal subspaces $v^{\mathrm{T}}w = 0$ ***for all*** v ***in*** V ***and all*** w ***in*** W.

Example 1 The floor of your room (extended to infinity) is a subspace V. The line where two walls meet is a subspace W (one-dimensional). Those subspaces are orthogonal. Every vector up the meeting line is perpendicular to every vector in the floor.

Example 2 Two walls look perpendicular but they are not orthogonal subspaces! The meeting line is in both V and W—and this line is not perpendicular to itself. Two planes (dimensions 2 and 2 in $\mathbf{R}^3$) cannot be orthogonal subspaces.

When a vector is in two orthogonal subspaces, it *must* be zero. It is perpendicular to itself. It is v and it is w, so $v^{\mathrm{T}}v = 0$. This has to be the zero vector.



orthogonal line and plane — non-orthogonal planes

Figure 4.1: Orthogonality is impossible when dim V + dim W > dimension of whole space.

The crucial examples for linear algebra come from the fundamental subspaces. Zero is the only point where the nullspace meets the row space. More than that, the nullspace and row space of A meet at 90°. This key fact comes directly from $Ax = \mathbf{0}$:

Every vector x in the nullspace is perpendicular to every row of A, because $Ax = \mathbf{0}$.
The nullspace $N(A)$ ***and the row space*** $C(A^{\mathrm{T}})$ ***are orthogonal subspaces of*** $\mathbf{R}^n$.

To see why x is perpendicular to the rows, look at $Ax = \mathbf{0}$. Each row multiplies x:

$$Ax = \begin{bmatrix} \text{row 1} \\ \vdots \\ \text{row } m \end{bmatrix} \begin{bmatrix} \\ x \\ \\ \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \quad \begin{matrix} \leftarrow \text{(row 1)} \cdot x \text{ is zero} \\ \\ \leftarrow \text{(row } m\text{)} \cdot x \text{ is zero} \end{matrix} \tag{1}$$

The first equation says that row 1 is perpendicular to x. The last equation says that row m is perpendicular to x. *Every row has a zero dot product with* x. Then x is also perpendicular to every *combination* of the rows. The whole row space $C(A^{\mathrm{T}})$ is orthogonal to $N(A)$.

Here is a second proof of that orthogonality for readers who like matrix shorthand. The vectors in the row space are combinations $A^T y$ of the rows. Take the dot product of $A^T y$ with any x in the nullspace. *These vectors are perpendicular*:

Nullspace and Row space $$x^T(A^T y) = (Ax)^T y = 0^T y = 0. \tag{2}$$

We like the first proof. You can see those rows of A multiplying x to produce zeros in equation (1). The second proof shows why A and A^T are both in the Fundamental Theorem. A^T goes with y and A goes with x. At the end we used $Ax = 0$.

Example 3 The rows of A are perpendicular to $x = (1, 1, -1)$ in the nullspace:

$$Ax = \begin{bmatrix} 1 & 3 & 4 \\ 5 & 2 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{gives the dot products} \quad \begin{matrix} 1+3-4=0 \\ 5+2-7=0 \end{matrix}$$

Now we turn to the other two subspaces. In this example, the column space is all of $\mathbf{R}^2$. The nullspace of A^T is only the zero vector (orthogonal to every vector). The columns of A and nullspace of A^T are always orthogonal subspaces.

Every vector y in the nullspace of A^T is perpendicular to every column of A. ***The left nullspace $N(A^T)$ and the column space $C(A)$ are orthogonal in $\mathbf{R}^m$.***

Apply the original proof to A^T. Its nullspace is orthogonal to its row space—and the row space of A^T is the column space of A. Q.E.D.

For a visual proof, look at $A^T y = 0$. Each column of A multiplies y to give 0:

$$C(A) \perp N(A^T) \qquad A^T y = \begin{bmatrix} (\text{column } 1)^T \\ \cdots \\ (\text{column } n)^T \end{bmatrix} \begin{bmatrix} \\ y \\ \\ \end{bmatrix} = \begin{bmatrix} 0 \\ \cdot \\ 0 \end{bmatrix}. \tag{3}$$

The dot product of y with every column of A is zero. Then y in the left nullspace is perpendicular to each column—and to the whole column space.

Orthogonal Complements

Important The fundamental subspaces are more than just orthogonal (in pairs). Their dimensions are also right. Two lines could be perpendicular in $\mathbf{R}^3$, but those lines *could not be* the row space and nullspace of a 3 by 3 matrix. The lines have dimensions 1 and 1, adding to 2. The correct dimensions r and $n - r$ must add to $n = 3$.

The fundamental subspaces have dimensions 2 and 1, or 3 and 0. Those subspaces are not only orthogonal, they are *orthogonal complements*.

DEFINITION The ***orthogonal complement*** of a subspace V contains ***every*** vector that is perpendicular to V. This orthogonal subspace is denoted by $V^\perp$ (pronounced "V perp").

By this definition, the nullspace is the orthogonal complement of the row space. ***Every*** x that is perpendicular to the rows satisfies $Ax = 0$.

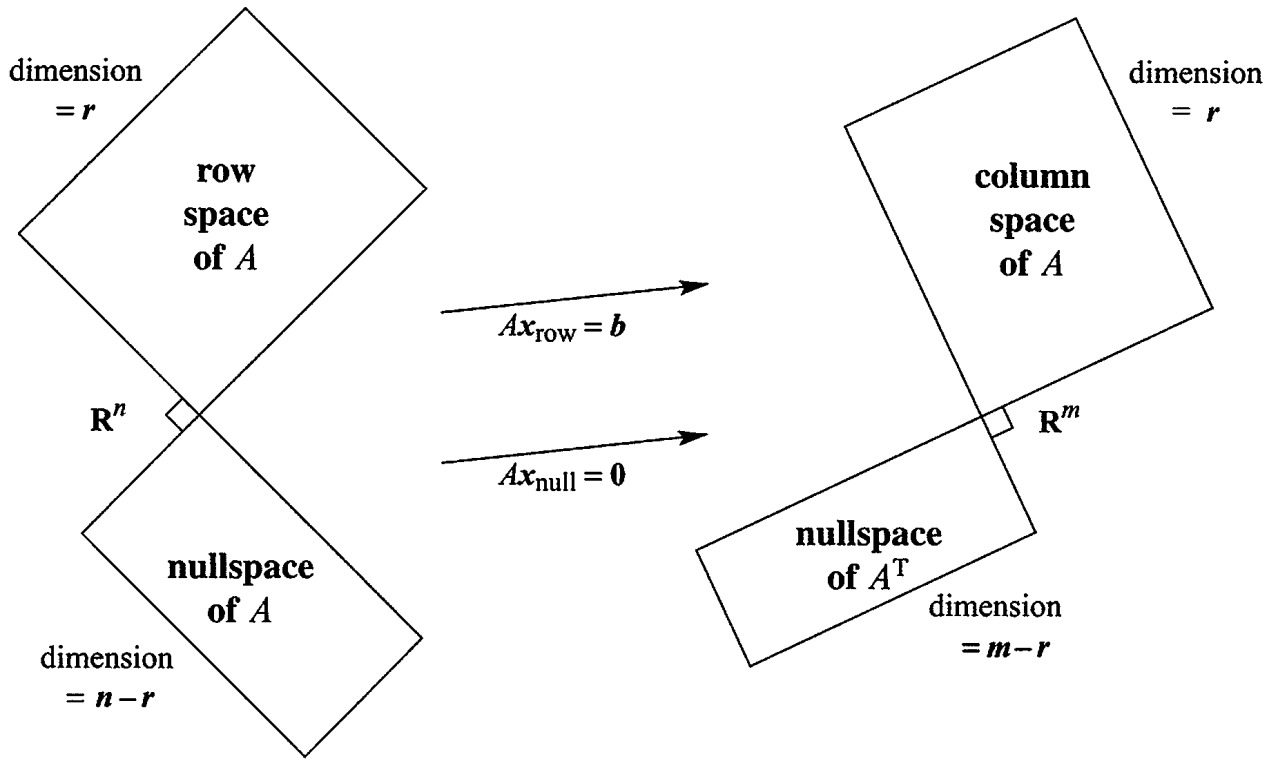


Figure 4.2: Two pairs of orthogonal subspaces. The dimensions add to n and add to m. **This is an important picture**—one pair of subspaces is in $\mathbf{R}^n$ and one pair is in $\mathbf{R}^m$.

The reverse is also true. ***If v is orthogonal to the nullspace, it must be in the row space.*** Otherwise we could add this v as an extra row of the matrix, without changing its nullspace. The row space would grow, which breaks the law $r + (n - r) = n$. We conclude that the nullspace complement $N(A)^\perp$ is exactly the row space $C(A^{\mathrm{T}})$.

The left nullspace and column space are orthogonal in $\mathbf{R}^m$, and they are orthogonal complements. Their dimensions r and $m - r$ add to the full dimension m.

***Fundamental Theorem of Linear Algebra*, Part 2**

$N(A)$ is the orthogonal complement of the row space $C(A^{\mathrm{T}})$ (in $\mathbf{R}^n$).

$N(A^{\mathrm{T}})$ is the orthogonal complement of the column space $C(A)$ (in $\mathbf{R}^m$).

Part 1 gave the dimensions of the subspaces. Part 2 gives the 90° angles between them. The point of "complements" is that every x can be split into a *row space component* x_r and a *nullspace component* x_n. When A multiplies $x = x_r + x_n$, Figure 4.3 shows what happens:

The nullspace component goes to zero: $Ax_n = 0$.

The row space component goes to the column space: $Ax_r = Ax$.

Every vector goes to the column space! Multiplying by A cannot do anything else.

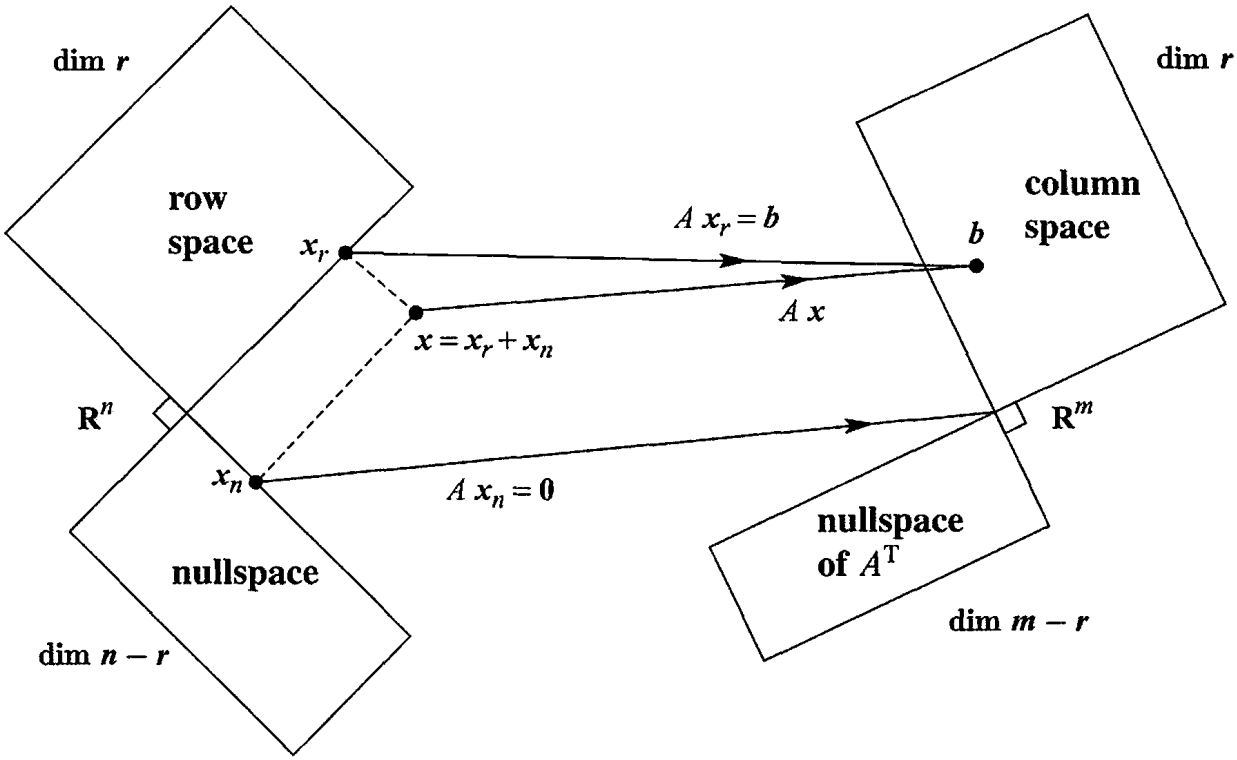


Figure 4.3: This update of Figure 4.2 shows the true action of A on $x = x_r + x_n$. Row space vector x_r to column space, nullspace vector x_n to zero.

More than that: *Every vector* $\boldsymbol{b}$ *in the column space comes from one and only one vector in the row space.* Proof: If $Ax_r = Ax'_r$, the difference $x_r - x'_r$ is in the nullspace. It is also in the row space, where x_r and x'_r came from. This difference must be the zero vector, because the nullspace and row space are perpendicular. Therefore $x_r = x'_r$.

There is an r by r invertible matrix hiding inside A, if we throw away the two nullspaces. ***From the row space to the column space,*** A ***is invertible.*** The "pseudoinverse" will invert it in Section 7.3.

Example 4 Every diagonal matrix has an r by r invertible submatrix:

$$A = \begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{contains the submatrix} \quad \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}.$$

The other eleven zeros are responsible for the nullspaces. The rank of B is also $r = 2$:

$$B = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 4 & 5 & 6 \\ 1 & 2 & 4 & 5 & 6 \end{bmatrix} \quad \text{contains} \quad \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix} \quad \text{in the pivot rows and columns.}$$

Every A becomes a diagonal matrix, when we choose the right bases for $\mathbf{R}^n$ and $\mathbf{R}^m$. This ***Singular Value Decomposition*** has become extremely important in applications.

Combining Bases from Subspaces

What follows are some valuable facts about bases. They were saved until now—when we are ready to use them. After a week you have a clearer sense of what a basis is (*linearly independent* vectors that *span* the space). Normally we have to check both of these properties. When the count is right, one property implies the other:

Any n independent vectors in $\mathbf{R}^n$ must span $\mathbf{R}^n$. So they are a basis.
Any n vectors that span $\mathbf{R}^n$ must be independent. So they are a basis.

Starting with the correct number of vectors, one property of a basis produces the other. This is true in any vector space, but we care most about $\mathbf{R}^n$. When the vectors go into the columns of an n by n *square* matrix A, here are the same two facts:

If the n columns of A are independent, they span $\mathbf{R}^n$. So $A\boldsymbol{x} = \boldsymbol{b}$ is solvable.
If the n columns span $\mathbf{R}^n$, they are independent. So $A\boldsymbol{x} = \boldsymbol{b}$ has only one solution.

Uniqueness implies existence and existence implies uniqueness. ***Then A is invertible.*** If there are no free variables, the solution $\boldsymbol{x}$ is unique. There must be n pivots. Then back substitution solves $A\boldsymbol{x} = \boldsymbol{b}$ (the solution exists).

Starting in the opposite direction, suppose $A\boldsymbol{x} = \boldsymbol{b}$ can be solved for every $\boldsymbol{b}$ (*existence of solutions*). Then elimination produced no zero rows. There are n pivots and no free variables. The nullspace contains only $\boldsymbol{x} = \mathbf{0}$ (*uniqueness* of solutions).

With bases for the row space and the nullspace, we have $r + (n - r) = n$ vectors, This is the right number. Those n vectors are independent.[2] *Therefore they span* $\mathbf{R}^n$.

Each $\boldsymbol{x}$ is the sum $\boldsymbol{x}_r + \boldsymbol{x}_n$ of a row space vector $\boldsymbol{x}_r$ and a nullspace vector $\boldsymbol{x}_n$.

The splitting in Figure 4.3 shows the key point of orthogonal complements—the dimensions add to n and all vectors are fully accounted for.

Example 5 For $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ split $\boldsymbol{x} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ into $\boldsymbol{x}_r + \boldsymbol{x}_n = \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

The vector $(2, 4)$ is in the row space. The orthogonal vector $(2, -1)$ is in the nullspace. The next section will compute this splitting for any A and $\boldsymbol{x}$, by a projection.

[2] If a combination of all n vectors gives $\boldsymbol{x}_r + \boldsymbol{x}_n = \mathbf{0}$, then $\boldsymbol{x}_r = -\boldsymbol{x}_n$ is in both subspaces. So $\boldsymbol{x}_r = \boldsymbol{x}_n = 0$. All coefficients of the row space basis and nullspace basis must be zero—which proves independence of the n vectors together.

■ REVIEW OF THE KEY IDEAS ■

1. Subspaces V and W are orthogonal if every v in V is orthogonal to every w in W.
2. V and W are "orthogonal complements" if W contains **all** vectors perpendicular to V (and vice versa). Inside $\mathbf{R}^n$, the dimensions of complements V and W add to n.
3. The nullspace $N(A)$ and the row space $C(A^{\mathrm{T}})$ are orthogonal complements, from $Ax = \mathbf{0}$. Similarly $N(A^{\mathrm{T}})$ and $C(A)$ are orthogonal complements.
4. Any n independent vectors in $\mathbf{R}^n$ will span $\mathbf{R}^n$.
5. Every x in $\mathbf{R}^n$ has a nullspace component x_n and a row space component x_r.

■ WORKED EXAMPLES ■

4.1 A Suppose S is a six-dimensional subspace of nine-dimensional space $\mathbf{R}^9$.

(a) What are the possible dimensions of subspaces orthogonal to S?

(b) What are the possible dimensions of the orthogonal complement $S^{\perp}$ of S?

(c) What is the smallest possible size of a matrix A that has row space S?

(d) What is the shape of its nullspace matrix N?

Solution

(a) If S is six-dimensional in $\mathbf{R}^9$, subspaces orthogonal to S can have dimensions 0, 1, 2, 3.

(b) The complement $S^{\perp}$ is the largest orthogonal subspace, with dimension 3.

(c) The smallest matrix A is 6 by 9 (its six rows are a basis for S).

(d) Its nullspace matrix N is 9 by 3. The columns of N contain a basis for $S^{\perp}$.

If a new row 7 of B is a combination of the six rows of A, then B has the same row space as A. It also has the same nullspace matrix N. The special solutions s_1, s_2, s_3 will be the same. Elimination will change row 7 of B to all zeros.

4.1 B The equation $x - 3y - 4z = 0$ describes a plane P in $\mathbf{R}^3$ (actually a subspace).

(a) The plane P is the nullspace $N(A)$ of what 1 by 3 matrix A?

(b) Find a basis s_1, s_2 of special solutions of $x - 3y - 4z = 0$ (these would be the columns of the nullspace matrix N).

(c) Also find a basis for the line $P^{\perp}$ that is perpendicular to P.

(d) Split $v = (6, 4, 5)$ into its nullspace component v_n in P and its row space component v_r in $P^{\perp}$.

Solution

(a) The equation $x - 3y - 4z = 0$ is $Ax = \mathbf{0}$ for the 1 by 3 matrix $A = [1 \ -3 \ -4]$.

(b) Columns 2 and 3 are free (the only pivot is 1). The special solutions with free variables 1 and 0 are $s_1 = (3, 1, 0)$ and $s_2 = (4, 0, 1)$ in the plane $P = N(A)$.

(c) The row space of A is the line $P^{\perp}$ in the direction of the row $z = (1, -3, -4)$.

(d) To split v into $v_n + v_r = (c_1 s_1 + c_2 s_2) + c_3 z$, solve for $c_1 = 1, c_2 = 1, c_3 = -1$.

$$\begin{bmatrix} 6 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 1 \\ 1 & 0 & -3 \\ 0 & 1 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \qquad \begin{array}{l} v_n = s_1 + s_2 = (7, 1, 1) \quad \text{is in} \quad P = N(A) \\ v_r = -s_3 = (-1, 3, 4) \quad \text{is in} \quad P^{\perp} = C(A^{\mathrm{T}}). \\ v \ = (6, 4, 5) \quad \text{equals} \quad (7, 1, 1) + (-1, 3, 4) \end{array}$$

This method used a basis for each subspace combined into an overall basis s_1, s_2, z. Section 4.2 will also project v onto a subspace S. There we will not need a basis for the perpendicular subspace $S^{\perp}$.

Problem Set 4.1

Questions 1–12 grow out of Figures 4.2 and 4.3 with four subspaces.

1 Construct any 2 by 3 matrix of rank one. Copy Figure 4.2 and put one vector in each subspace (two in the nullspace). Which vectors are orthogonal?

2 Redraw Figure 4.3 for a 3 by 2 matrix of rank $r = 2$. Which subspace is Z (zero vector only)? The nullspace part of any vector x in $\mathbf{R}^2$ is $x_n =$ _____.

3 Construct a matrix with the required property or say why that is impossible:

(a) Column space contains $\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$, nullspace contains $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

(b) Row space contains $\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$, nullspace contains $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

(c) $Ax = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ has a solution and $A^{\mathrm{T}} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(d) Every row is orthogonal to every column (A is not the zero matrix)

(e) Columns add up to a column of zeros, rows add to a row of 1's.

4 If $AB = 0$ then the columns of B are in the _____ of A. The rows of A are in the _____ of B. Why can't A and B be 3 by 3 matrices of rank 2?

5 (a) If $Ax = b$ has a solution and $A^T y = 0$, *is* $(y^T x = 0)$ *or* $(y^T b = 0)$?

(b) If $A^T y = (1, 1, 1)$ has a solution and $Ax = 0$, then _____.

6 This system of equations $Ax = b$ has *no solution* (they lead to $0 = 1$):

$$\begin{aligned} x + 2y + 2z &= 5 \\ 2x + 2y + 3z &= 5 \\ 3x + 4y + 5z &= 9 \end{aligned}$$

Find numbers y_1, y_2, y_3 to multiply the equations so they add to $0 = 1$. You have found a vector y in which subspace? Its dot product $y^T b$ is 1, so no solution x.

7 Every system with no solution is like the one in Problem 6. There are numbers $y_1, \ldots, y_m$ that multiply the m equations so they add up to $0 = 1$. This is called **Fredholm's Alternative**:

Exactly one of these problems has a solution

$$Ax = b \quad \textbf{OR} \quad A^T y = 0 \quad \text{with} \quad y^T b = 1.$$

If b is not in the column space of A, it is not orthogonal to the nullspace of A^T. Multiply the equations $x_1 - x_2 = 1$ and $x_2 - x_3 = 1$ and $x_1 - x_3 = 1$ by numbers y_1, y_2, y_3 chosen so that the equations add up to $0 = 1$.

8 In Figure 4.3, how do we know that Ax_r is equal to Ax? How do we know that this vector is in the column space? If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ what is x_r?

9 If $A^T Ax = 0$ then $Ax = 0$. Reason: Ax is in the nullspace of A^T and also in the _____ of A and those spaces are _____. *Conclusion*: $A^T A$ *has the same nullspace as* A. *This key fact is repeated in the next section.*

10 Suppose A is a symmetric matrix $(A^T = A)$.

(a) Why is its column space perpendicular to its nullspace?

(b) If $Ax = 0$ and $Az = 5z$, which subspaces contain these "eigenvectors" x and z? **Symmetric matrices have perpendicular eigenvectors** $x^T z = 0$.

11 (Recommended) Draw Figure 4.2 to show each subspace correctly for

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}.$$

12 Find the pieces x_r and x_n and draw Figure 4.3 properly if

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad x = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Questions 13–23 are about orthogonal subspaces.

13 Put bases for the subspaces V and W into the columns of matrices V and W. Explain why the test for orthogonal subspaces can be written $V^{\mathrm{T}}W$ = zero matrix. This matches $v^{\mathrm{T}}w = 0$ for orthogonal vectors.

14 The floor V and the wall W are not orthogonal subspaces, because they share a nonzero vector (along the line where they meet). No planes V and W in $\mathbf{R}^3$ can be orthogonal! Find a vector in the column spaces of both matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 5 & 4 \\ 6 & 3 \\ 5 & 1 \end{bmatrix}$$

This will be a vector Ax and also $B\widehat{x}$. Think 3 by 4 with the matrix $[\,A \;\; B\,]$.

15 Extend Problem 14 to a p-dimensional subspace V and a q-dimensional subspace W of $\mathbf{R}^n$. What inequality on $p + q$ guarantees that V intersects W in a nonzero vector? These subspaces cannot be orthogonal.

16 Prove that every y in $N(A^{\mathrm{T}})$ is perpendicular to every Ax in the column space, using the matrix shorthand of equation (2). Start from $A^{\mathrm{T}}y = \mathbf{0}$.

17 If S is the subspace of $\mathbf{R}^3$ containing only the zero vector, what is $S^{\perp}$? If S is spanned by $(1, 1, 1)$, what is $S^{\perp}$? If S is spanned by $(1, 1, 1)$ and $(1, 1, -1)$, what is a basis for $S^{\perp}$?

18 Suppose S only contains two vectors $(1, 5, 1)$ and $(2, 2, 2)$ (not a subspace). Then $S^{\perp}$ is the nullspace of the matrix $A =$ _____. $S^{\perp}$ is a subspace even if S is not.

19 Suppose L is a one-dimensional subspace (a line) in $\mathbf{R}^3$. Its orthogonal complement $L^{\perp}$ is the _____ perpendicular to L. Then $(L^{\perp})^{\perp}$ is a _____ perpendicular to $L^{\perp}$. In fact $(L^{\perp})^{\perp}$ is the same as _____.

20 Suppose V is the whole space $\mathbf{R}^4$. Then $V^{\perp}$ contains only the vector _____. Then $(V^{\perp})^{\perp}$ is _____. So $(V^{\perp})^{\perp}$ is the same as _____.

21 Suppose S is spanned by the vectors $(1, 2, 2, 3)$ and $(1, 3, 3, 2)$. Find two vectors that span $S^{\perp}$. This is the same as solving $Ax = \mathbf{0}$ for which A?

22 If P is the plane of vectors in $\mathbf{R}^4$ satisfying $x_1 + x_2 + x_3 + x_4 = 0$, write a basis for $P^{\perp}$. Construct a matrix that has P as its nullspace.

23 If a subspace S is contained in a subspace V, prove that $S^{\perp}$ contains $V^{\perp}$.

Questions 24–30 are about perpendicular columns and rows.

24 Suppose an n by n matrix is invertible: $AA^{-1} = I$. Then the first column of A^{-1} is orthogonal to the space spanned by which rows of A?

25 Find $A^{\mathrm{T}}A$ if the columns of A are unit vectors, all mutually perpendicular.

26 Construct a 3 by 3 matrix A with no zero entries whose columns are mutually perpendicular. Compute $A^{\mathrm{T}}A$. Why is it a diagonal matrix?

27 The lines $3x + y = b_1$ and $6x + 2y = b_2$ are _____. They are the same line if _____. In that case (b_1, b_2) is perpendicular to the vector _____. The nullspace of the matrix is the line $3x + y =$ _____. One particular vector in that nullspace is _____.

28 Why is each of these statements false?

(a) $(1, 1, 1)$ is perpendicular to $(1, 1, -2)$ so the planes $x + y + z = 0$ and $x + y - 2z = 0$ are orthogonal subspaces.

(b) The subspace spanned by $(1, 1, 0, 0, 0)$ and $(0, 0, 0, 1, 1)$ is the orthogonal complement of the subspace spanned by $(1, -1, 0, 0, 0)$ and $(2, -2, 3, 4, -4)$.

(c) Two subspaces that meet only in the zero vector are orthogonal.

29 Find a matrix with $v = (1, 2, 3)$ in the row space and column space. Find another matrix with v in the nullspace and column space. Which pairs of subspaces can v *not* be in?

Challenge Problems

30 Suppose A is 3 by 4 and B is 4 by 5 and $AB = 0$. So $N(A)$ contains $C(B)$. Prove from the dimensions of $N(A)$ and $C(B)$ that $\text{rank}(A) + \text{rank}(B) \leq 4$.

31 The command $N = \text{null}(A)$ will produce a basis for the nullspace of A. Then the command $B = \text{null}(N')$ will produce a basis for the _____ of A.

32 Suppose I give you four nonzero vectors $\boldsymbol{r}, \boldsymbol{n}, \boldsymbol{c}, \boldsymbol{l}$ in $\mathbf{R}^2$.

(a) What are the conditions for those to be bases for the four fundamental subspaces $C(A^{\mathrm{T}}), N(A), C(A), N(A^{\mathrm{T}})$ of a 2 by 2 matrix?

(b) What is one possible matrix A?

33 Suppose I give you eight vectors $\boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{n}_1, \boldsymbol{n}_2, \boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{l}_1, \boldsymbol{l}_2$ in $\mathbf{R}^4$.

(a) What are the conditions for those pairs to be bases for the four fundamental subspaces of a 4 by 4 matrix?

(b) What is one possible matrix A?

4.2 Projections

May we start this section with two questions? (In addition to that one.) The first question aims to show that projections are easy to visualize. The second question is about "projection matrices"—symmetric matrices with $P^2 = P$. *The projection of* $\boldsymbol{b}$ *is* $P\boldsymbol{b}$.

1 What are the projections of $\boldsymbol{b} = (2, 3, 4)$ onto the z axis and the xy plane?

2 What matrices produce those projections onto a line and a plane?

When $\boldsymbol{b}$ is projected onto a line, ***its projection*** $\boldsymbol{p}$ ***is the part of*** $\boldsymbol{b}$ ***along that line.*** If $\boldsymbol{b}$ is projected onto a plane, $\boldsymbol{p}$ is the part in that plane. ***The projection*** $\boldsymbol{p}$ ***is*** $P\boldsymbol{b}$.

The projection matrix P multiplies $\boldsymbol{b}$ to give $\boldsymbol{p}$. This section finds $\boldsymbol{p}$ and P.

The projection onto the z axis we call $\boldsymbol{p}_1$. The second projection drops straight down to the xy plane. The picture in your mind should be Figure 4.4. Start with $\boldsymbol{b} = (2, 3, 4)$. One projection gives $\boldsymbol{p}_1 = (0, 0, 4)$ and the other gives $\boldsymbol{p}_2 = (2, 3, 0)$. Those are the parts of $\boldsymbol{b}$ along the z axis and in the xy plane.

The projection matrices P_1 and P_2 are 3 by 3. They multiply $\boldsymbol{b}$ with 3 components to produce $\boldsymbol{p}$ with 3 components. Projection onto a line comes from a rank one matrix. Projection onto a plane comes from a rank two matrix:

$$\text{Onto the } z \text{ axis: } P_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \text{Onto the } xy \text{ plane: } P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

P_1 picks out the z component of every vector. P_2 picks out the x and y components. To find the projections $\boldsymbol{p}_1$ and $\boldsymbol{p}_2$ of $\boldsymbol{b}$, multiply $\boldsymbol{b}$ by P_1 and P_2 (small $\boldsymbol{p}$ for the vector, capital P for the matrix that produces it):

$$\boldsymbol{p}_1 = P_1\boldsymbol{b} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ z \end{bmatrix} \qquad \boldsymbol{p}_2 = P_2\boldsymbol{b} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}.$$

In this case the projections p_1 and p_2 are perpendicular. The xy plane and the z axis are ***orthogonal subspaces***, like the floor of a room and the line between two walls.

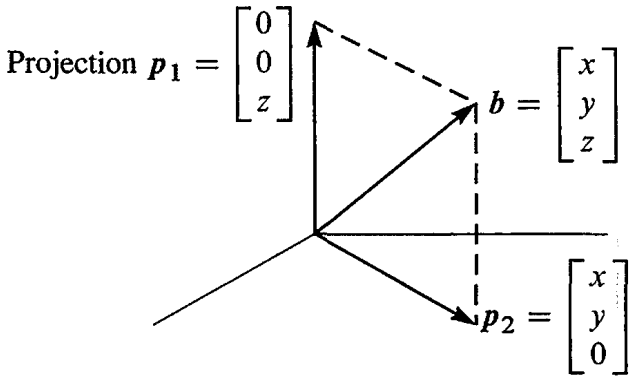


Figure 4.4: The projections $\boldsymbol{p}_1 = P_1\boldsymbol{b}$ and $\boldsymbol{p}_2 = P_2\boldsymbol{b}$ onto the z axis and the xy plane.

More than that, the line and plane are orthogonal ***complements***. Their dimensions add to $1+2=3$. Every vector $\boldsymbol{b}$ in the whole space is the sum of its parts in the two subspaces. The projections $\boldsymbol{p}_1$ and $\boldsymbol{p}_2$ are exactly those parts:

$$\text{The vectors give } \boldsymbol{p}_1 + \boldsymbol{p}_2 = \boldsymbol{b}. \qquad \text{The matrices give } P_1 + P_2 = I. \tag{1}$$

This is perfect. Our goal is reached—for this example. We have the same goal for any line and any plane and any n-dimensional subspace. The object is to find the part $\boldsymbol{p}$ in each subspace, and the projection matrix P that produces that part $\boldsymbol{p} = P\boldsymbol{b}$. Every subspace of $\mathbf{R}^m$ has its own m by m projection matrix. To compute P, we absolutely need a good description of the subspace that it projects onto.

The best description of a subspace is a basis. We put the basis vectors into the columns of A. ***Now we are projecting onto the column space of A!*** Certainly the z axis is the column space of the 3 by 1 matrix A_1. The xy plane is the column space of A_2. That plane is also the column space of A_3 (a subspace has many bases):

$$A_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad A_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad A_3 = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 0 & 0 \end{bmatrix}.$$

Our problem is ***to project any $\boldsymbol{b}$ onto the column space of any m by n matrix.*** Start with a line (dimension $n = 1$). The matrix A has only one column. Call it $\boldsymbol{a}$.

Projection Onto a Line

A line goes through the origin in the direction of $\boldsymbol{a} = (a_1, \ldots, a_m)$. Along that line, we want the point $\boldsymbol{p}$ closest to $\boldsymbol{b} = (b_1, \ldots, b_m)$. The key to projection is orthogonality: ***The line from $\boldsymbol{b}$ to $\boldsymbol{p}$ is perpendicular to the vector $\boldsymbol{a}$.*** This is the dotted line marked $\boldsymbol{e}$ for error in Figure 4.5—which we now compute by algebra.

The projection $\boldsymbol{p}$ is some multiple of $\boldsymbol{a}$. Call it $\boldsymbol{p} = \widehat{x}\boldsymbol{a}$ = "x hat" times $\boldsymbol{a}$. Computing this number $\widehat{x}$ will give the vector $\boldsymbol{p}$. Then from the formula for $\boldsymbol{p}$, we read off the projection matrix P. These three steps will lead to all projection matrices: *find $\widehat{x}$, then find the vector $\boldsymbol{p}$, then find the matrix P.*

The dotted line $\boldsymbol{b} - \boldsymbol{p}$ is $\boldsymbol{e} = \boldsymbol{b} - \widehat{x}\boldsymbol{a}$. It is perpendicular to $\boldsymbol{a}$—this will determine $\widehat{x}$. Use the fact that $\boldsymbol{b} - \boldsymbol{p}$ is perpendicular to $\boldsymbol{a}$ when their dot product is zero:

Projecting $\boldsymbol{b}$ onto $\boldsymbol{a}$, error $\boldsymbol{e} = \boldsymbol{b} - \widehat{x}\boldsymbol{a}$

$$\boldsymbol{a} \cdot (\boldsymbol{b} - \widehat{x}\boldsymbol{a}) = 0 \quad \text{or} \quad \boldsymbol{a} \cdot \boldsymbol{b} - \widehat{x}\boldsymbol{a} \cdot \boldsymbol{a} = 0$$

$$\widehat{x} = \frac{\boldsymbol{a} \cdot \boldsymbol{b}}{\boldsymbol{a} \cdot \boldsymbol{a}} = \frac{\boldsymbol{a}^{\mathrm{T}}\boldsymbol{b}}{\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}}. \tag{2}$$

The multiplication $\boldsymbol{a}^{\mathrm{T}}\boldsymbol{b}$ is the same as $\boldsymbol{a} \cdot \boldsymbol{b}$. Using the transpose is better, because it applies also to matrices. Our formula $\widehat{x} = \boldsymbol{a}^{\mathrm{T}}\boldsymbol{b}/\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}$ gives the projection $\boldsymbol{p} = \widehat{x}\boldsymbol{a}$.

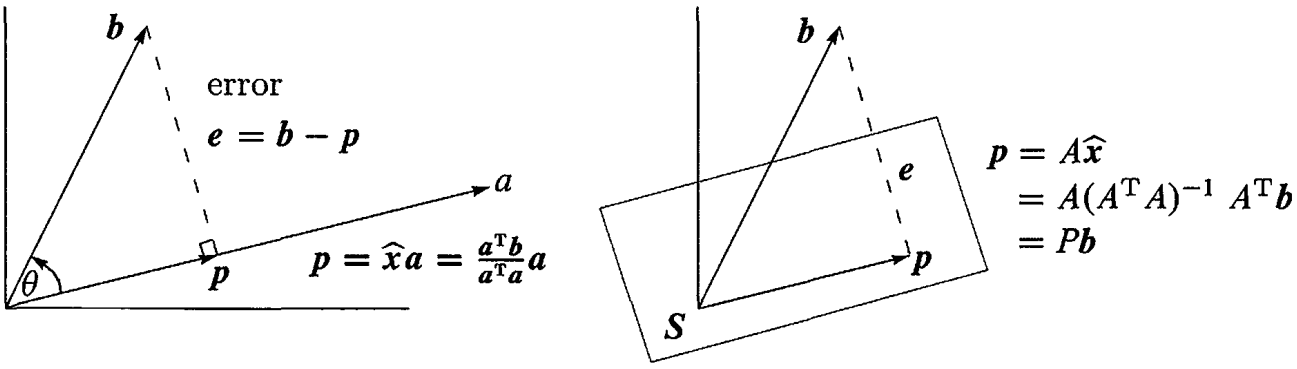


Figure 4.5: The projection p of b onto a line and onto S = column space of A.

The projection of b onto the line through a is the vector $p = \widehat{x}a = \dfrac{a^{\mathrm{T}}b}{a^{\mathrm{T}}a}a$.

Special case 1: If $b = a$ then $\widehat{x} = 1$. The projection of a onto a is itself. $Pa = a$.

Special case 2: If b is perpendicular to a then $a^{\mathrm{T}}b = 0$. The projection is $p = \mathbf{0}$.

Example 1 Project $b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ onto $a = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ to find $p = \widehat{x}a$ in Figure 4.5.

Solution The number $\widehat{x}$ is the ratio of $a^{\mathrm{T}}b = 5$ to $a^{\mathrm{T}}a = 9$. So the projection is $p = \frac{5}{9}a$. The error vector between b and p is $e = b - p$. Those vectors p and e will add to $b = (1, 1, 1)$:

$$p = \frac{5}{9}a = \left(\frac{5}{9}, \frac{10}{9}, \frac{10}{9}\right) \quad \text{and} \quad e = b - p = \left(\frac{4}{9}, -\frac{1}{9}, -\frac{1}{9}\right).$$

The error e should be perpendicular to $a = (1, 2, 2)$ and it is: $e^{\mathrm{T}}a = \frac{4}{9} - \frac{2}{9} - \frac{2}{9} = 0$.

Look at the right triangle of b, p, and e. The vector b is split into two parts—its component along the line is p, its perpendicular part is e. Those two sides of a right triangle have length $\|b\| \cos\theta$ and $\|b\| \sin\theta$. Trigonometry matches the dot product:

$$p = \frac{a^{\mathrm{T}}b}{a^{\mathrm{T}}a}a \quad \text{has length} \quad \|p\| = \frac{\|a\|\,\|b\| \cos\theta}{\|a\|^2}\|a\| = \|b\| \cos\theta. \tag{3}$$

The dot product is a lot simpler than getting involved with $\cos\theta$ and the length of b. The example has square roots in $\cos\theta = 5/3\sqrt{3}$ and $\|b\| = \sqrt{3}$. There are no square roots in the projection $p = 5a/9$. The good way to $5/9$ is $b^{\mathrm{T}}a/a^{\mathrm{T}}a$.

Now comes the ***projection matrix***. In the formula for p, what matrix is multiplying b? You can see the matrix better if the number $\widehat{x}$ is on the right side of a:

Projection matrix P $\quad p = a\widehat{x} = a\dfrac{a^{\mathrm{T}}b}{a^{\mathrm{T}}a} = Pb$ ***when the matrix is*** $\quad P = \dfrac{aa^{\mathrm{T}}}{a^{\mathrm{T}}a}$.

P is a column times a row! The column is $\boldsymbol{a}$, the row is $\boldsymbol{a}^{\mathrm{T}}$. Then divide by the number $\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}$. The projection matrix P is m by m, but ***its rank is one***. We are projecting onto a one-dimensional subspace, the line through $\boldsymbol{a}$. That is the column space of P.

Example 2 Find the projection matrix $P = \dfrac{\boldsymbol{a}\boldsymbol{a}^{\mathrm{T}}}{\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}}$ onto the line through $\boldsymbol{a} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$.

Solution Multiply column $\boldsymbol{a}$ times row $\boldsymbol{a}^{\mathrm{T}}$ and divide by $\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a} = 9$:

$$\text{Projection matrix} \qquad P = \frac{\boldsymbol{a}\boldsymbol{a}^{\mathrm{T}}}{\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}} = \frac{1}{9}\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}\begin{bmatrix} 1 & 2 & 2 \end{bmatrix} = \frac{1}{9}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}.$$

This matrix projects *any* vector $\boldsymbol{b}$ onto $\boldsymbol{a}$. Check $\boldsymbol{p} = P\boldsymbol{b}$ for $\boldsymbol{b} = (1, 1, 1)$ in Example 1:

$$\boldsymbol{p} = P\boldsymbol{b} = \frac{1}{9}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{9}\begin{bmatrix} 5 \\ 10 \\ 10 \end{bmatrix} \quad \text{which is correct.}$$

If the vector $\boldsymbol{a}$ is doubled, the matrix P stays the same. It still projects onto the same line. If the matrix is squared, P^2 equals P. ***Projecting a second time doesn't change anything***, so $\boldsymbol{P^2 = P}$. The diagonal entries of P add up to $\frac{1}{9}(1 + 4 + 4) = 1$.

The matrix $I - P$ should be a projection too. It produces the other side $\boldsymbol{e}$ of the triangle—the perpendicular part of $\boldsymbol{b}$. Note that $(I - P)\boldsymbol{b}$ equals $\boldsymbol{b} - \boldsymbol{p}$ which is $\boldsymbol{e}$ in the left nullspace. ***When P projects onto one subspace, $I - P$ projects onto the perpendicular subspace.*** Here $I - P$ projects onto the plane perpendicular to $\boldsymbol{a}$.

Now we move beyond projection onto a line. Projecting onto an n-dimensional subspace of $\mathbf{R}^m$ takes more effort. The crucial formulas will be collected in equations (5)–(6)–(7). Basically you need to remember those three equations.

Projection Onto a Subspace

Start with n vectors $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ in $\mathbf{R}^m$. Assume that these $\boldsymbol{a}$'s are linearly independent.

Problem: Find the combination $\boldsymbol{p} = \widehat{x}_1\boldsymbol{a}_1 + \cdots + \widehat{x}_n\boldsymbol{a}_n$ closest to a given vector $\boldsymbol{b}$. We are projecting each $\boldsymbol{b}$ in $\mathbf{R}^m$ onto the subspace spanned by the $\boldsymbol{a}$'s, to get $\boldsymbol{p}$.

With $n = 1$ (only one vector $\boldsymbol{a}_1$) this is projection onto a line. The line is the column space of A, which has just one column. In general the matrix A has n columns $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$.

The combinations in $\mathbf{R}^m$ are the vectors $A\boldsymbol{x}$ in the column space. We are looking for the particular combination $\boldsymbol{p} = A\widehat{\boldsymbol{x}}$ (***the projection***) that is closest to $\boldsymbol{b}$. The hat over $\widehat{\boldsymbol{x}}$ indicates the *best* choice $\widehat{\boldsymbol{x}}$, to give the closest vector in the column space. That choice is $\boldsymbol{a}^{\mathrm{T}}\boldsymbol{b}/\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}$ when $n = 1$. For $n > 1$, the best $\widehat{\boldsymbol{x}}$ is to be found now.

We compute projections onto n-dimensional subspaces in three steps as before: ***Find the vector $\widehat{\boldsymbol{x}}$, find the projection $\boldsymbol{p} = A\widehat{\boldsymbol{x}}$, find the matrix P.***

The key is in the geometry! The dotted line in Figure 4.5 goes from $\boldsymbol{b}$ to the nearest point $A\widehat{\boldsymbol{x}}$ in the subspace. ***This error vector $\boldsymbol{b} - A\widehat{\boldsymbol{x}}$ is perpendicular to the subspace.***

The error $\boldsymbol{b} - A\widehat{\boldsymbol{x}}$ makes a right angle with all the vectors $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$. The n right angles give the n equations for $\widehat{\boldsymbol{x}}$:

$$\begin{matrix} \boldsymbol{a}_1^{\mathrm{T}}(\boldsymbol{b} - A\widehat{\boldsymbol{x}}) = 0 \\ \vdots \\ \boldsymbol{a}_n^{\mathrm{T}}(\boldsymbol{b} - A\widehat{\boldsymbol{x}}) = 0 \end{matrix} \quad \text{or} \quad \begin{bmatrix} -\boldsymbol{a}_1^{\mathrm{T}}- \\ \vdots \\ -\boldsymbol{a}_n^{\mathrm{T}}- \end{bmatrix} \begin{bmatrix} \\ \boldsymbol{b} - A\widehat{\boldsymbol{x}} \\ \\ \end{bmatrix} = \begin{bmatrix} \\ \boldsymbol{0} \\ \\ \end{bmatrix}. \tag{4}$$

The matrix with those rows $\boldsymbol{a}_i^{\mathrm{T}}$ is A^{T}. The n equations are exactly $A^{\mathrm{T}}(\boldsymbol{b} - A\widehat{\boldsymbol{x}}) = \boldsymbol{0}$.

Rewrite $A^{\mathrm{T}}(\boldsymbol{b} - A\widehat{\boldsymbol{x}}) = \boldsymbol{0}$ in its famous form $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$. This is the equation for $\widehat{\boldsymbol{x}}$, and the coefficient matrix is $A^{\mathrm{T}}A$. Now we can find $\widehat{\boldsymbol{x}}$ and $\boldsymbol{p}$ and P, in that order:

The combination $\boldsymbol{p} = \widehat{x}_1\boldsymbol{a}_1 + \cdots + \widehat{x}_n\boldsymbol{a}_n = A\widehat{\boldsymbol{x}}$ that is closest to $\boldsymbol{b}$ comes from

$$A^{\mathrm{T}}(\boldsymbol{b} - A\widehat{\boldsymbol{x}}) = \boldsymbol{0} \quad \text{or} \quad A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}. \tag{5}$$

This symmetric matrix $A^{\mathrm{T}}A$ is n by n. It is invertible if the $\boldsymbol{a}$'s are independent. The solution is $\widehat{\boldsymbol{x}} = (A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}\boldsymbol{b}$. The ***projection*** of $\boldsymbol{b}$ onto the subspace is $\boldsymbol{p}$:

$$\boldsymbol{p} = A\widehat{\boldsymbol{x}} = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}\boldsymbol{b}. \tag{6}$$

This formula shows the n by n ***projection matrix*** that produces $\boldsymbol{p} = P\boldsymbol{b}$:

$$P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}. \tag{7}$$

Compare with projection onto a line, when the matrix A has only one column $\boldsymbol{a}$:

For $n = 1$ $\quad \widehat{x} = \dfrac{\boldsymbol{a}^{\mathrm{T}}\boldsymbol{b}}{\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}}$ **and** $\boldsymbol{p} = \boldsymbol{a}\dfrac{\boldsymbol{a}^{\mathrm{T}}\boldsymbol{b}}{\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}}$ **and** $P = \dfrac{\boldsymbol{a}\boldsymbol{a}^{\mathrm{T}}}{\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}}$.

Those formulas are identical with (5) and (6) and (7). The number $\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}$ becomes the matrix $A^{\mathrm{T}}A$. When it is a number, we divide by it. When it is a matrix, we invert it. The new formulas contain $(A^{\mathrm{T}}A)^{-1}$ instead of $1/\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}$. The linear independence of the columns $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ will guarantee that this inverse matrix exists.

The key step was $A^{\mathrm{T}}(\boldsymbol{b} - A\widehat{\boldsymbol{x}}) = \boldsymbol{0}$. We used geometry ($\boldsymbol{e}$ is perpendicular to all the $\boldsymbol{a}$'s). Linear algebra gives this "normal equation" too, in a very quick way:

1. Our subspace is the column space of A.
2. The error vector $\boldsymbol{b} - A\widehat{\boldsymbol{x}}$ is perpendicular to that column space.
3. Therefore $\boldsymbol{b} - A\widehat{\boldsymbol{x}}$ is in the nullspace of A^{T}. This means $A^{\mathrm{T}}(\boldsymbol{b} - A\widehat{\boldsymbol{x}}) = \boldsymbol{0}$.

The left nullspace is important in projections. That nullspace of A^{T} contains the error vector $\boldsymbol{e} = \boldsymbol{b} - A\widehat{\boldsymbol{x}}$. The vector $\boldsymbol{b}$ is being split into the projection $\boldsymbol{p}$ and the error $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$. Projection produces a right triangle (Figure 4.5) with sides $\boldsymbol{p}$, $\boldsymbol{e}$, and $\boldsymbol{b}$.

Example 3 If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$ and $\boldsymbol{b} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$ find $\widehat{\boldsymbol{x}}$ and $\boldsymbol{p}$ and P.

Solution Compute the square matrix $A^{\mathrm{T}}A$ and also the vector $A^{\mathrm{T}}\boldsymbol{b}$:

$$A^{\mathrm{T}}A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \quad \text{and} \quad A^{\mathrm{T}}\boldsymbol{b} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}.$$

Now solve the normal equation $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$ to find $\widehat{\boldsymbol{x}}$:

$$\begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} \widehat{x}_1 \\ \widehat{x}_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix} \quad \text{gives} \quad \widehat{\boldsymbol{x}} = \begin{bmatrix} \widehat{x}_1 \\ \widehat{x}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{5} \\ \mathbf{-3} \end{bmatrix}. \tag{8}$$

The combination $\boldsymbol{p} = A\widehat{\boldsymbol{x}}$ is the projection of $\boldsymbol{b}$ onto the column space of A:

$$\boldsymbol{p} = \mathbf{5}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \mathbf{3}\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}. \quad \text{The error is} \quad \boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}. \tag{9}$$

Two checks on the calculation. First, the error $\boldsymbol{e} = (1, -2, 1)$ is perpendicular to both columns $(1, 1, 1)$ and $(0, 1, 2)$. Second, the final P times $\boldsymbol{b} = (6, 0, 0)$ correctly gives $\boldsymbol{p} = (5, 2, -1)$. That solves the problem for one particular $\boldsymbol{b}$.

To find $\boldsymbol{p} = P\boldsymbol{b}$ for every $\boldsymbol{b}$, compute $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$. The determinant of $A^{\mathrm{T}}A$ is $15 - 9 = 6$; then $(A^{\mathrm{T}}A)^{-1}$ is easy. Multiply A times $(A^{\mathrm{T}}A)^{-1}$ times A^{T} to reach P:

$$(A^{\mathrm{T}}A)^{-1} = \frac{1}{6}\begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix} \quad \text{and} \quad P = \frac{1}{6}\begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}. \tag{10}$$

We must have $P^2 = P$, because a second projection doesn't change the first projection.

Warning The matrix $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ is deceptive. You might try to split $(A^{\mathrm{T}}A)^{-1}$ into A^{-1} times $(A^{\mathrm{T}})^{-1}$. If you make that mistake, and substitute it into P, you will find $P = AA^{-1}(A^{\mathrm{T}})^{-1}A^{\mathrm{T}}$. Apparently everything cancels. This looks like $P = I$, the identity matrix. We want to say why this is wrong.

The matrix A is rectangular. It has no inverse matrix. We cannot split $(A^{\mathrm{T}}A)^{-1}$ into A^{-1} times $(A^{\mathrm{T}})^{-1}$ because there is no A^{-1} in the first place.

In our experience, a problem that involves a rectangular matrix almost always leads to $A^{\mathrm{T}}A$. When A has independent columns, $A^{\mathrm{T}}A$ is invertible. This fact is so crucial that we state it clearly and give a proof.

$A^{\mathrm{T}}A$ is invertible if and only if A has linearly independent columns.

Proof $A^{\mathrm{T}}A$ is a square matrix (n by n). For every matrix A, we will now show that $A^{\mathrm{T}}A$ ***has the same nullspace as*** A. When the columns of A are linearly independent, its nullspace contains only the zero vector. Then $A^{\mathrm{T}}A$, with this same nullspace, is invertible.

Let A be any matrix. If $\boldsymbol{x}$ is in its nullspace, then $A\boldsymbol{x} = \boldsymbol{0}$. Multiplying by A^{T} gives $A^{\mathrm{T}}A\boldsymbol{x} = \boldsymbol{0}$. So $\boldsymbol{x}$ is also in the nullspace of $A^{\mathrm{T}}A$.

Now start with the nullspace of $A^{\mathrm{T}}A$. **From $A^{\mathrm{T}}A\boldsymbol{x} = \boldsymbol{0}$ we must prove $A\boldsymbol{x} = \boldsymbol{0}$.** We can't multiply by $(A^{\mathrm{T}})^{-1}$, which generally doesn't exist. Just multiply by $\boldsymbol{x}^{\mathrm{T}}$:

$$(\boldsymbol{x}^{\mathrm{T}})A^{\mathrm{T}}A\boldsymbol{x} = 0 \quad \text{or} \quad (A\boldsymbol{x})^{\mathrm{T}}(A\boldsymbol{x}) = 0 \quad \text{or} \quad \|A\boldsymbol{x}\|^2 = 0.$$

This says: If $A^{\mathrm{T}}A\boldsymbol{x} = \boldsymbol{0}$ then $A\boldsymbol{x}$ has length zero. Therefore $A\boldsymbol{x} = \boldsymbol{0}$. Every vector $\boldsymbol{x}$ in one nullspace is in the other nullspace. If $A^{\mathrm{T}}A$ has dependent columns, so has A. If $A^{\mathrm{T}}A$ has independent columns, so has A. This is the good case:

When A has independent columns, $A^{\mathrm{T}}A$ is square, symmetric, and invertible.

To repeat for emphasis: $A^{\mathrm{T}}A$ is (n by m) times (m by n). Then $A^{\mathrm{T}}A$ is square (n by n). It is symmetric, because its transpose is $(A^{\mathrm{T}}A)^{\mathrm{T}} = A^{\mathrm{T}}(A^{\mathrm{T}})^{\mathrm{T}}$ which equals $A^{\mathrm{T}}A$. We just proved that $A^{\mathrm{T}}A$ is invertible—provided A has independent columns. Watch the difference between dependent and independent columns:

$$\begin{matrix} A^{\mathrm{T}} & & A & & A^{\mathrm{T}}A \\ \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 0 \end{bmatrix} & \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \end{bmatrix} & = & \begin{bmatrix} 2 & 4 \\ 4 & 8 \end{bmatrix} \\ & \text{dependent} & & \text{singular} \end{matrix} \qquad \begin{matrix} A^{\mathrm{T}} & & A & & A^{\mathrm{T}}A \\ \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 1 \end{bmatrix} & = & \begin{bmatrix} 2 & 4 \\ 4 & 9 \end{bmatrix} \\ & \text{indep.} & & \text{invertible} \end{matrix}$$

Very brief summary To find the projection $\boldsymbol{p} = \widehat{x}_1\boldsymbol{a}_1 + \cdots + \widehat{x}_n\boldsymbol{a}_n$, solve $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$. This gives $\widehat{\boldsymbol{x}}$. The projection is $A\widehat{\boldsymbol{x}}$ and the error is $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = \boldsymbol{b} - A\widehat{\boldsymbol{x}}$. The projection matrix $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ gives $\boldsymbol{p} = P\boldsymbol{b}$.

This matrix satisfies $P^2 = P$. *The distance from $\boldsymbol{b}$ to the subspace is $\|\boldsymbol{e}\|$.*

■ REVIEW OF THE KEY IDEAS ■

1. The projection of $\boldsymbol{b}$ onto the line through $\boldsymbol{a}$ is $\boldsymbol{p} = \boldsymbol{a}\widehat{x} = \boldsymbol{a}(\boldsymbol{a}^{\mathrm{T}}\boldsymbol{b}/\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a})$.
2. The rank one projection matrix $P = \boldsymbol{a}\boldsymbol{a}^{\mathrm{T}}/\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}$ multiplies $\boldsymbol{b}$ to produce $\boldsymbol{p}$.
3. Projecting $\boldsymbol{b}$ onto a subspace leaves $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$ perpendicular to the subspace.
4. When A has full rank n, the equation $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$ leads to $\widehat{\boldsymbol{x}}$ and $\boldsymbol{p} = A\widehat{\boldsymbol{x}}$.
5. The projection matrix $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ has $P^{\mathrm{T}} = P$ and $P^2 = P$.

■ WORKED EXAMPLES ■

4.2 A Project the vector $\boldsymbol{b} = (3,4,4)$ onto the line through $\boldsymbol{a} = (2,2,1)$ and then onto the plane that also contains $\boldsymbol{a}^* = (1,0,0)$. Check that the first error vector $\boldsymbol{b} - \boldsymbol{p}$ is perpendicular to $\boldsymbol{a}$, and the second error vector $\boldsymbol{e}^* = \boldsymbol{b} - \boldsymbol{p}^*$ is also perpendicular to $\boldsymbol{a}^*$.

Find the 3 by 3 projection matrix P onto that plane of $\boldsymbol{a}$ and $\boldsymbol{a}^*$. Find a vector whose projection onto the plane is the zero vector.

Solution The projection of $\boldsymbol{b} = (3,4,4)$ onto the line through $\boldsymbol{a} = (2,2,1)$ is $\boldsymbol{p} = 2\boldsymbol{a}$:

Onto a line
$$\boldsymbol{p} = \frac{\boldsymbol{a}^{\mathrm{T}}\boldsymbol{b}}{\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}}\boldsymbol{a} = \frac{18}{9}(2,2,1) = (4,4,2).$$

The error vector $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = (-1,0,2)$ is perpendicular to $\boldsymbol{a}$. So $\boldsymbol{p}$ is correct.

The plane of $\boldsymbol{a} = (2,2,1)$ and $\boldsymbol{a}^* = (1,0,0)$ is the column space of $A = [\boldsymbol{a}\ \boldsymbol{a}^*]$:

$$A = \begin{bmatrix} 2 & 1 \\ 2 & 0 \\ 1 & 0 \end{bmatrix} \quad A^{\mathrm{T}}A = \begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix} \quad (A^{\mathrm{T}}A)^{-1} = \frac{1}{5}\begin{bmatrix} 1 & -2 \\ -2 & 9 \end{bmatrix} \quad P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & .8 & .4 \\ 0 & .4 & .2 \end{bmatrix}$$

Then $\boldsymbol{p}^* = P\boldsymbol{b} = (3, 4.8, 2.4)$. The error $\boldsymbol{e}^* = \boldsymbol{b} - \boldsymbol{p}^* = (0, -.8, 1.6)$ is perpendicular to $\boldsymbol{a}$ and $\boldsymbol{a}^*$. This $\boldsymbol{e}^*$ is in the nullspace of P and its projection is zero! Note $P^2 = P$.

4.2 B Suppose your pulse is measured at $x = 70$ beats per minute, then at $x = 80$, then at $x = 120$. Those three equations $Ax = \boldsymbol{b}$ in one unknown have $A^{\mathrm{T}} = [1\ 1\ 1]$ and $\boldsymbol{b} = (70, 80, 120)$. ***The best $\widehat{x}$ is the ____ of 70, 80, 120.*** Use calculus and projection:

1. Minimize $E = (x-70)^2 + (x-80)^2 + (x-120)^2$ by solving $dE/dx = 0$.
2. Project $\boldsymbol{b} = (70, 80, 120)$ onto $\boldsymbol{a} = (1,1,1)$ to find $\widehat{x} = \boldsymbol{a}^{\mathrm{T}}\boldsymbol{b}/\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}$.

Solution The closest horizontal line to the heights 70, 80, 120 is the *average* $\widehat{x} = 90$:

$$\frac{dE}{dx} = 2(x-70) + 2(x-80) + 2(x-120) = 0 \quad \text{gives} \quad \widehat{x} = \frac{70+80+120}{3}$$

Projection :
$$\widehat{x} = \frac{\boldsymbol{a}^{\mathrm{T}}\boldsymbol{b}}{\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}} = \frac{(1,1,1)^{\mathrm{T}}(70,80,120)}{(1,1,1)^{\mathrm{T}}(1,1,1)} = \frac{70+80+120}{3} = 90.$$

4.2 C In *recursive* least squares, a fourth measurement 130 changes $\widehat{x}_{\text{old}}$ to $\widehat{x}_{\text{new}}$. Compute $\widehat{x}_{\text{new}}$ and verify the *update formula* $\widehat{x}_{\text{new}} = \widehat{x}_{\text{old}} + \frac{1}{4}(130 - \widehat{x}_{\text{old}})$.

Going from 999 to 1000 measurements, $\widehat{x}_{\text{new}} = \widehat{x}_{\text{old}} + \frac{1}{1000}(b_{1000} - \widehat{x}_{\text{old}})$ would only need $\widehat{x}_{\text{old}}$ and the latest value b_{1000}. We don't have to average all 1000 numbers!

Solution The new measurement $b_4 = 130$ adds a fourth equation and $\widehat{x}$ is updated to 100. You can average b_1, b_2, b_3, b_4 or combine the average of b_1, b_2, b_3 with b_4:

$$\frac{70+80+120+130}{4} = 100 \quad \text{is also} \quad \widehat{x}_{\text{old}} + \frac{1}{4}(b_4 - \widehat{x}_{\text{old}}) = 90 + \frac{1}{4}(40).$$

The update from 999 to 1000 measurements shows the "gain matrix" $\frac{1}{1000}$ in a **Kalman filter** multiplying the prediction error $b_{\text{new}} - \widehat{x}_{\text{old}}$. Notice $\frac{1}{1000} = \frac{1}{999} - \frac{1}{999000}$:

$$\widehat{x}_{\text{new}} = \frac{b_1 + \cdots + b_{1000}}{1000} = \frac{b_1 + \cdots + b_{999}}{999} + \frac{1}{1000}\left(b_{1000} - \frac{b_1 + \cdots + b_{999}}{999}\right).$$

Problem Set 4.2

Questions 1–9 ask for projections onto lines. Also errors $e = b - p$ and matrices P.

1 Project the vector $\boldsymbol{b}$ onto the line through $\boldsymbol{a}$. Check that $\boldsymbol{e}$ is perpendicular to $\boldsymbol{a}$:

(a) $\boldsymbol{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ and $\boldsymbol{a} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ (b) $\boldsymbol{b} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$ and $\boldsymbol{a} = \begin{bmatrix} -1 \\ -3 \\ -1 \end{bmatrix}$.

2 *Draw* the projection of $\boldsymbol{b}$ onto $\boldsymbol{a}$ and also compute it from $\boldsymbol{p} = \widehat{x}\boldsymbol{a}$:

(a) $\boldsymbol{b} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$ and $\boldsymbol{a} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ (b) $\boldsymbol{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\boldsymbol{a} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

3 In Problem 1, find the projection matrix $P = \boldsymbol{a}\boldsymbol{a}^{\mathrm{T}}/\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}$ onto the line through each vector $\boldsymbol{a}$. Verify in both cases that $P^2 = P$. Multiply $P\boldsymbol{b}$ in each case to compute the projection $\boldsymbol{p}$.

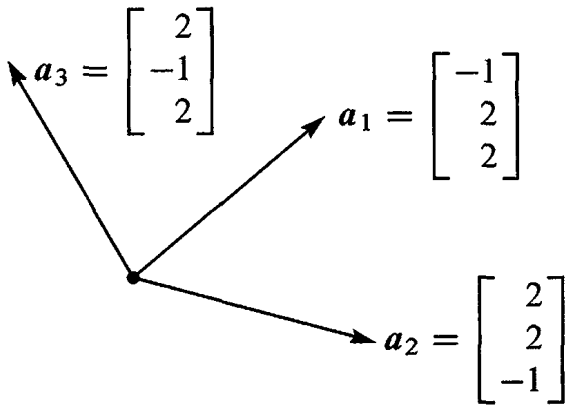
4 Construct the projection matrices P_1 and P_2 onto the lines through the $\boldsymbol{a}$'s in Problem 2. Is it true that $(P_1 + P_2)^2 = P_1 + P_2$? This *would* be true if $P_1 P_2 = 0$.

5 Compute the projection matrices $\boldsymbol{a}\boldsymbol{a}^{\mathrm{T}}/\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}$ onto the lines through $\boldsymbol{a}_1 = (-1, 2, 2)$ and $\boldsymbol{a}_2 = (2, 2, -1)$. Multiply those projection matrices and explain why their product $P_1 P_2$ is what it is.

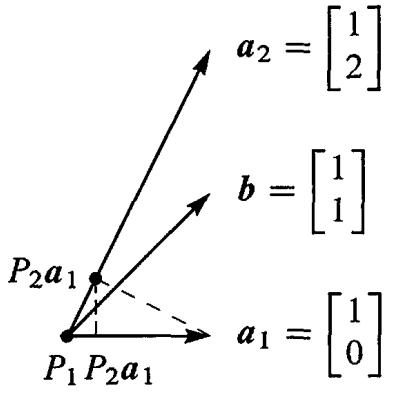
6 Project $\boldsymbol{b} = (1, 0, 0)$ onto the lines through $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$ in Problem 5 and also onto $\boldsymbol{a}_3 = (2, -1, 2)$. Add up the three projections $\boldsymbol{p}_1 + \boldsymbol{p}_2 + \boldsymbol{p}_3$.

7 Continuing Problems 5–6, find the projection matrix P_3 onto $\boldsymbol{a}_3 = (2, -1, 2)$. Verify that $P_1 + P_2 + P_3 = I$. The basis $\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3$ is orthogonal!

8 Project the vector $\boldsymbol{b} = (1, 1)$ onto the lines through $\boldsymbol{a}_1 = (1, 0)$ and $\boldsymbol{a}_2 = (1, 2)$. Draw the projections $\boldsymbol{p}_1$ and $\boldsymbol{p}_2$ and add $\boldsymbol{p}_1 + \boldsymbol{p}_2$. The projections do not add to $\boldsymbol{b}$ because the $\boldsymbol{a}$'s are not orthogonal.



Questions 5–6–7



Questions 8–9–10

9 In Problem 8, the projection of $\boldsymbol{b}$ onto the *plane* of $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$ will equal $\boldsymbol{b}$. Find $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ for $A = \begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$.

10 Project $\boldsymbol{a}_1 = (1, 0)$ onto $\boldsymbol{a}_2 = (1, 2)$. Then project the result back onto $\boldsymbol{a}_1$. Draw these projections and multiply the projection matrices P_1P_2: Is this a projection?

Questions 11–20 ask for projections, and projection matrices, onto subspaces.

11 Project $\boldsymbol{b}$ onto the column space of A by solving $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}\boldsymbol{b}$ and $\boldsymbol{p} = A\widehat{x}$:

(a) $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $\boldsymbol{b} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ (b) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\boldsymbol{b} = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}$.

Find $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$. It should be perpendicular to the columns of A.

12 Compute the projection matrices P_1 and P_2 onto the column spaces in Problem 11. Verify that $P_1\boldsymbol{b}$ gives the first projection $\boldsymbol{p}_1$. Also verify $P_2^2 = P_2$.

13 (Quick and Recommended) Suppose A is the 4 by 4 identity matrix with its last column removed. A is 4 by 3. Project $\boldsymbol{b} = (1, 2, 3, 4)$ onto the column space of A. What shape is the projection matrix P and what is P?

14 Suppose $\boldsymbol{b}$ equals 2 times the first column of A. What is the projection of $\boldsymbol{b}$ onto the column space of A? Is $P = I$ for sure in this case? Compute $\boldsymbol{p}$ and P when $\boldsymbol{b} = (0, 2, 4)$ and the columns of A are $(0, 1, 2)$ and $(1, 2, 0)$.

15 If A is doubled, then $P = 2A(4A^{\mathrm{T}}A)^{-1}2A^{\mathrm{T}}$. This is the same as $A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$. The column space of $2A$ is the same as _____. Is $\widehat{x}$ the same for A and $2A$?

16 What linear combination of $(1, 2, -1)$ and $(1, 0, 1)$ is closest to $\boldsymbol{b} = (2, 1, 1)$?

17 (*Important*) If $P^2 = P$ show that $(I - P)^2 = I - P$. When P projects onto the column space of A, $I - P$ projects onto the _____.

18 (a) If P is the 2 by 2 projection matrix onto the line through $(1, 1)$, then $I - P$ is the projection matrix onto _____.

(b) If P is the 3 by 3 projection matrix onto the line through $(1, 1, 1)$, then $I - P$ is the projection matrix onto _____.

19 To find the projection matrix onto the plane $x - y - 2z = 0$, choose two vectors in that plane and make them the columns of A. The plane should be the column space. Then compute $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$.

20 To find the projection matrix P onto the same plane $x - y - 2z = 0$, write down a vector $\boldsymbol{e}$ that is perpendicular to that plane. Compute the projection $Q = \boldsymbol{e}\boldsymbol{e}^{\mathrm{T}}/\boldsymbol{e}^{\mathrm{T}}\boldsymbol{e}$ and then $P = I - Q$.

Questions 21–26 show that projection matrices satisfy $P^2 = P$ and $P^{\mathrm{T}} = P$.

21 Multiply the matrix $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ by itself. Cancel to prove that $P^2 = P$. Explain why $P(P\boldsymbol{b})$ always equals $P\boldsymbol{b}$: The vector $P\boldsymbol{b}$ is in the column space so its projection is _____.

22 Prove that $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ is symmetric by computing P^{T}. Remember that the inverse of a symmetric matrix is symmetric.

23 If A is square and invertible, the warning against splitting $(A^{\mathrm{T}}A)^{-1}$ does not apply. It is true that $AA^{-1}(A^{\mathrm{T}})^{-1}A^{\mathrm{T}} = I$. *When A is invertible, why is $P = I$?* ***What is the error $\boldsymbol{e}$?***

24 The nullspace of A^{T} is _____ to the column space $\boldsymbol{C}(A)$. So if $A^{\mathrm{T}}\boldsymbol{b} = \boldsymbol{0}$, the projection of $\boldsymbol{b}$ onto $\boldsymbol{C}(A)$ should be $\boldsymbol{p}$ = _____. Check that $P = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ gives this answer.

25 The projection matrix P onto an n-dimensional subspace has rank $r = n$. ***Reason:*** The projections $P\boldsymbol{b}$ fill the subspace S. So S is the _____ of P.

26 If an m by m matrix has $A^2 = A$ and its rank is m, prove that $A = I$.

27 The important fact that ends the section is this: ***If $A^{\mathrm{T}}A\boldsymbol{x} = \boldsymbol{0}$ then $A\boldsymbol{x} = \boldsymbol{0}$.*** *New Proof*: The vector $A\boldsymbol{x}$ is in the nullspace of _____. $A\boldsymbol{x}$ is always in the column space of _____. To be in both of those perpendicular spaces, $A\boldsymbol{x}$ must be zero.

28 Use $P^{\mathrm{T}} = P$ and $P^2 = P$ to prove that the length squared of column 2 always equals the diagonal entry P_{22}. This number is $\frac{2}{6} = \frac{4}{36} + \frac{4}{36} + \frac{4}{36}$ for

$$P = \frac{1}{6}\begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}.$$

29 If B has rank m (full row rank, independent rows) show that BB^{T} is invertible.

Challenge Problems

30 (a) Find the projection matrix P_C onto the column space of A (after looking closely at the matrix!)

$$A = \begin{bmatrix} 3 & 6 & 6 \\ 4 & 8 & 8 \end{bmatrix}$$

(b) Find the 3 by 3 projection matrix P_R onto the row space of A. Multiply $B = P_C A P_R$. Your answer B should be a little surprising—can you explain it?

31 In $\mathbf{R}^m$, suppose I give you $\boldsymbol{b}$ and $\boldsymbol{p}$, and $\boldsymbol{p}$ is a combination of $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$. How would you test to see if $\boldsymbol{p}$ is the projection of $\boldsymbol{b}$ onto the subspace spanned by the $\boldsymbol{a}$'s?

32 Suppose P_1 is the projection matrix onto the 1-dimensional subspace spanned by the first column of A. Suppose P_2 is the projection matrix onto the 2-dimensional column space of A. After thinking a little, compute the product $P_2 P_1$.

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}.$$

33 P_1 and P_2 are projections onto subspaces $\boldsymbol{S}$ and $\boldsymbol{T}$. What is the requirement on those subspaces to have $P_1 P_2 = P_2 P_1$?

34 **If A has r independent columns and B has r independent rows, AB is invertible.**

Proof: When A is m by r with independent columns, we know that $A^{\mathrm{T}}A$ is invertible. If B is r by n with independent rows, show that BB^{T} is invertible. (Take $A = B^{\mathrm{T}}$.)

Now show that AB has rank r. Hint: Why does $A^{\mathrm{T}}ABB^{\mathrm{T}}$ have rank r? That matrix multiplication by A^{T} and B^{T} cannot increase the rank of AB, by Problem 3.6:26.

4.3 Least Squares Approximations

It often happens that $Ax = b$ has no solution. The usual reason is: *too many equations.* The matrix has more rows than columns. There are more equations than unknowns (m is greater than n). The n columns span a small part of m-dimensional space. Unless all measurements are perfect, b is outside that column space. Elimination reaches an impossible equation and stops. But we can't stop just because measurements include noise.

To repeat: We cannot always get the error $e = b - Ax$ down to zero. When e is zero, x is an exact solution to $Ax = b$. *When the length of e is as small as possible, $\widehat{x}$ is a least squares solution.* Our goal in this section is to compute $\widehat{x}$ and use it. These are real problems and they need an answer.

The previous section emphasized p (the projection). This section emphasizes $\widehat{x}$ (the least squares solution). They are connected by $p = A\widehat{x}$. The fundamental equation is still $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$. Here is a short unofficial way to reach this equation:

When $Ax = b$ has no solution, multiply by A^{T} and solve $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$.

Example 1 A crucial application of least squares is fitting a straight line to m points. Start with three points: *Find the closest line to the points* $(0,6), (1,0)$, *and* $(2,0)$.

No straight line $b = C + Dt$ goes through those three points. We are asking for two numbers C and D that satisfy three equations. Here are the equations at $t = 0, 1, 2$ to match the given values $b = 6, 0, 0$:

$t = 0$	The first point is on the line $b = C + Dt$ if	$C + D \cdot 0 = 6$
$t = 1$	The second point is on the line $b = C + Dt$ if	$C + D \cdot 1 = 0$
$t = 2$	The third point is on the line $b = C + Dt$ if	$C + D \cdot 2 = 0.$

This 3 by 2 system has *no solution*: $b = (6,0,0)$ is not a combination of the columns $(1,1,1)$ and $(0,1,2)$. Read off A, x, and b from those equations:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \quad x = \begin{bmatrix} C \\ D \end{bmatrix} \quad b = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} \quad Ax = b \text{ is } \textit{not} \text{ solvable.}$$

The same numbers were in Example 3 in the last section. We computed $\widehat{x} = (5, -3)$. Those numbers are the best C and D, so $5 - 3t$ will be the best line for the 3 points. We must connect projections to least squares, by explaining why $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$.

In practical problems, there could easily be $m = 100$ points instead of $m = 3$. They don't exactly match any straight line $C + Dt$. Our numbers $6, 0, 0$ exaggerate the error so you can see e_1, e_2, and e_3 in Figure 4.6.

Minimizing the Error

How do we make the error $e = b - Ax$ as small as possible? This is an important question with a beautiful answer. The best x (called $\widehat{x}$) can be found by geometry or algebra or calculus: 90° angle or project using P or set the derivative of the error to zero.

By geometry Every Ax lies in the plane of the columns $(1,1,1)$ and $(0,1,2)$. In that plane, we look for the point closest to $\boldsymbol{b}$. *The nearest point is the projection* $\boldsymbol{p}$.

The best choice for $A\widehat{x}$ is $\boldsymbol{p}$. The smallest possible error is $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$. The three points at heights (p_1, p_2, p_3) *do lie on a line*, because $\boldsymbol{p}$ is in the column space. In fitting a straight line, $\widehat{x}$ gives the best choice for (C, D).

By algebra Every vector $\boldsymbol{b}$ splits into two parts. The part in the column space is $\boldsymbol{p}$. The perpendicular part in the nullspace of A^{T} is $\boldsymbol{e}$. There is an equation we cannot solve $(Ax = \boldsymbol{b})$. There is an equation $A\widehat{x} = \boldsymbol{p}$ we do solve (by removing $\boldsymbol{e}$):

$$Ax = \boldsymbol{b} = \boldsymbol{p} + \boldsymbol{e} \quad \text{is impossible;} \qquad A\widehat{x} = \boldsymbol{p} \quad \text{is solvable.} \tag{1}$$

The solution to $A\widehat{x} = \boldsymbol{p}$ leaves the least possible error (which is $\boldsymbol{e}$):

$$\textbf{Squared length for any } x \qquad \|Ax - \boldsymbol{b}\|^2 = \|Ax - \boldsymbol{p}\|^2 + \|\boldsymbol{e}\|^2. \tag{2}$$

This is the law $c^2 = a^2 + b^2$ for a right triangle. The vector $Ax - \boldsymbol{p}$ in the column space is perpendicular to $\boldsymbol{e}$ in the left nullspace. We reduce $Ax - \boldsymbol{p}$ to zero by choosing $\boldsymbol{x}$ to be $\widehat{x}$. That leaves the smallest possible error $\boldsymbol{e} = (e_1, e_2, e_3)$.

Notice what "smallest" means. The *squared length* of $Ax - \boldsymbol{b}$ is minimized:

The least squares solution $\widehat{x}$ makes $E = \|Ax - \boldsymbol{b}\|^2$ as small as possible.

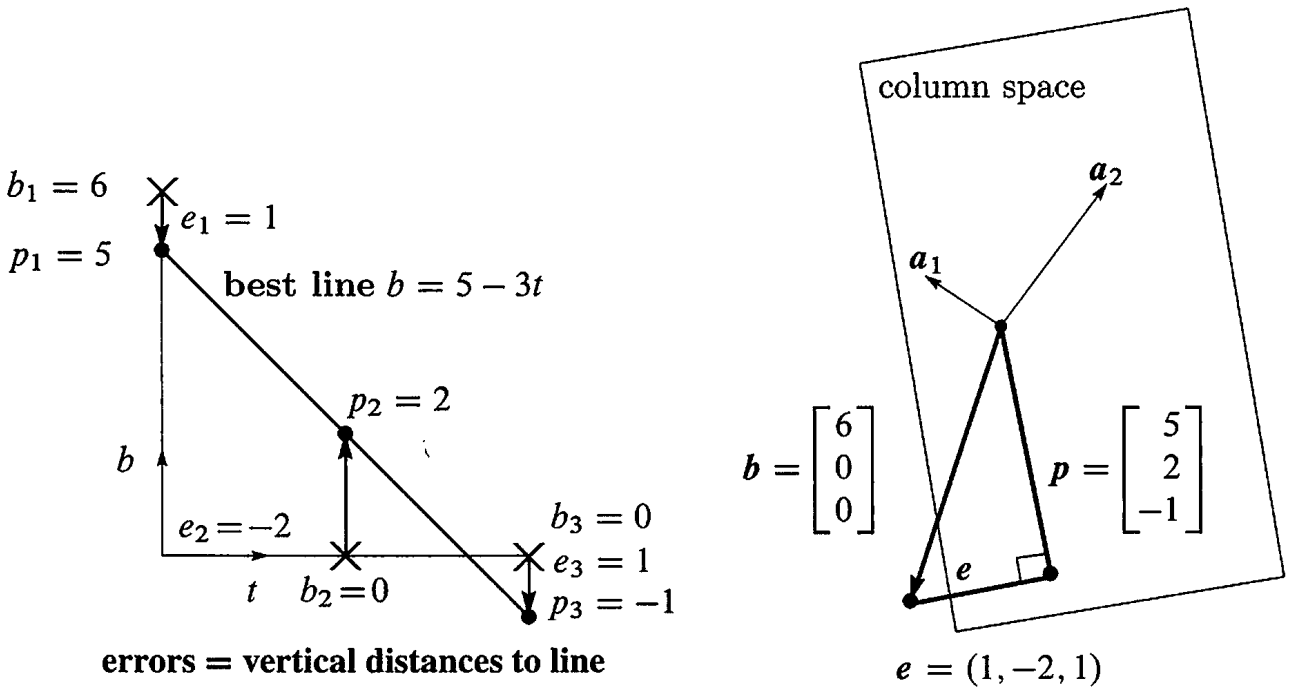


Figure 4.6: **Best line and projection: Two pictures, same problem.** The line has heights $\boldsymbol{p} = (5, 2, -1)$ with errors $\boldsymbol{e} = (1, -2, 1)$. The equations $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}\boldsymbol{b}$ give $\widehat{x} = (5, -3)$. The best line is $b = 5 - 3t$ and the projection is $\boldsymbol{p} = 5\boldsymbol{a}_1 - 3\boldsymbol{a}_2$.

Figure 4.6a shows the closest line. It misses by distances $e_1, e_2, e_3 = 1, -2, 1$. *Those are vertical distances.* The least squares line minimizes $E = e_1^2 + e_2^2 + e_3^2$.

Figure 4.6b shows the same problem in 3-dimensional space ($\boldsymbol{b}\ \boldsymbol{p}\ \boldsymbol{e}$ space). The vector $\boldsymbol{b}$ is not in the column space of A. That is why we could not solve $A\boldsymbol{x} = \boldsymbol{b}$. No line goes through the three points. The smallest possible error is the perpendicular vector $\boldsymbol{e}$. This is $\boldsymbol{e} = \boldsymbol{b} - A\widehat{\boldsymbol{x}}$, the vector of errors $(1, -2, 1)$ in the three equations. Those are the distances from the best line. Behind both figures is the fundamental equation $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$.

Notice that the errors $1, -2, 1$ add to zero. The error $\boldsymbol{e} = (e_1, e_2, e_3)$ is perpendicular to the first column $(1, 1, 1)$ in A. The dot product gives $e_1 + e_2 + e_3 = 0$.

By calculus Most functions are minimized by calculus! The graph bottoms out and the derivative in every direction is zero. Here the error function E to be minimized is a *sum of squares* $e_1^2 + e_2^2 + e_3^2$ (the square of the error in each equation):

$$E = \|A\boldsymbol{x} - \boldsymbol{b}\|^2 = (C + D \cdot 0 - 6)^2 + (C + D \cdot 1)^2 + (C + D \cdot 2)^2. \tag{3}$$

The unknowns are C and D. With two unknowns there are *two derivatives*—both zero at the minimum. They are "partial derivatives" because $\partial E/\partial C$ treats D as constant and $\partial E/\partial D$ treats C as constant:

$$\partial E/\partial C = 2(C + D \cdot 0 - 6) \quad + 2(C + D \cdot 1) \quad + 2(C + D \cdot 2) \quad = 0$$

$$\partial E/\partial D = 2(C + D \cdot 0 - 6)(\mathbf{0}) + 2(C + D \cdot 1)(\mathbf{1}) + 2(C + D \cdot 2)(\mathbf{2}) = 0.$$

$\partial E/\partial D$ contains the extra factors $\mathbf{0}, \mathbf{1}, \mathbf{2}$ from the chain rule. (The last derivative from $(C + 2\mathrm{D})^2$ was 2 times $C + 2\mathrm{D}$ times that extra 2.) In the C derivative the corresponding factors are $1, 1, 1$, because C is always multiplied by 1. It is no accident that 1, 1, 1 and 0, 1, 2 are the columns of A.

Now cancel 2 from every term and collect all C's and all D's:

$$\begin{array}{ll} \text{The } C \text{ derivative is zero:} & 3C + 3D = 6 \\ \text{The } D \text{ derivative is zero:} & 3C + 5D = 0 \end{array} \quad \textbf{This matrix} \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \textbf{ is } A^{\mathrm{T}}A \tag{4}$$

These equations are identical with $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$. The best C and D are the components of $\widehat{\boldsymbol{x}}$. The equations from calculus are the same as the "normal equations" from linear algebra. These are the key equations of least squares:

The partial derivatives of $\|A\boldsymbol{x} - \boldsymbol{b}\|^2$ ***are zero when*** $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$.

The solution is $C = 5$ and $D = -3$. Therefore $b = 5 - 3t$ is the best line—it comes closest to the three points. At $t = 0, 1, 2$ this line goes through $\boldsymbol{p} = 5, 2, -1$. It could not go through $\boldsymbol{b} = 6, 0, 0$. The errors are $1, -2, 1$. This is the vector $\boldsymbol{e}$!

The Big Picture

The key figure of this book shows the four subspaces and the true action of a matrix. The vector $\boldsymbol{x}$ on the left side of Figure 4.3 went to $\boldsymbol{b} = A\boldsymbol{x}$ on the right side. In that figure $\boldsymbol{x}$ was split into $\boldsymbol{x}_r + \boldsymbol{x}_n$. There were many solutions to $A\boldsymbol{x} = \boldsymbol{b}$.

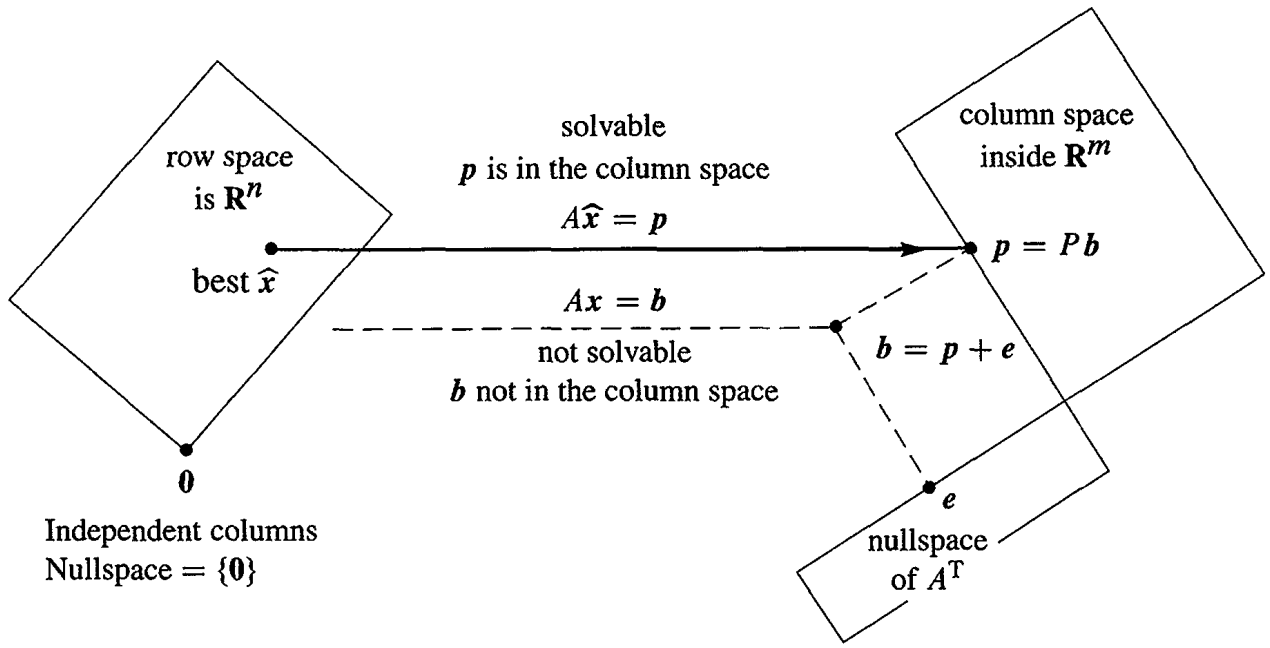


Figure 4.7: The projection $p = A\widehat{x}$ is closest to b, so $\widehat{x}$ minimizes $E = \|b - Ax\|^2$.

In this section the situation is just the opposite. There are *no* solutions to $Ax = b$. *Instead of splitting up* x *we are splitting up* b. Figure 4.3 shows the big picture for least squares. Instead of $Ax = b$ we solve $A\widehat{x} = p$. The error $e = b - p$ is unavoidable.

Notice how the nullspace $N(A)$ is very small—just one point. With independent columns, the only solution to $Ax = 0$ is $x = 0$. Then A^TA is invertible. The equation $A^TA\widehat{x} = A^Tb$ fully determines the best vector $\widehat{x}$. The error has $A^Te = 0$.

Chapter 7 will have the complete picture—all four subspaces included. Every x splits into $x_r + x_n$, and every b splits into $p + e$. The best solution is $\widehat{x}_r$ in the row space. We can't help e and we don't want x_n—this leaves $A\widehat{x} = p$.

Fitting a Straight Line

Fitting a line is the clearest application of least squares. It starts with $m > 2$ points, hopefully near a straight line. At times $t_1, \ldots, t_m$ those m points are at heights $b_1, \ldots, b_m$. The best line $C + Dt$ misses the points by vertical distances $e_1, \ldots, e_m$. No line is perfect, and the least squares line minimizes $E = e_1^2 + \cdots + e_m^2$.

The first example in this section had three points in Figure 4.6. Now we allow m points (and m can be large). The two components of $\widehat{x}$ are still C and D.

A line goes through the m points when we exactly solve $Ax = b$. Generally we can't do it. Two unknowns C and D determine a line, so A has only $n = 2$ columns. To fit the m points, we are trying to solve m equations (and we only want two!):

$$Ax = b \quad \text{is} \quad \begin{aligned} C + Dt_1 &= b_1 \\ C + Dt_2 &= b_2 \\ &\vdots \\ C + Dt_m &= b_m \end{aligned} \qquad \text{with} \qquad A = \begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}. \tag{5}$$

The column space is so thin that almost certainly $\boldsymbol{b}$ is outside of it. When $\boldsymbol{b}$ happens to lie in the column space, the points happen to lie on a line. In that case $\boldsymbol{b} = \boldsymbol{p}$. Then $A\boldsymbol{x} = \boldsymbol{b}$ is solvable and the errors are $\boldsymbol{e} = (0, \ldots, 0)$.

The closest line $C + Dt$ has heights $p_1, \ldots, p_m$ with errors $e_1, \ldots, e_m$.

Solve $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$ for $\widehat{x} = (C, D)$. The errors are $e_i = b_i - C - Dt_i$.

Fitting points by a straight line is so important that we give the two equations $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}\boldsymbol{b}$, once and for all. The two columns of A are independent (unless all times t_i are the same). So we turn to least squares and solve $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}\boldsymbol{b}$.

Dot-product matrix
$$A^{\mathrm{T}}A = \begin{bmatrix} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{bmatrix} \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix} = \begin{bmatrix} m & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix}. \tag{6}$$

On the right side of the normal equation is the 2 by 1 vector $A^{\mathrm{T}}\boldsymbol{b}$:

$$A^{\mathrm{T}}\boldsymbol{b} = \begin{bmatrix} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{bmatrix} \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} = \begin{bmatrix} \sum b_i \\ \sum t_i b_i \end{bmatrix}. \tag{7}$$

In a specific problem, these numbers are given. The best $\widehat{x} = (C, D)$ is in equation (9).

The line $C + Dt$ minimizes $e_1^2 + \cdots + e_m^2 = \|A\boldsymbol{x} - \boldsymbol{b}\|^2$ when $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}\boldsymbol{b}$:

$$\begin{bmatrix} m & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} \sum b_i \\ \sum t_i b_i \end{bmatrix}. \tag{8}$$

The vertical errors at the m points on the line are the components of $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$. This error vector (the *residual*) $\boldsymbol{b} - A\widehat{x}$ is perpendicular to the columns of A (geometry). The error is in the nullspace of A^{T} (linear algebra). The best $\widehat{x} = (C, D)$ minimizes the total error E, the sum of squares:

$$E(\boldsymbol{x}) = \|A\boldsymbol{x} - \boldsymbol{b}\|^2 = (C + Dt_1 - b_1)^2 + \cdots + (C + Dt_m - b_m)^2.$$

When calculus sets the derivatives $\partial E/\partial C$ and $\partial E/\partial D$ to zero, it produces $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}\boldsymbol{b}$.

Other least squares problems have more than two unknowns. Fitting by the best parabola has $n = 3$ coefficients C, D, E (see below). In general we are fitting m data points by n parameters $x_1, \ldots, x_n$. The matrix A has n columns and $n < m$. The derivatives of $\|A\boldsymbol{x} - \boldsymbol{b}\|^2$ give the n equations $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}\boldsymbol{b}$. **The derivative of a square is linear.** This is why the method of least squares is so popular.

Example 2 A has *orthogonal columns* when the measurement times t_i add to zero.

Suppose $b = 1, 2, 4$ at times $t = -2, 0, 2$. Those times add to zero. The columns of A have *zero dot product*:

$$\begin{array}{l} C + D(-2) = 1 \\ C + D(0) = 2 \\ C + D(2) = 4 \end{array} \quad \text{or} \quad A\boldsymbol{x} = \begin{bmatrix} 1 & -2 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

Look at the zeros in $A^{\mathrm{T}}A$:

$$A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b} \quad \text{is} \quad \begin{bmatrix} 3 & 0 \\ 0 & 8 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}.$$

Main point: Now $A^{\mathrm{T}}A$ *is diagonal.* We can solve separately for $C = \frac{7}{3}$ and $D = \frac{6}{8}$. The zeros in $A^{\mathrm{T}}A$ are dot products of perpendicular columns in A. The diagonal matrix $A^{\mathrm{T}}A$, with entries $m = 3$ and $t_1^2 + t_2^2 + t_3^2 = 8$, is virtually as good as the identity matrix.

Orthogonal columns are so helpful that it is worth moving the time origin to produce them. To do that, subtract away the average time $\widehat{t} = (t_1 + \cdots + t_m)/m$. The shifted times $T_i = t_i - \widehat{t}$ add to $\sum T_i = m\widehat{t} - m\widehat{t} = 0$. With the columns now orthogonal, $A^{\mathrm{T}}A$ is diagonal. Its entries are m and $T_1^2 + \cdots + T_m^2$. The best C and D have direct formulas:

$$\boldsymbol{T} \textbf{ is } \boldsymbol{t - \widehat{t}} \qquad C = \frac{b_1 + \cdots + b_m}{m} \qquad \text{and} \qquad D = \frac{b_1 T_1 + \cdots + b_m T_m}{T_1^2 + \cdots + T_m^2}. \tag{9}$$

The best line is $C + DT$ ***or*** $C + D(t - \widehat{t})$. The time shift that makes $A^{\mathrm{T}}A$ diagonal is an example of the Gram-Schmidt process: *orthogonalize the columns in advance.*

Fitting by a Parabola

If we throw a ball, it would be crazy to fit the path by a straight line. A parabola $b = C + Dt + Et^2$ allows the ball to go up and come down again (b is the height at time t). The actual path is not a perfect parabola, but the whole theory of projectiles starts with that approximation.

When Galileo dropped a stone from the Leaning Tower of Pisa, it accelerated. The distance contains a quadratic term $\frac{1}{2}gt^2$. (Galileo's point was that the stone's mass is not involved.) Without that t^2 term we could never send a satellite into the right orbit. But even with a nonlinear function like t^2, the unknowns C, D, E appear linearly! Choosing the best parabola is still a problem in linear algebra.

Problem Fit heights $b_1, \ldots, b_m$ at times $t_1, \ldots, t_m$ by a parabola $C + Dt + Et^2$.

Solution With $m > 3$ points, the m equations for an exact fit are generally unsolvable:

$$\begin{array}{l} C + Dt_1 + Et_1^2 = b_1 \\ \qquad\vdots \\ C + Dt_m + Et_m^2 = b_m \end{array} \quad \text{has the } m \text{ by 3 matrix} \quad A = \begin{bmatrix} 1 & t_1 & t_1^2 \\ \vdots & \vdots & \vdots \\ 1 & t_m & t_m^2 \end{bmatrix}. \tag{10}$$

Least squares The closest parabola $C + Dt + Et^2$ chooses $\widehat{\boldsymbol{x}} = (C, D, E)$ to satisfy the three normal equations $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$.

May I ask you to convert this to a problem of projection? The column space of A has dimension _____. The projection of $\boldsymbol{b}$ is $\boldsymbol{p} = A\widehat{\boldsymbol{x}}$, which combines the three columns using the coefficients C, D, E. The error at the first data point is $e_1 = b_1 - C - Dt_1 - Et_1^2$. The total squared error is $e_1^2 +$ _____. If you prefer to minimize by calculus, take the partial derivatives of E with respect to _____, _____, _____. These three derivatives will be zero when $\widehat{\boldsymbol{x}} = (C, D, E)$ solves the 3 by 3 system of equations _____.

Section 8.5 has more least squares applications. The big one is Fourier series—approximating functions instead of vectors. The function to be minimized changes from a sum of squared errors $e_1^2 + \cdots + e_m^2$ to an integral of the squared error.

Example 3 For a parabola $b = C + Dt + Et^2$ to go through the three heights $b = 6, 0, 0$ when $t = 0, 1, 2$, the equations are

$$\begin{aligned} C + D \cdot 0 + E \cdot 0^2 &= 6 \\ C + D \cdot 1 + E \cdot 1^2 &= 0 \\ C + D \cdot 2 + E \cdot 2^2 &= 0. \end{aligned} \tag{11}$$

This is $A\boldsymbol{x} = \boldsymbol{b}$. We can solve it exactly. Three data points give three equations and a square matrix. The solution is $\boldsymbol{x} = (C, D, E) = (6, -9, 3)$. The parabola through the three points in Figure 4.8a is $b = 6 - 9t + 3t^2$.

What does this mean for projection? The matrix has three columns, which span the whole space $\mathbf{R}^3$. The projection matrix is the identity. The projection of $\boldsymbol{b}$ is $\boldsymbol{b}$. The error is zero. We didn't need $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$, because we solved $A\boldsymbol{x} = \boldsymbol{b}$. Of course we could multiply by A^{T}, but there is no reason to do it.

Figure 4.8 also shows a fourth point b_4 at time t_4. If that falls on the parabola, the new $A\boldsymbol{x} = \boldsymbol{b}$ (four equations) is still solvable. When the fourth point is not on the parabola, we turn to $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$. Will the least squares parabola stay the same, with all the error at the fourth point? Not likely!

The smallest error vector (e_1, e_2, e_3, e_4) is perpendicular to $(1, 1, 1, 1)$, the first column of A. Least squares balances out the four errors, and they add to zero.

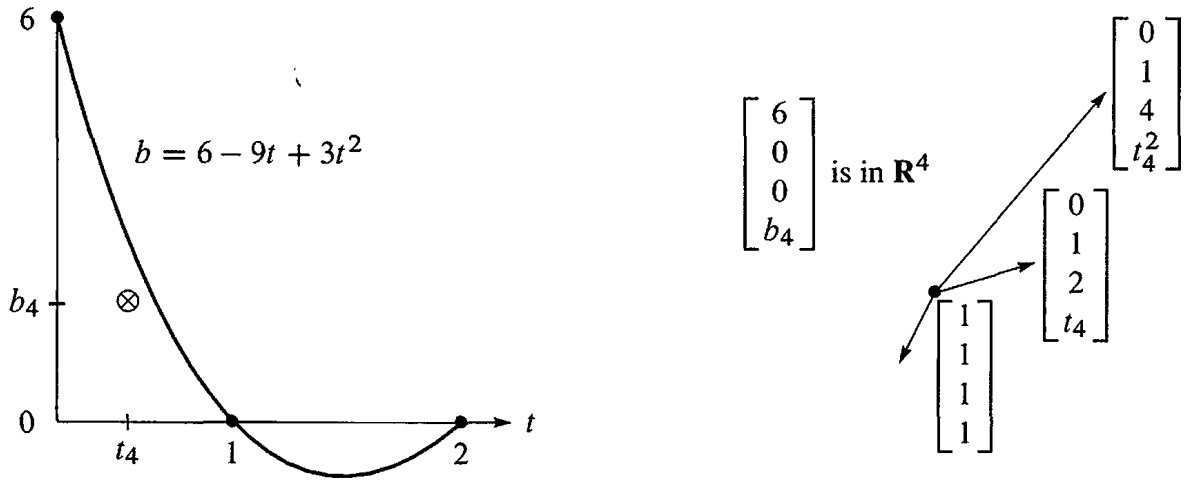


Figure 4.8: From Example 3: An exact fit of the parabola at $t = 0, 1, 2$ means that $\boldsymbol{p} = \boldsymbol{b}$ and $\boldsymbol{e} = \boldsymbol{0}$. The point b_4 off the parabola makes $m > n$ and we need least squares.

■ REVIEW OF THE KEY IDEAS ■

1. The least squares solution $\widehat{x}$ minimizes $E = \|Ax - b\|^2$. This is the sum of squares of the errors in the m equations ($m > n$).
2. The best $\widehat{x}$ comes from the normal equations $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$.
3. To fit m points by a line $b = C + Dt$, the normal equations give C and D.
4. The heights of the best line are $p = (p_1, \ldots, p_m)$. The vertical distances to the data points are the errors $e = (e_1, \ldots, e_m)$.
5. If we try to fit m points by a combination of $n < m$ functions, the m equations $Ax = b$ are generally unsolvable. The n equations $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$ give the least squares solution—the combination with smallest MSE (mean square error).

■ WORKED EXAMPLES ■

4.3 A Start with nine measurements b_1 to b_9, *all zero*, at times $t = 1, \ldots, 9$. The tenth measurement $b_{10} = 40$ is an outlier. Find the best *horizontal line* $y = C$ to fit the ten points $(1,0), (2,0), \ldots, (9,0), (10,40)$ using three measures for the error E:

(1) Least squares $E_2 = e_1^2 + \cdots + e_{10}^2$ (then the normal equation for C is linear)

(2) Least maximum error $E_\infty = |e_{\max}|$ **(3)** Least sum of errors $E_1 = |e_1| + \cdots + |e_{10}|$.

Solution **(1)** The least squares fit to $0, 0, \ldots, 0, 40$ by a horizontal line is $C = 4$:

$$A = \text{column of 1's} \quad A^{\mathrm{T}}A = 10 \quad A^{\mathrm{T}}b = \text{sum of } b_i = 40. \quad \text{So } 10C = 40.$$

(2) The least maximum error requires $C = 20$, halfway between 0 and 40.

(3) The least sum requires $C = 0$ (!!). The sum of errors $9|C| + |40 - C|$ would increase if C moves up from zero.

The least sum comes from the *median* measurement (the median of $0, \ldots, 0, 40$ is zero). Many statisticians feel that the least squares solution is too heavily influenced by outliers like $b_{10} = 40$, and they prefer least sum. But the equations become nonlinear.

Now find the least squares straight line $C + Dt$ through those ten points.

$$A^{\mathrm{T}}A = \begin{bmatrix} 10 & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix} = \begin{bmatrix} 10 & 55 \\ 55 & 385 \end{bmatrix} \qquad A^{\mathrm{T}}b = \begin{bmatrix} \sum b_i \\ \sum t_i b_i \end{bmatrix} = \begin{bmatrix} 40 \\ 400 \end{bmatrix}$$

Those come from equation (8). Then $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}b$ gives $C = -8$ and $D = 24/11$.

What happens to C and D if you multiply the b_i by 3 and then add 30 to get $b_{\text{new}} = (30, 30, \ldots, 150)$? Linearity allows us to rescale $b = (0, 0, \ldots, 40)$. Multiplying b by 3 will multiply C and D by 3. Adding 30 to all b_i will add 30 to C.

4.3 B Find the parabola $C + Dt + Et^2$ that comes closest (least squares error) to the values $\boldsymbol{b} = (0, 0, 1, 0, 0)$ at the times $t = -2, -1, 0, 1, 2$. First write down the five equations $A\boldsymbol{x} = \boldsymbol{b}$ in three unknowns $\boldsymbol{x} = (C, D, E)$ for a parabola to go through the five points. No solution because no such parabola exists. Solve $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$.

I would predict $D = 0$. Why should the best parabola be symmetric around $t = 0$? In $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$, equation 2 for D should uncouple from equations 1 and 3.

Solution The five equations $A\boldsymbol{x} = \boldsymbol{b}$ have a rectangular "Vandermonde" matrix A:

$$\begin{array}{rcrcrcl} C & + & D\,(-2) & + & E\,(-2)^2 & = & 0 \\ C & + & D\,(-1) & + & E\,(-1)^2 & = & 0 \\ C & + & D\,(0) & + & E\,(0)^2 & = & 1 \\ C & + & D\,(1) & + & E\,(1)^2 & = & 0 \\ C & + & D\,(2) & + & E\,(2)^2 & = & 0 \end{array} \qquad A = \begin{bmatrix} 1 & -2 & 4 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \qquad A^{\mathrm{T}}A = \begin{bmatrix} 5 & \mathbf{0} & 10 \\ \mathbf{0} & 10 & \mathbf{0} \\ 10 & \mathbf{0} & 34 \end{bmatrix}$$

Those zeros in $A^{\mathrm{T}}A$ mean that column 2 of A is orthogonal to columns 1 and 3. We see this directly in A (the times $-2, -1, 0, 1, 2$ are symmetric). The best C, D, E in the parabola $C + Dt + Et^2$ come from $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$, and D is uncoupled:

$$\begin{bmatrix} 5 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 34 \end{bmatrix} \begin{bmatrix} C \\ D \\ E \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{leads to} \quad \begin{array}{l} C = 34/70 \\ D = 0 \text{ as predicted} \\ E = -10/70 \end{array}$$

Problem Set 4.3

Problems 1–11 use four data points $\boldsymbol{b} = (0, 8, 8, 20)$ to bring out the key ideas.

1 With $b = 0, 8, 8, 20$ at $t = 0, 1, 3, 4$, set up and solve the normal equations $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$. For the best straight line in Figure 4.9a, find its four heights p_i and four errors e_i. What is the minimum value $E = e_1^2 + e_2^2 + e_3^2 + e_4^2$?

2 (Line $C + Dt$ does go through p's) With $b = 0, 8, 8, 20$ at times $t = 0, 1, 3, 4$, write down the four equations $A\boldsymbol{x} = \boldsymbol{b}$ (unsolvable). Change the measurements to $p = 1, 5, 13, 17$ and find an exact solution to $A\widehat{\boldsymbol{x}} = \boldsymbol{p}$.

3 Check that $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p} = (-1, 3, -5, 3)$ is perpendicular to both columns of the same matrix A. What is the shortest distance $\|\boldsymbol{e}\|$ from $\boldsymbol{b}$ to the column space of A?

4 (By calculus) Write down $E = \|A\boldsymbol{x} - \boldsymbol{b}\|^2$ as a sum of four squares—the last one is $(C + 4D - 20)^2$. Find the derivative equations $\partial E/\partial C = 0$ and $\partial E/\partial D = 0$. Divide by 2 to obtain the normal equations $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$.

5 Find the height C of the best *horizontal line* to fit $\boldsymbol{b} = (0, 8, 8, 20)$. An exact fit would solve the unsolvable equations $C = 0$, $C = 8$, $C = 8$, $C = 20$. Find the 4 by 1 matrix A in these equations and solve $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$. Draw the horizontal line at height $\widehat{\boldsymbol{x}} = C$ and the four errors in $\boldsymbol{e}$.

6 Project $\boldsymbol{b} = (0, 8, 8, 20)$ onto the line through $\boldsymbol{a} = (1, 1, 1, 1)$. Find $\widehat{x} = \boldsymbol{a}^{\mathrm{T}}\boldsymbol{b}/\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}$ and the projection $\boldsymbol{p} = \widehat{x}\boldsymbol{a}$. Check that $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$ is perpendicular to $\boldsymbol{a}$, and find the shortest distance $\|\boldsymbol{e}\|$ from $\boldsymbol{b}$ to the line through $\boldsymbol{a}$.

7 Find the closest line $b = Dt$, *through the origin*, to the same four points. An exact fit would solve $D \cdot 0 = 0$, $D \cdot 1 = 8$, $D \cdot 3 = 8$, $D \cdot 4 = 20$. Find the 4 by 1 matrix and solve $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}\boldsymbol{b}$. Redraw Figure 4.9a showing the best line $b = Dt$ and the e's.

8 Project $\boldsymbol{b} = (0, 8, 8, 20)$ onto the line through $\boldsymbol{a} = (0, 1, 3, 4)$. Find $\widehat{x} = D$ and $\boldsymbol{p} = \widehat{x}\boldsymbol{a}$. The best C in Problems 5–6 and the best D in Problems 7–8 do *not* agree with the best (C, D) in Problems 1–4. That is because $(1, 1, 1, 1)$ and $(0, 1, 3, 4)$ are _____ perpendicular.

9 For the closest parabola $b = C + Dt + Et^2$ to the same four points, write down the unsolvable equations $A\boldsymbol{x} = \boldsymbol{b}$ in three unknowns $\boldsymbol{x} = (C, D, E)$. Set up the three normal equations $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$ (solution not required). In Figure 4.9a you are now fitting a parabola to 4 points—what is happening in Figure 4.9b?

10 For the closest cubic $b = C + Dt + Et^2 + Ft^3$ to the same four points, write down the four equations $A\boldsymbol{x} = \boldsymbol{b}$. Solve them by elimination. In Figure 4.9a this cubic now goes exactly through the points. What are $\boldsymbol{p}$ and $\boldsymbol{e}$?

11 The average of the four times is $\widehat{t} = \frac{1}{4}(0 + 1 + 3 + 4) = 2$. The average of the four b's is $\widehat{b} = \frac{1}{4}(0 + 8 + 8 + 20) = 9$.

(a) Verify that the best line goes through the center point $(\widehat{t}, \widehat{b}) = (2, 9)$.

(b) Explain why $C + D\widehat{t} = \widehat{b}$ comes from the first equation in $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$.

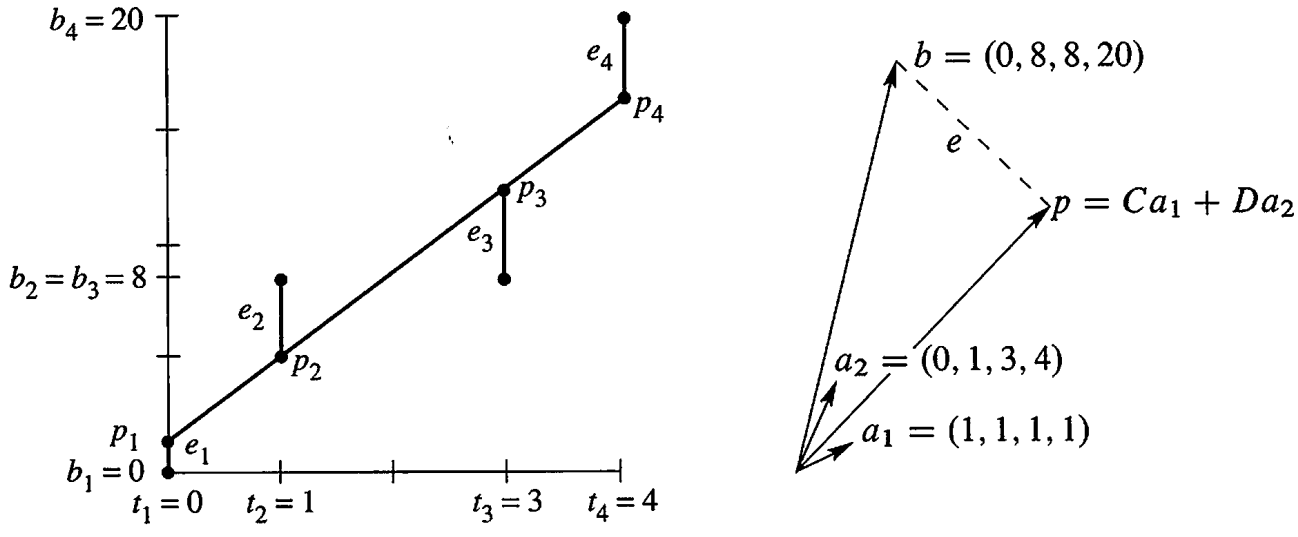


Figure 4.9: **Problems 1–11**: The closest line $C + Dt$ matches $Ca_1 + Da_2$ in $\mathbf{R}^4$.

Questions 12–16 introduce basic ideas of statistics—the foundation for least squares.

12 (Recommended) This problem projects $\boldsymbol{b} = (b_1, \ldots, b_m)$ onto the line through $\boldsymbol{a} = (1, \ldots, 1)$. We solve m equations $\boldsymbol{a}x = \boldsymbol{b}$ in 1 unknown (by least squares).

(a) Solve $\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}\widehat{x} = \boldsymbol{a}^{\mathrm{T}}\boldsymbol{b}$ to show that $\widehat{x}$ is the *mean* (the average) of the $\boldsymbol{b}$'s.

(b) Find $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{a}\widehat{x}$ and the *variance* $\|\boldsymbol{e}\|^2$ and the *standard deviation* $\|\boldsymbol{e}\|$.

(c) The horizontal line $\widehat{\boldsymbol{b}} = 3$ is closest to $\boldsymbol{b} = (1, 2, 6)$. Check that $\boldsymbol{p} = (3, 3, 3)$ is perpendicular to $\boldsymbol{e}$ and find the 3 by 3 projection matrix P.

13 First assumption behind least squares: $A\boldsymbol{x} = \boldsymbol{b}$ ***(noise e with mean zero)***. Multiply the error vectors $\boldsymbol{e} = \boldsymbol{b} - A\boldsymbol{x}$ by $(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ to get $\widehat{\boldsymbol{x}} - \boldsymbol{x}$ on the right. The estimation errors $\widehat{\boldsymbol{x}} - \boldsymbol{x}$ also average to zero. The estimate $\widehat{\boldsymbol{x}}$ is *unbiased*.

14 Second assumption behind least squares: The m errors $\boldsymbol{e}_i$ are independent with variance σ^2, so the average of $(\boldsymbol{b} - A\boldsymbol{x})(\boldsymbol{b} - A\boldsymbol{x})^{\mathrm{T}}$ is $\sigma^2 I$. Multiply on the left by $(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ and on the right by $A(A^{\mathrm{T}}A)^{-1}$ to show that the average matrix $(\widehat{\boldsymbol{x}} - \boldsymbol{x})(\widehat{\boldsymbol{x}} - \boldsymbol{x})^{\mathrm{T}}$ is $\sigma^2(A^{\mathrm{T}}A)^{-1}$. This is the ***covariance matrix*** P in section 8.6.

15 A doctor takes 4 readings of your heart rate. The best solution to $x = b_1, \ldots, x = b_4$ is the average $\widehat{x}$ of $b_1, \ldots, b_4$. The matrix A is a column of 1's. Problem 14 gives the expected error $(\widehat{x} - x)^2$ as $\sigma^2(A^{\mathrm{T}}A)^{-1} =$ _____. ***By averaging, the variance drops from*** σ^2 ***to*** $\sigma^2/4$.

16 If you know the average $\widehat{x}_9$ of 9 numbers $b_1, \ldots, b_9$, how can you quickly find the average $\widehat{x}_{10}$ with one more number b_{10}? The idea of *recursive* least squares is to avoid adding 10 numbers. What number multiplies $\widehat{x}_9$ in computing $\widehat{x}_{10}$?

$$\widehat{x}_{10} = \tfrac{1}{10}b_{10} + ____\widehat{x}_9 = \tfrac{1}{10}(b_1 + \cdots + b_{10}) \quad \text{as in Worked Example 4.2 C.}$$

Questions 17–24 give more practice with $\widehat{\boldsymbol{x}}$ and $\boldsymbol{p}$ and $\boldsymbol{e}$.

17 Write down three equations for the line $b = C + Dt$ to go through $b = 7$ at $t = -1$, $b = 7$ at $t = 1$, and $b = 21$ at $t = 2$. Find the least squares solution $\widehat{\boldsymbol{x}} = (C, D)$ and draw the closest line.

18 Find the projection $\boldsymbol{p} = A\widehat{\boldsymbol{x}}$ in Problem 17. This gives the three heights of the closest line. Show that the error vector is $\boldsymbol{e} = (2, -6, 4)$. Why is $P\boldsymbol{e} = \boldsymbol{0}$?

19 Suppose the measurements at $t = -1, 1, 2$ are the errors $2, -6, 4$ in Problem 18. Compute $\widehat{\boldsymbol{x}}$ and the closest line to these new measurements. Explain the answer: $\boldsymbol{b} = (2, -6, 4)$ is perpendicular to _____ so the projection is $\boldsymbol{p} = \boldsymbol{0}$.

20 Suppose the measurements at $t = -1, 1, 2$ are $\boldsymbol{b} = (5, 13, 17)$. Compute $\widehat{\boldsymbol{x}}$ and the closest line and $\boldsymbol{e}$. The error is $\boldsymbol{e} = \boldsymbol{0}$ because this $\boldsymbol{b}$ is _____.

21 Which of the four subspaces contains the error vector $\boldsymbol{e}$? Which contains $\boldsymbol{p}$? Which contains $\widehat{\boldsymbol{x}}$? What is the nullspace of A?

22 Find the best line $C + Dt$ to fit $b = 4, 2, -1, 0, 0$ at times $t = -2, -1, 0, 1, 2$.

23 Is the error vector $\boldsymbol{e}$ orthogonal to $\boldsymbol{b}$ or $\boldsymbol{p}$ or $\boldsymbol{e}$ or $\widehat{\boldsymbol{x}}$? Show that $\|\boldsymbol{e}\|^2$ equals $\boldsymbol{e}^{\mathrm{T}}\boldsymbol{b}$ which equals $\boldsymbol{b}^{\mathrm{T}}\boldsymbol{b} - \boldsymbol{p}^{\mathrm{T}}\boldsymbol{b}$. This is the smallest total error E.

24 The partial derivatives of $\|A\boldsymbol{x}\|^2$ with respect to $x_1, \ldots, x_n$ fill the vector $2A^{\mathrm{T}}A\boldsymbol{x}$. The derivatives of $2\boldsymbol{b}^{\mathrm{T}}A\boldsymbol{x}$ fill the vector $2A^{\mathrm{T}}\boldsymbol{b}$. So the derivatives of $\|A\boldsymbol{x} - \boldsymbol{b}\|^2$ are zero when _____.

Challenge Problems

25 *What condition on* $(t_1, b_1), (t_2, b_2), (t_3, b_3)$ *puts those three points onto a straight line*? A column space answer is: (b_1, b_2, b_3) must be a combination of (1, 1, 1) and (t_1, t_2, t_3). Try to reach a specific equation connecting the t's and b's. I should have thought of this question sooner!

26 Find the *plane* that gives the best fit to the 4 values $\boldsymbol{b} = (0, 1, 3, 4)$ at the corners $(1, 0)$ and $(0, 1)$ and $(-1, 0)$ and $(0, -1)$ of a square. The equations $C + Dx + Ey = b$ at those 4 points are $A\boldsymbol{x} = \boldsymbol{b}$ with 3 unknowns $\boldsymbol{x} = (C, D, E)$. What is A? At the center $(0, 0)$ of the square, show that $C + Dx + Ey =$ average of the b's.

27 (Distance between lines) The points $P = (x, x, x)$ and $Q = (y, 3y, -1)$ are on two lines in space that don't meet. Choose x and y to minimize the squared distance $\|P - Q\|^2$. The line connecting the closest P and Q is perpendicular to _____.

28 Suppose the columns of A are not independent. How could you find a matrix B so that $P = B(B^{\mathrm{T}}B)^{-1}B^{\mathrm{T}}$ does give the projection onto the column space of A? (The usual formula will fail when $A^{\mathrm{T}}A$ is not invertible.)

29 Usually there will be exactly one hyperplane in $\mathbf{R}^n$ that contains the n given points $\boldsymbol{x} = \boldsymbol{0}, \boldsymbol{a}_1, \ldots, \boldsymbol{a}_{n-1}$. (Example for $n = 3$: There will be one plane containing $0, \boldsymbol{a}_1, \boldsymbol{a}_2$ unless _____.) What is the test to have exactly one plane in $\mathbf{R}^n$?

4.4 Orthogonal Bases and Gram-Schmidt

This section has two goals. The first is to see how orthogonality makes it easy to find $\widehat{x}$ and p and P. Dot products are zero—so $A^{\mathrm{T}}A$ becomes a diagonal matrix. ***The second goal is to construct orthogonal vectors.*** We will pick combinations of the original vectors to produce right angles. Those original vectors are the columns of A, probably *not* orthogonal. ***The orthogonal vectors will be the columns of a new matrix Q.***

From Chapter 3, a basis consists of independent vectors that span the space. The basis vectors could meet at any angle (except 0° and 180°). But every time we visualize axes, they are perpendicular. *In our imagination, the coordinate axes are practically always orthogonal.* This simplifies the picture and it greatly simplifies the computations.

The vectors $q_1, \ldots, q_n$ are ***orthogonal*** when their dot products $q_i \cdot q_j$ are zero. More exactly $q_i^{\mathrm{T}} q_j = 0$ whenever $i \neq j$. With one more step—just divide each vector by its length—the vectors become ***orthogonal unit vectors***. Their lengths are all 1. Then the basis is called ***orthonormal***.

DEFINITION The vectors $q_1, \ldots, q_n$ are ***orthonormal*** if

$$q_i^{\mathrm{T}} q_j = \begin{cases} 0 & \text{when } i \neq j \quad (\textit{orthogonal} \text{ vectors}) \\ 1 & \text{when } i = j \quad (\textit{unit} \text{ vectors: } \|q_i\| = 1) \end{cases}$$

A matrix with orthonormal columns is assigned the special letter Q.

The matrix Q is easy to work with because $Q^{\mathrm{T}}Q = I$. This repeats in matrix language that the columns $q_1, \ldots, q_n$ are orthonormal. Q is not required to be square.

A matrix Q with orthonormal columns satisfies $Q^{\mathrm{T}}Q = I$:

$$Q^{\mathrm{T}}Q = \begin{bmatrix} -q_1^{\mathrm{T}}- \\ -q_2^{\mathrm{T}}- \\ \\ -q_n^{\mathrm{T}}- \end{bmatrix} \begin{bmatrix} | & | & & | \\ q_1 & q_2 & & q_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = I. \quad (1)$$

When row i of Q^{T} multiplies column j of Q, the dot product is $q_i^{\mathrm{T}} q_j$. Off the diagonal ($i \neq j$) that dot product is zero by orthogonality. On the diagonal ($i = j$) the unit vectors give $q_i^{\mathrm{T}} q_i = \|q_i\|^2 = 1$. Often Q is rectangular ($m > n$). Sometimes $m = n$.

When Q is square, $Q^{\mathrm{T}}Q = I$ means that $Q^{\mathrm{T}} = Q^{-1}$: transpose = inverse.

If the columns are only orthogonal (not unit vectors), dot products still give a diagonal matrix (not the identity matrix). But this matrix is almost as good. The important thing is orthogonality—then it is easy to produce unit vectors.

To repeat: $Q^TQ = I$ even when Q is rectangular. In that case Q^T is only an inverse from the left. For square matrices we also have $QQ^T = I$, so Q^T is the two-sided inverse of Q. The rows of a square Q are orthonormal like the columns. ***The inverse is the transpose.*** In this square case we call Q an ***orthogonal matrix.***[1]

Here are three examples of orthogonal matrices—rotation and permutation and reflection. The quickest test is to check $Q^TQ = I$.

Example 1 **(Rotation)** Q rotates every vector in the plane clockwise by the angle θ:

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \quad \text{and} \quad Q^T = Q^{-1} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}.$$

The columns of Q are orthogonal (take their dot product). They are unit vectors because $\sin^2\theta + \cos^2\theta = 1$. Those columns give an ***orthonormal basis*** for the plane $\mathbf{R}^2$. The standard basis vectors $\boldsymbol{i}$ and $\boldsymbol{j}$ are rotated through θ (see Figure 4.10a). Q^{-1} rotates vectors back through $-\theta$. It agrees with Q^T, because the cosine of $-\theta$ is the cosine of θ, and $\sin(-\theta) = -\sin\theta$. We have $Q^TQ = I$ and $QQ^T = I$.

Example 2 **(Permutation)** These matrices change the order to (y, z, x) and (y, x):

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y \\ z \\ x \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}.$$

All columns of these Q's are unit vectors (their lengths are obviously 1). They are also orthogonal (the 1's appear in different places). *The inverse of a permutation matrix is its transpose.* The inverse puts the components back into their original order:

Inverse = transpose: $$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} y \\ z \\ x \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y \\ x \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

Every permutation matrix is an orthogonal matrix.

Example 3 **(Reflection)** If $\boldsymbol{u}$ is any unit vector, set $Q = I - 2\boldsymbol{u}\boldsymbol{u}^T$. Notice that $\boldsymbol{u}\boldsymbol{u}^T$ is a matrix while $\boldsymbol{u}^T\boldsymbol{u}$ is the number $\|\boldsymbol{u}\|^2 = 1$. Then Q^T and Q^{-1} both equal Q:

$$Q^T = I - 2\boldsymbol{u}\boldsymbol{u}^T = Q \quad \text{and} \quad Q^TQ = I - 4\boldsymbol{u}\boldsymbol{u}^T + 4\boldsymbol{u}\boldsymbol{u}^T\boldsymbol{u}\boldsymbol{u}^T = I. \tag{2}$$

Reflection matrices $I - 2\boldsymbol{u}\boldsymbol{u}^T$ are symmetric and also orthogonal. If you square them, you get the identity matrix: $Q^2 = Q^TQ = I$. Reflecting twice through a mirror brings back the original. Notice $\boldsymbol{u}^T\boldsymbol{u} = 1$ inside $4\boldsymbol{u}\boldsymbol{u}^T\boldsymbol{u}\boldsymbol{u}^T$ in equation (2).

[1] "Orthonormal matrix" would have been a better name for Q, but it's not used. Any matrix with orthonormal columns has the letter Q, but we only call it an *orthogonal matrix* when it is square.

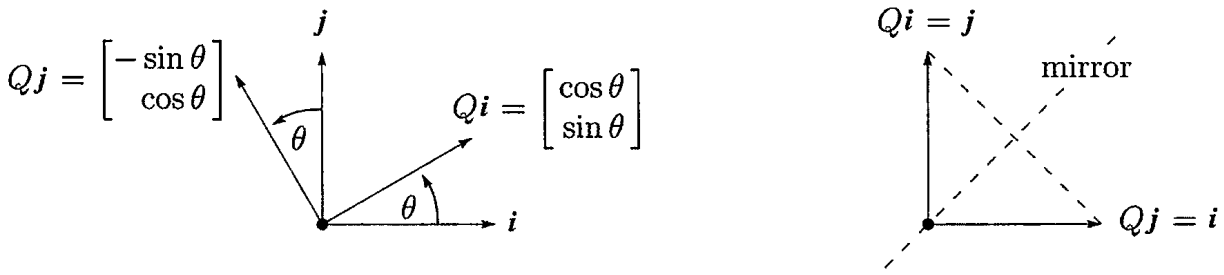


Figure 4.10: Rotation by $Q = \begin{bmatrix} c & -s \\ s & c \end{bmatrix}$ and reflection across 45° by $Q = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

As examples choose two unit vectors, $\boldsymbol{u} = (1, 0)$ and then $\boldsymbol{u} = (1/\sqrt{2}, -1/\sqrt{2})$. Compute $2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$ (column times row) and subtract from I to get reflections Q_1 and Q_2:

$$Q_1 = I - 2\begin{bmatrix} 1 \\ 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad Q_2 = I - 2\begin{bmatrix} .5 & -.5 \\ -.5 & .5 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Q_1 reflects $(x, 0)$ across the y axis to $(-x, 0)$. Every vector (x, y) goes into its image $(-x, y)$, and the y axis is the mirror. Q_2 is reflection across the 45° line:

$$\textbf{Reflections} \quad \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -x \\ y \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}.$$

When (x, y) goes to (y, x), a vector like $(3, 3)$ doesn't move. It is on the mirror line. Figure 4.10b shows the 45° mirror.

Rotations preserve the length of a vector. So do reflections. So do permutations. So does multiplication by any orthogonal matrix—***lengths and angles don't change.***

If Q has orthonormal columns ($Q^{\mathrm{T}}Q = I$), it leaves lengths unchanged:

Same length $\|Qx\| = \|x\|$ **for every vector** x. (3)

Q also preserves dot products: $(Qx)^{\mathrm{T}}(Qy) = x^{\mathrm{T}}Q^{\mathrm{T}}Qy = x^{\mathrm{T}}y$. Just use $Q^{\mathrm{T}}Q = I$!

Proof $\|Qx\|^2$ equals $\|x\|^2$ because $(Qx)^{\mathrm{T}}(Qx) = x^{\mathrm{T}}Q^{\mathrm{T}}Qx = x^{\mathrm{T}}Ix = x^{\mathrm{T}}x$. Orthogonal matrices are excellent for computations—numbers can never grow too large when lengths of vectors are fixed. Stable computer codes use Q's as much as possible.

Projections Using Orthogonal Bases: Q Replaces A

This chapter is about projections onto subspaces. We developed the equations for $\widehat{x}$ and $\boldsymbol{p}$ and the matrix P. When the columns of A were a basis for the subspace, all formulas involved $A^{\mathrm{T}}A$. The entries of $A^{\mathrm{T}}A$ are the dot products $\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{a}_j$.

Suppose the basis vectors are actually orthonormal. The a's become q's. Then A^TA simplifies to $Q^TQ = I$. Look at the improvements in $\hat{x}$ and p and P. Instead of Q^TQ we print a blank for the identity matrix:

$$____\hat{x} = Q^Tb \quad \text{and} \quad p = Q\hat{x} \quad \text{and} \quad P = Q____Q^T. \tag{4}$$

The least squares solution of $Qx = b$ is $\hat{x} = Q^Tb$. The projection matrix is $P = QQ^T$.

There are no matrices to invert. This is the point of an orthonormal basis. The best $\hat{x} = Q^Tb$ just has dot products of $q_1, \dots, q_n$ with b. We have n 1-dimensional projections! The "coupling matrix" or "correlation matrix" A^TA is now $Q^TQ = I$. There is no coupling. When A is Q, with orthonormal columns, here is $p = Q\hat{x} = QQ^Tb$:

Projection onto q's

$$p = \begin{bmatrix} | & & | \\ q_1 & \cdots & q_n \\ | & & | \end{bmatrix} \begin{bmatrix} q_1^Tb \\ \vdots \\ q_n^Tb \end{bmatrix} = q_1(q_1^Tb) + \cdots + q_n(q_n^Tb). \tag{5}$$

Important case: When Q is square and $m = n$, the subspace is the whole space. Then $Q^T = Q^{-1}$ and $\hat{x} = Q^Tb$ is the same as $x = Q^{-1}b$. The solution is exact! The projection of b onto the whole space is b itself. In this case $P = QQ^T = I$.

You may think that projection onto the whole space is not worth mentioning. But when $p = b$, our formula assembles b out of its 1-dimensional projections. If $q_1, \dots, q_n$ is an orthonormal basis for the whole space, so Q is square, then every $b = QQ^Tb$ is the sum of its components along the q's:

$$b = q_1(q_1^Tb) + q_2(q_2^Tb) + \cdots + q_n(q_n^Tb). \tag{6}$$

That is $QQ^T = I$. It is the foundation of Fourier series and all the great "transforms" of applied mathematics. They break vectors or functions into perpendicular pieces. Then by adding the pieces, the inverse transform puts the function back together.

Example 4 The columns of this orthogonal Q are orthonormal vectors q_1, q_2, q_3:

$$Q = \frac{1}{3}\begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix} \quad \text{has} \quad Q^TQ = QQ^T = I.$$

The separate projections of $b = (0, 0, 1)$ onto q_1 and q_2 and q_3 are p_1 and p_2 and p_3:

$$q_1(q_1^Tb) = \tfrac{2}{3}q_1 \quad \text{and} \quad q_2(q_2^Tb) = \tfrac{2}{3}q_2 \quad \text{and} \quad q_3(q_3^Tb) = -\tfrac{1}{3}q_3.$$

The sum of the first two is the projection of b onto the *plane* of q_1 and q_2. The sum of all three is the projection of b onto the *whole space*—which is b itself:

Reconstruct $b = p_1 + p_2 + p_3$

$$\tfrac{2}{3}q_1 + \tfrac{2}{3}q_2 - \tfrac{1}{3}q_3 = \frac{1}{9}\begin{bmatrix} -2+4-2 \\ 4-2-2 \\ 4+4+1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = b.$$

The Gram-Schmidt Process

The point of this section is that "orthogonal is good." Projections and least squares always involve A^TA. When this matrix becomes $Q^TQ = I$, the inverse is no problem. The one-dimensional projections are uncoupled. The best $\widehat{x}$ is Q^Tb (just n separate dot products). For this to be true, we had to say "*If* the vectors are orthonormal". ***Now we find a way to create orthonormal vectors.***

Start with three independent vectors a, b, c. We intend to construct three orthogonal vectors A, B, C. Then (at the end is easiest) we divide A, B, C by their lengths. That produces three orthonormal vectors $q_1 = A/\|A\|$, $q_2 = B/\|B\|$, $q_3 = C/\|C\|$.

Gram-Schmidt Begin by choosing $A = a$. This first direction is accepted. The next direction B must be perpendicular to A. ***Start with b and subtract its projection along A.*** This leaves the perpendicular part, which is the orthogonal vector B:

First Gram-Schmidt step
$$B = b - \frac{A^Tb}{A^TA}A. \tag{7}$$

A and B are orthogonal in Figure 4.11. Take the dot product with A to verify that $A^TB = A^Tb - A^Tb = 0$. This vector B is what we have called the error vector e, perpendicular to A. Notice that B in equation (7) is not zero (otherwise a and b would be dependent). The directions A and B are now set.

The third direction starts with c. This is not a combination of A and B (because c is not a combination of a and b). But most likely c is not perpendicular to A and B. So subtract off its components in those two directions to get C:

Next Gram-Schmidt step
$$C = c - \frac{A^Tc}{A^TA}A - \frac{B^Tc}{B^TB}B. \tag{8}$$

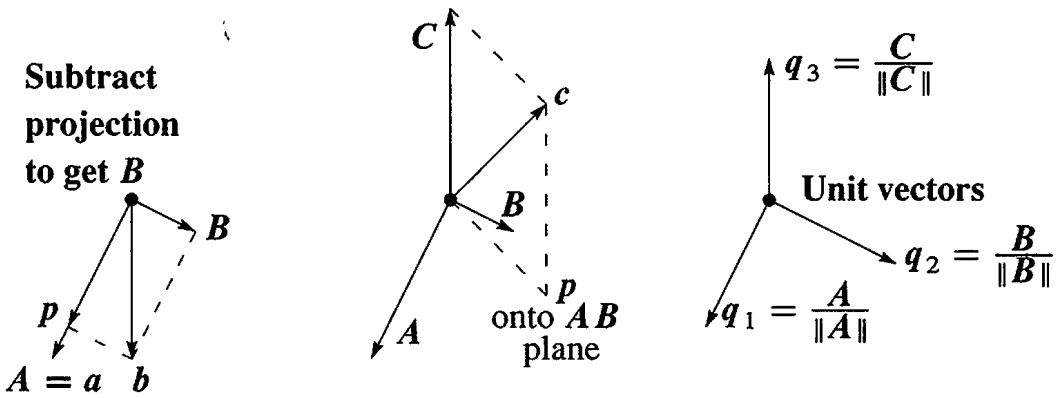


Figure 4.11: First project b onto the line through a and find the orthogonal B as $b - p$. Then project c onto the AB plane and find C as $c - p$. Divide by $\|A\|, \|B\|, \|C\|$.

This is the one and only idea of the Gram-Schmidt process. ***Subtract from every new vector its projections in the directions already set.*** That idea is repeated at every step.[2] If we had a fourth vector d, we would subtract three projections onto A, B, C to get D. At the end, *or immediately when each one is found*, divide the orthogonal vectors A, B, C, D by their lengths. The resulting vectors q_1, q_2, q_3, q_4 are orthonormal.

Example 5 Suppose the independent non-orthogonal vectors a, b, c are

$$a = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix} \quad \text{and} \quad c = \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix}.$$

Then $A = a$ has $A^{\mathrm{T}}A = 2$. Subtract from b its projection along $A = (1, -1, 0)$:

First step
$$B = b - \frac{A^{\mathrm{T}}b}{A^{\mathrm{T}}A}A = b - \tfrac{2}{2}A = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}.$$

Check: $A^{\mathrm{T}}B = 0$ as required. Now subtract two projections from c to get C:

Next step
$$C = c - \frac{A^{\mathrm{T}}c}{A^{\mathrm{T}}A}A - \frac{B^{\mathrm{T}}c}{B^{\mathrm{T}}B}B = c - \tfrac{6}{2}A + \tfrac{6}{6}B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Check: $C = (1, 1, 1)$ is perpendicular to A and B. Finally convert A, B, C to unit vectors (length 1, orthonormal). The lengths of A, B, C are $\sqrt{2}$ and $\sqrt{6}$ and $\sqrt{3}$. Divide by those lengths, for an orthonormal basis:

$$q_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad \text{and} \quad q_2 = \frac{1}{\sqrt{6}}\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \quad \text{and} \quad q_3 = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Usually A, B, C contain fractions. Almost always q_1, q_2, q_3 contain square roots.

The Factorization $A = QR$

We started with a matrix A, whose columns were a, b, c. We ended with a matrix Q, whose columns are q_1, q_2, q_3. How are those matrices related? Since the vectors a, b, c are combinations of the q's (and vice versa), there must be a third matrix connecting A to Q. This third matrix is the triangular R in $A = QR$.

The first step was $q_1 = a/\|a\|$ (other vectors not involved). The second step was equation (7), where b is a combination of A and B. At that stage C and q_3 were not involved. This non-involvement of later vectors is the key point of Gram-Schmidt:

[2] I think Gram had the idea. I don't really know where Schmidt came in.

- The vectors $\boldsymbol{a}$ and $\boldsymbol{A}$ and $\boldsymbol{q}_1$ are all along a single line.
- The vectors $\boldsymbol{a}, \boldsymbol{b}$ and $\boldsymbol{A}, \boldsymbol{B}$ and $\boldsymbol{q}_1, \boldsymbol{q}_2$ are all in the same plane.
- The vectors $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ and $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ and $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$ are in one subspace (dimension 3).

At every step $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_k$ are combinations of $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k$. Later $\boldsymbol{q}$'s are not involved. The connecting matrix R is ***triangular***, and we have $A = QR$:

$$\begin{bmatrix} & & \\ \boldsymbol{a} & \boldsymbol{b} & \boldsymbol{c} \\ & & \end{bmatrix} = \begin{bmatrix} & & \\ \boldsymbol{q}_1 & \boldsymbol{q}_2 & \boldsymbol{q}_3 \\ & & \end{bmatrix} \begin{bmatrix} \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{a} & \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{b} & \boldsymbol{q}_1^{\mathrm{T}}\boldsymbol{c} \\ & \boldsymbol{q}_2^{\mathrm{T}}\boldsymbol{b} & \boldsymbol{q}_2^{\mathrm{T}}\boldsymbol{c} \\ & & \boldsymbol{q}_3^{\mathrm{T}}\boldsymbol{c} \end{bmatrix} \quad \text{or} \quad A = QR. \tag{9}$$

$A = QR$ is Gram-Schmidt in a nutshell. Multiply by Q^{T} to see why $R = Q^{\mathrm{T}}A$.

(Gram-Schmidt) From independent vectors $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$, Gram-Schmidt constructs orthonormal vectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$. The matrices with these columns satisfy $A = QR$. Then $R = Q^{\mathrm{T}}A$ is ***upper triangular*** because later $\boldsymbol{q}$'s are orthogonal to earlier $\boldsymbol{a}$'s.

Here are the $\boldsymbol{a}$'s and $\boldsymbol{q}$'s from the example. The i, j entry of $R = Q^{\mathrm{T}}A$ is row i of Q^{T} times column j of A. This is the dot product of $\boldsymbol{q}_i$ with $\boldsymbol{a}_j$:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & -3 \\ 0 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ -1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ 0 & -2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{2} & \sqrt{2} & \sqrt{18} \\ \mathbf{0} & \sqrt{6} & -\sqrt{6} \\ \mathbf{0} & \mathbf{0} & \sqrt{3} \end{bmatrix} = QR.$$

The lengths of $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ are the numbers $\sqrt{2}, \sqrt{6}, \sqrt{3}$ on the diagonal of R. Because of the square roots, QR looks less beautiful than LU. Both factorizations are absolutely central to calculations in linear algebra.

Any m by n matrix A with independent columns can be factored into QR. The m by n matrix Q has orthonormal columns, and the square matrix R is upper triangular with positive diagonal. We must not forget why this is useful for least squares: $\boldsymbol{A^{\mathrm{T}}A}$ ***equals*** $\boldsymbol{R^{\mathrm{T}}Q^{\mathrm{T}}QR = R^{\mathrm{T}}R}$. The least squares equation $A^{\mathrm{T}}A\widehat{x} = A^{\mathrm{T}}\boldsymbol{b}$ simplifies to $R\boldsymbol{x} = Q^{\mathrm{T}}\boldsymbol{b}$:

Least squares $\quad R^{\mathrm{T}}R\widehat{\boldsymbol{x}} = R^{\mathrm{T}}Q^{\mathrm{T}}\boldsymbol{b}$ or $R\widehat{\boldsymbol{x}} = Q^{\mathrm{T}}\boldsymbol{b}$ or $\widehat{\boldsymbol{x}} = R^{-1}Q^{\mathrm{T}}\boldsymbol{b}$ (10)

Instead of solving $A\boldsymbol{x} = \boldsymbol{b}$, which is impossible, we solve $R\widehat{\boldsymbol{x}} = Q^{\mathrm{T}}\boldsymbol{b}$ by back substitution—which is very fast. The real cost is the mn^2 multiplications in the Gram-Schmidt process, which are needed to construct the orthogonal Q and the triangular R.

Below is an informal code. It executes equations (11) and (12), for $k = 1$ then $k = 2$ and eventually $k = n$. The last line of that code normalizes to unit vectors $\boldsymbol{q}_j$:

Divide by length $\boldsymbol{q_j}$ = unit vector

$$r_{jj} = \left(\sum_{i=1}^{m} v_{ij}^2\right)^{1/2} \quad \textbf{and} \quad q_{ij} = \frac{v_{ij}}{r_{jj}} \quad \textbf{for} \quad i = 1, \ldots, m. \tag{11}$$

The important lines subtract from $v = a_j$ its projection onto each q_i:

$$r_{kj} = \sum_{i=1}^{m} q_{ik} v_{ij} \qquad \text{and} \qquad v_{ij} = v_{ij} - q_{ik} r_{kj}. \tag{12}$$

Starting from $a, b, c = a_1, a_2, a_3$ this code will construct q_1, B, q_2, C, q_3:

$$q_1 = a_1/\|a_1\| \qquad B = a_2 - (q_1^{\mathrm{T}} a_2) q_1 \qquad q_2 = B/\|B\|$$

$$C^* = a_3 - (q_1^{\mathrm{T}} a_3) q_1 \qquad C = C^* - (q_2^{\mathrm{T}} C^*) q_2 \qquad q_3 = C/\|C\|$$

Equation (12) subtracts off projections as soon as the new vector q_k is found. This change to "subtract one projection at a time" is called ***modified Gram-Schmidt***. That is numerically more stable than equation (8) which subtracts all projections at once.

```
for j = 1:n
   v = A(:, j);
   for i = 1:j-1
      R(i, j) = Q(:, i)' * v;
      v = v - R(i, j) * Q(:, i);
   end
   R(j, j) = norm(v);
   Q(:, j) = v/R(j, j);
end
```

```
% modified Gram-Schmidt
% v begins as column j of A
% columns up to j - 1, already settled in Q
% compute r_ij = q_i^T a_j which is q_i^T v
% subtract the projection (q_i^T a_j) q_i = (q_i^T v) q_i
% v is now perpendicular to all of q_1, ..., q_{j-1}
% diagonal entries of R
% normalize v to be the next unit vector q_j
```

To recover column j of A, undo the last step and the middle steps of the code:

$$R(j, j) q_j = (v \textit{ minus its projections}) = (\text{column } j \text{ of } A) - \sum_{i=1}^{j-1} R(i, j) q_i \,. \tag{13}$$

Moving the sum to the far left, this is column j in the multiplication $A = QR$.

Confession Good software like LAPACK, used in good systems like MATLAB and Octave and Python, will not use this Gram-Schmidt code. There is now a better way. "Householder reflections" produce the upper triangular R, one column at a time, exactly as elimination produces the upper triangular U.

Those reflection matrices $I - 2uu^{\mathrm{T}}$ will be described in Chapter 9 on numerical linear algebra. If A is tridiagonal we can simplify even more to use 2 by 2 rotations. The result is always $A = QR$ and the MATLAB command is $[Q, R] = \mathrm{qr}(A)$. I believe that Gram-Schmidt is still the good process to understand, even if the reflections or rotations lead to a more perfect Q.

■ REVIEW OF THE KEY IDEAS ■

1. If the orthonormal vectors $q_1, \ldots, q_n$ are the columns of Q, then $q_i^T q_j = 0$ and $q_i^T q_i = 1$ translate into $Q^T Q = I$.
2. If Q is square (an ***orthogonal matrix***) then $Q^T = Q^{-1}$: ***transpose = inverse.***
3. The length of Qx equals the length of x: $\|Qx\| = \|x\|$.
4. The projection onto the column space spanned by the q's is $P = QQ^T$.
5. If Q is square then $P = I$ and every $b = q_1(q_1^T b) + \cdots + q_n(q_n^T b)$.
6. Gram-Schmidt produces orthonormal vectors q_1, q_2, q_3 from independent a, b, c. In matrix form this is the factorization $A = QR =$ (orthogonal Q)(triangular R).

■ WORKED EXAMPLES ■

4.4 A Add two more columns with all entries 1 or -1, so the columns of this 4 by 4 "Hadamard matrix" are orthogonal. How do you turn H_4 into an *orthogonal matrix* Q?

$$H_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \qquad H_4 = \begin{bmatrix} 1 & 1 & x & x \\ 1 & -1 & x & x \\ 1 & 1 & x & x \\ 1 & -1 & x & x \end{bmatrix} \quad \text{and} \quad Q_4 = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$$

The block matrix $H_8 = \begin{bmatrix} H_4 & H_4 \\ H_4 & -H_4 \end{bmatrix}$ is the next Hadamard matrix with 1's and -1's. What is the product $H_8^T H_8$?

The projection of $b = (6, 0, 0, 2)$ onto the first column of H_4 is $p_1 = (2, 2, 2, 2)$. The projection onto the second column is $p_2 = (1, -1, 1, -1)$. What is the projection $p_{1,2}$ of b onto the 2-dimensional space spanned by the first two columns?

Solution H_4 can be built from H_2 just as H_8 is built from H_4:

$$H_4 = \begin{bmatrix} H_2 & H_2 \\ H_2 & -H_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \text{ has orthogonal columns.}$$

Then $Q = H/2$ has orthonormal columns. Dividing by 2 gives unit vectors in Q. Orthogonality for 5 by 5 is impossible because the dot product of columns would have five 1's

and/or -1's and could not add to zero. H_8 has orthogonal columns of length $\sqrt{8}$.

$$H_8^{\mathrm{T}} H_8 = \begin{bmatrix} H^{\mathrm{T}} & H^{\mathrm{T}} \\ H^{\mathrm{T}} & -H^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} H & H \\ H & -H \end{bmatrix} = \begin{bmatrix} 2H^{\mathrm{T}}H & 0 \\ 0 & 2H^{\mathrm{T}}H \end{bmatrix} = \begin{bmatrix} 8I & 0 \\ 0 & 8I \end{bmatrix} \cdot Q_8 = \frac{H_8}{\sqrt{8}}$$

Key point of orthogonal columns: We can project $(6,0,0,2)$ onto $(1,1,1,1)$ and $(1,-1,1,-1)$ and **add**. There is no $A^{\mathrm{T}}A$ matrix to invert:

Add p's Projection $p_{1,2} = p_1 + p_2 = (2,2,2,2) + (1,-1,1,-1) = (3,1,3,1)$.

Check that columns a_1 and a_2 of H are perpendicular to the error $e = b - p_1 - p_2$:

$$e = b - \frac{a_1^{\mathrm{T}} b}{a_1^{\mathrm{T}} a_1} a_1 - \frac{a_2^{\mathrm{T}} b}{a_2^{\mathrm{T}} a_2} a_2 \quad \text{and} \quad a_1^{\mathrm{T}} e = a_1^{\mathrm{T}} b - \frac{a_1^{\mathrm{T}} b}{a_1^{\mathrm{T}} a_1} a_1^{\mathrm{T}} a_1 = 0 \quad \text{and also} \quad a_2^{\mathrm{T}} e = 0.$$

So $p_1 + p_2$ is in the space of a_1 and a_2, and *its error e is perpendicular to that space.*

The Gram-Schmidt process on those orthogonal columns a_1 and a_2 would not change their directions. It would only divide by their lengths. *But if a_1 and a_2 are not orthogonal, the projection $p_{1,2}$ is not generally $p_1 + p_2$.* For example, if b is the same as a_1, then $p_1 = b$ and $p_{1,2} = b$ but $p_2 \neq 0$.

Problem Set 4.4

Problems 1–12 are about orthogonal vectors and orthogonal matrices.

1 Are these pairs of vectors orthonormal or only orthogonal or only independent?

(a) $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ (b) $\begin{bmatrix} .6 \\ .8 \end{bmatrix}$ and $\begin{bmatrix} .4 \\ -.3 \end{bmatrix}$ (c) $\begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$ and $\begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$.

Change the second vector when necessary to produce orthonormal vectors.

2 The vectors $(2,2,-1)$ and $(-1,2,2)$ are orthogonal. Divide them by their lengths to find orthonormal vectors q_1 and q_2. Put those into the columns of Q and multiply $Q^{\mathrm{T}}Q$ and QQ^{T}.

3 (a) If A has three orthogonal columns each of length 4, what is $A^{\mathrm{T}}A$?

(b) If A has three orthogonal columns of lengths $1,2,3$, what is $A^{\mathrm{T}}A$?

4 Give an example of each of the following:

(a) A matrix Q that has orthonormal columns but $QQ^{\mathrm{T}} \neq I$.

(b) Two orthogonal vectors that are not linearly independent.

(c) An orthonormal basis for $\mathbf{R}^3$, including the vector $q_1 = (1,1,1)/\sqrt{3}$.

5 Find two orthogonal vectors in the plane $x + y + 2z = 0$. Make them orthonormal.

6 If Q_1 and Q_2 are orthogonal matrices, show that their product Q_1Q_2 is also an orthogonal matrix. (Use $Q^TQ = I$.)

7 If Q has orthonormal columns, what is the least squares solution $\hat{x}$ to $Qx = b$?

8 If q_1 and q_2 are orthonormal vectors in $\mathbf{R}^5$, what combination $____ q_1 + ____ q_2$ is closest to a given vector b?

9 (a) Compute $P = QQ^T$ when $q_1 = (.8, .6, 0)$ and $q_2 = (-.6, .8, 0)$. Verify that $P^2 = P$.

(b) Prove that always $(QQ^T)^2 = QQ^T$ by using $Q^TQ = I$. Then $P = QQ^T$ is the projection matrix onto the column space of Q.

10 Orthonormal vectors are automatically linearly independent.

(a) Vector proof: When $c_1q_1 + c_2q_2 + c_3q_3 = \mathbf{0}$, what dot product leads to $c_1 = 0$? Similarly $c_2 = 0$ and $c_3 = 0$. Thus the q's are independent.

(b) Matrix proof: Show that $Qx = \mathbf{0}$ leads to $x = \mathbf{0}$. Since Q may be rectangular, you can use Q^T but not Q^{-1}.

11 (a) Gram-Schmidt: Find orthonormal vectors q_1 and q_2 in the plane spanned by $a = (1, 3, 4, 5, 7)$ and $b = (-6, 6, 8, 0, 8)$.

(b) Which vector in this plane is closest to $(1, 0, 0, 0, 0)$?

12 If a_1, a_2, a_3 is a basis for $\mathbf{R}^3$, any vector b can be written as

$$b = x_1a_1 + x_2a_2 + x_3a_3 \qquad \text{or} \qquad \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = b.$$

(a) Suppose the a's are orthonormal. Show that $x_1 = a_1^Tb$.

(b) Suppose the a's are orthogonal. Show that $x_1 = a_1^Tb/a_1^Ta_1$.

(c) If the a's are independent, x_1 is the first component of $____$ times b.

Problems 13–25 are about the Gram-Schmidt process and $A = QR$.

13 What multiple of $a = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ should be subtracted from $b = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$ to make the result B orthogonal to a? Sketch a figure to show a, b, and B.

14 Complete the Gram-Schmidt process in Problem 13 by computing $q_1 = a/\|a\|$ and $q_2 = B/\|B\|$ and factoring into QR:

$$\begin{bmatrix} 1 & 4 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} q_1 & q_2 \end{bmatrix} \begin{bmatrix} \|a\| & ? \\ 0 & \|B\| \end{bmatrix}.$$

15 (a) Find orthonormal vectors q_1, q_2, q_3 such that q_1, q_2 span the column space of

$$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}.$$

(b) Which of the four fundamental subspaces contains q_3?

(c) Solve $Ax = (1, 2, 7)$ by least squares.

16 What multiple of $a = (4, 5, 2, 2)$ is closest to $b = (1, 2, 0, 0)$? Find orthonormal vectors q_1 and q_2 in the plane of a and b.

17 Find the projection of b onto the line through a:

$$a = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \quad \text{and} \quad p = ? \quad \text{and} \quad e = b - p = ?$$

Compute the orthonormal vectors $q_1 = a/\|a\|$ and $q_2 = e/\|e\|$.

18 (Recommended) Find orthogonal vectors A, B, C by Gram-Schmidt from a, b, c:

$$a = (1, -1, 0, 0) \qquad b = (0, 1, -1, 0) \qquad c = (0, 0, 1, -1).$$

A, B, C and a, b, c are bases for the vectors perpendicular to $d = (1, 1, 1, 1)$.

19 If $A = QR$ then $A^{\mathrm{T}}A = R^{\mathrm{T}}R =$ _____ triangular times _____ triangular. *Gram-Schmidt on A corresponds to elimination on $A^{\mathrm{T}}A$.* The pivots for $A^{\mathrm{T}}A$ must be the squares of diagonal entries of R. Find Q and R by Gram-Schmidt for this A:

$$A = \begin{bmatrix} -1 & 1 \\ 2 & 1 \\ 2 & 4 \end{bmatrix} \quad \text{and} \quad A^{\mathrm{T}}A = \begin{bmatrix} 9 & 9 \\ 9 & 18 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 9 & \\ & 9 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

20 True or false (give an example in either case):

(a) Q^{-1} is an orthogonal matrix when Q is an orthogonal matrix.

(b) If Q (3 by 2) has orthonormal columns then $\|Qx\|$ always equals $\|x\|$.

21 Find an orthonormal basis for the column space of A:

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 0 \\ 1 & 1 \\ 1 & 3 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} -4 \\ -3 \\ 3 \\ 0 \end{bmatrix}.$$

Then compute the projection of b onto that column space.

22 Find orthogonal vectors A, B, C by Gram-Schmidt from

$$a = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad \text{and} \quad c = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}.$$

23 Find q_1, q_2, q_3 (orthonormal) as combinations of a, b, c (independent columns). Then write A as QR:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix}.$$

24 (a) Find a basis for the subspace S in $\mathbf{R}^4$ spanned by all solutions of

$$x_1 + x_2 + x_3 - x_4 = 0.$$

(b) Find a basis for the orthogonal complement $S^\perp$.

(c) Find b_1 in S and b_2 in $S^\perp$ so that $b_1 + b_2 = b = (1, 1, 1, 1)$.

25 If $ad - bc > 0$, the entries in $A = QR$ are

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \frac{\begin{bmatrix} a & -c \\ c & a \end{bmatrix}}{\sqrt{a^2 + c^2}} \frac{\begin{bmatrix} a^2 + c^2 & ab + cd \\ 0 & ad - bc \end{bmatrix}}{\sqrt{a^2 + c^2}}.$$

Write $A = QR$ when $a, b, c, d = 2, 1, 1, 1$ and also $1, 1, 1, 1$. Which entry of R becomes zero when the columns are dependent and Gram-Schmidt breaks down?

Problems 26–29 use the QR code in equations (11–12). It executes Gram-Schmidt.

26 Show why C (found via C^* in the steps after (12)) is equal to C in equation (8).

27 Equation (8) subtracts from c its components along A and B. Why not subtract the components along a and along b?

28 Where are the mn^2 multiplications in equations (11) and (12)?

29 Apply the MATLAB qr code to $a = (2, 2, -1)$, $b = (0, -3, 3)$, $c = (1, 0, 0)$. What are the q's?

Problems 30–35 involve orthogonal matrices that are special.

30 The first four ***wavelets*** are in the columns of this wavelet matrix W:

$$W = \frac{1}{2} \begin{bmatrix} 1 & 1 & \sqrt{2} & 0 \\ 1 & 1 & -\sqrt{2} & 0 \\ 1 & -1 & 0 & \sqrt{2} \\ 1 & -1 & 0 & -\sqrt{2} \end{bmatrix}.$$

What is special about the columns? Find the inverse wavelet transform W^{-1}.

31 (a) Choose c so that Q is an orthogonal matrix:

$$Q = c\begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}.$$

Project $\boldsymbol{b} = (1, 1, 1, 1)$ onto the first column. Then project $\boldsymbol{b}$ onto the plane of the first two columns.

32 If $\boldsymbol{u}$ is a unit vector, then $Q = I - 2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$ is a reflection matrix (Example 3). Find Q_1 from $\boldsymbol{u} = (0, 1)$ and Q_2 from $\boldsymbol{u} = (0, \sqrt{2}/2, \sqrt{2}/2)$. Draw the reflections when Q_1 and Q_2 multiply the vectors $(1, 2)$ and $(1, 1, 1)$.

33 Find all matrices that are both orthogonal and lower triangular.

34 $Q = I - 2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$ is a reflection matrix when $\boldsymbol{u}^{\mathrm{T}}\boldsymbol{u} = 1$. Two reflections give $Q^2 = I$.

(a) Show that $Q\boldsymbol{u} = -\boldsymbol{u}$. The mirror is perpendicular to $\boldsymbol{u}$.

(b) Find $Q\boldsymbol{v}$ when $\boldsymbol{u}^{\mathrm{T}}\boldsymbol{v} = 0$. The mirror contains $\boldsymbol{v}$. It reflects to itself.

Challenge Problems

35 (MATLAB) Factor $[Q, R] = \mathbf{qr}(A)$ for $A = \mathbf{eye}(4) - \mathbf{diag}([1\ \ 1\ \ 1], -1)$. You are orthogonalizing the columns $(1, -1, 0, 0)$ and $(0, 1, -1, 0)$ and $(0, 0, 1, -1)$ and $(0, 0, 0, 1)$ of A. Can you scale the orthogonal columns of Q to get nice integer components?

36 If A is m by n with rank n, $\mathbf{qr}(A)$ produces a *square* Q and zeros below R:

The factors from MATLAB are (m by m)(m by n) $\qquad A = [Q_1 \quad Q_2]\begin{bmatrix} R \\ 0 \end{bmatrix}.$

The n columns of Q_1 are an orthonormal basis for which fundamental subspace? The $m-n$ columns of Q_2 are an orthonormal basis for which fundamental subspace?

37 We know that $P = QQ^{\mathrm{T}}$ is the projection onto the column space of Q(m by n). Now add another column $\boldsymbol{a}$ to produce $A = [Q \quad \boldsymbol{a}]$. What is the new orthonormal vector $\boldsymbol{q}$ from Gram-Schmidt: start with $\boldsymbol{a}$, subtract _____, divide by _____.

6.5 Positive Definite Matrices

This section concentrates on ***symmetric matrices that have positive eigenvalues.*** If symmetry makes a matrix important, this extra property (*all* $\lambda > 0$) makes it truly special. When we say special, we don't mean rare. Symmetric matrices with positive eigenvalues are at the center of all kinds of applications. They are called ***positive definite.***

The first problem is to recognize these matrices. You may say, just find the eigenvalues and test $\lambda > 0$. That is exactly what we want to avoid. Calculating eigenvalues is work. When the λ's are needed, we can compute them. But if we just want to know that they are positive, there are faster ways. Here are two goals of this section:

- To find *quick tests* on a symmetric matrix that guarantee *positive eigenvalues.*
- To explain important applications of positive definiteness.

The λ's are automatically real because the matrix is symmetric.

Start with 2 by 2. When does $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ ***have*** $\lambda_1 > 0$ ***and*** $\lambda_2 > 0$?

The eigenvalues of A are positive if and only if $a > 0$ ***and*** $ac - b^2 > 0$.

$A_1 = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ is *not* positive definite because $ac - b^2 = 1 - 4 < 0$

$A_2 = \begin{bmatrix} 1 & -2 \\ -2 & 6 \end{bmatrix}$ is positive definite because $a = 1$ and $ac - b^2 = 6 - 4 > 0$

$A_3 = \begin{bmatrix} -1 & 2 \\ 2 & -6 \end{bmatrix}$ is *not* positive definite (even with $\det A = +2$) because $a = -1$

Notice that we didn't compute the eigenvalues 3 and -1 of A_1. Positive trace $3 - 1 = 2$, negative determinant $(3)(-1) = -3$. And $A_3 = -A_2$ is *negative* definite. The positive eigenvalues for A_2, two negative eigenvalues for A_3.

Proof that the 2 by 2 test is passed when $\lambda_1 > 0$ *and* $\lambda_2 > 0$. Their product $\lambda_1\lambda_2$ is the determinant so $ac - b^2 > 0$. Their sum is the trace so $a + c > 0$. Then a and c are both positive (if one of them is not positive, $ac - b^2 > 0$ will fail). Problem 1 reverses the reasoning to show that the tests guarantee $\lambda_1 > 0$ and $\lambda_2 > 0$.

This test uses the 1 by 1 determinant a and the 2 by 2 determinant $ac - b^2$. When A is 3 by 3, $\det A > 0$ is the third part of the test. The next test requires *positive pivots.*

The eigenvalues of $A = A^{\mathrm{T}}$ ***are positive if and only if the pivots are positive:***

$$a > 0 \quad \text{and} \quad \frac{ac - b^2}{a} > 0.$$

$a > 0$ is required in both tests. So $ac > b^2$ is also required, for the determinant test and now the pivot. The point is to recognize that ratio as the *second pivot* of A:

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix} \quad \begin{matrix} \text{The first pivot is } a \\ \longrightarrow \\ \text{The multiplier is } b/a \end{matrix} \quad \begin{bmatrix} a & b \\ 0 & c - \frac{b}{a}b \end{bmatrix} \quad \textbf{\textit{The second pivot is}} \; c - \frac{b^2}{a} = \frac{ac - b^2}{a}$$

This connects two big parts of linear algebra. ***Positive eigenvalues mean positive pivots and vice versa.*** We gave a proof for symmetric matrices of any size in the last section. The pivots give a quick test for $\lambda > 0$, and they are a lot faster to compute than the eigenvalues. It is very satisfying to see pivots and determinants and eigenvalues come together in this course.

$$A_1 = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \qquad A_2 = \begin{bmatrix} 1 & -2 \\ -2 & 6 \end{bmatrix} \qquad A_3 = \begin{bmatrix} -1 & 2 \\ 2 & -6 \end{bmatrix}$$

pivots 1 and -3	pivots 1 and 2	pivots -1 and -2
(indefinite)	**(positive definite)**	**(negative definite)**

Here is a different way to look at symmetric matrices with positive eigenvalues.

Energy-based Definition

From $Ax = \lambda x$, multiply by x^T to get $x^TAx = \lambda x^Tx$. The right side is a positive λ times a positive number $x^Tx = \|x\|^2$. So x^TAx is positive for any eigenvector.

The new idea is that ***x^TAx is positive for all nonzero vectors x***, not just the eigenvectors. In many applications this number x^TAx (or $\frac{1}{2}x^TAx$) is the **energy** in the system. The requirement of positive energy gives *another definition* of a positive definite matrix. I think this energy-based definition is the fundamental one.

Eigenvalues and pivots are two equivalent ways to test the new requirement $x^TAx > 0$.

Definition ***A is positive definite if $x^TAx > 0$ for every nonzero vector x:***

$$x^TAx = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = ax^2 + 2bxy + cy^2 > 0. \tag{1}$$

The four entries a, b, b, c give the four parts of x^TAx. From a and c come the pure squares ax^2 and cy^2. From b and b off the diagonal come the cross terms bxy and byx (the same). Adding those four parts gives x^TAx. This energy-based definition leads to a basic fact:

If A and B are symmetric positive definite, so is $A + B$.

Reason: $x^T(A + B)x$ is simply $x^TAx + x^TBx$. Those two terms are positive (for $x \neq 0$) so $A + B$ is also positive definite. The pivots and eigenvalues are not easy to follow when matrices are added, but the energies just add.

$x^T Ax$ also connects with our final way to recognize a positive definite matrix. Start with any matrix R, possibly rectangular. We know that $A = R^T R$ is square and symmetric. More than that, A will be positive definite when R has independent columns:

If the columns of R are independent, then $A = R^T R$ is positive definite.

Again eigenvalues and pivots are not easy. But the number $x^T Ax$ is the same as $x^T R^T Rx$. That is exactly $(Rx)^T(Rx)$—another important proof by parenthesis! That vector Rx is not zero when $x \neq 0$ (this is the meaning of independent columns). Then $x^T Ax$ is the positive number $\|Rx\|^2$ and the matrix A is positive definite.

Let me collect this theory together, into five equivalent statements of positive definiteness. You will see how that key idea connects the whole subject of linear algebra: pivots, determinants, eigenvalues, and least squares (from $R^T R$). Then come the applications.

When a symmetric matrix has one of these five properties, it has them all:

1. All ***n pivots*** are positive.
2. All ***n upper left determinants*** are positive.
3. All ***n eigenvalues*** are positive.
4. $x^T Ax$ is positive except at $x = 0$. This is the ***energy-based*** definition.
5. A equals $R^T R$ for a matrix R with ***independent columns.***

The "upper left determinants" are 1 by 1, 2 by 2, . . ., n by n. The last one is the determinant of the complete matrix A. This remarkable theorem ties together the whole linear algebra course—at least for symmetric matrices. We believe that two examples are more helpful than a detailed proof (we nearly have a proof already).

Example 1 Test these matrices A and B for positive definiteness:

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & -1 & b \\ -1 & 2 & -1 \\ b & -1 & 2 \end{bmatrix}.$$

Solution The pivots of A are 2 and $\frac{3}{2}$ and $\frac{4}{3}$, all positive. Its upper left determinants are 2 and 3 and 4, all positive. The eigenvalues of A are $2 - \sqrt{2}$ and 2 and $2 + \sqrt{2}$, all positive. That completes tests **1**, **2**, and **3**.

We can write $x^T Ax$ as a sum of three squares. The pivots 2, $\frac{3}{2}$, $\frac{4}{3}$ appear outside the squares. The multipliers $-\frac{1}{2}$ and $-\frac{2}{3}$ from elimination are inside the squares:

$$x^T Ax = 2(x_1^2 - x_1x_2 + x_2^2 - x_2x_3 + x_3^2) \qquad \textbf{Rewrite with squares}$$

$$= 2(x_1 - \tfrac{1}{2}x_2)^2 + \tfrac{3}{2}(x_2 - \tfrac{2}{3}x_3)^2 + \tfrac{4}{3}(x_3)^2. \qquad \textbf{This sum is positive.}$$

I have two candidates to suggest for R. Either one will show that $A = R^{\mathrm{T}}R$ is positive definite. R can be a rectangular first difference matrix, 4 by 3, to produce those second differences $-1, 2, -1$ in A:

$$A = R^{\mathrm{T}}R \quad \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

The three columns of this R are independent. A is positive definite.

Another R comes from $A = LDL^{\mathrm{T}}$ (the symmetric version of $A = LU$). Elimination gives the pivots $2, \frac{3}{2}, \frac{4}{3}$ in D and the multipliers $-\frac{1}{2}, 0, -\frac{2}{3}$ in L. **Just put $\sqrt{D}$ with L.**

$$LDL^{\mathrm{T}} = \begin{bmatrix} 1 & & \\ -\frac{1}{2} & 1 & \\ 0 & -\frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 2 & & \\ & \frac{3}{2} & \\ & & \frac{4}{3} \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{2} & \\ & 1 & -\frac{2}{3} \\ & & 1 \end{bmatrix} = (L\sqrt{D})(L\sqrt{D})^{\mathrm{T}} = R^{\mathrm{T}}R. \quad (2)$$

R is the Cholesky factor

This choice of R has square roots (not so beautiful). But it is the only R that is 3 by 3 and upper triangular. It is the "Cholesky factor" of A and it is computed by MATLAB's command $R = \mathrm{chol}(A)$. In applications, the rectangular R is how we build A and this Cholesky R is how we break it apart.

Eigenvalues give the symmetric choice $R = Q\sqrt{\Lambda}Q^{\mathrm{T}}$. This is also successful with $R^{\mathrm{T}}R = Q\Lambda Q^{\mathrm{T}} = A$. All these tests show that the $-1, 2, -1$ matrix A is positive definite.

Now turn to B, where the $(1,3)$ and $(3,1)$ entries move away from 0 to b. This b must not be too large! *The determinant test is easiest.* The 1 by 1 determinant is 2, the 2 by 2 determinant is still 3. The 3 by 3 determinant involves b:

$$\det B = 4 + 2b - 2b^2 = (1 + b)(4 - 2b) \quad \text{must be positive.}$$

At $b = -1$ and $b = 2$ we get $\det B = 0$. *Between $b = -1$ and $b = 2$ the matrix is positive definite.* The corner entry $b = 0$ in the first matrix A was safely between.

Positive Semidefinite Matrices

Often we are at the edge of positive definiteness. The determinant is zero. The smallest eigenvalue is zero. The energy in its eigenvector is $x^{\mathrm{T}}Ax = x^{\mathrm{T}}0x = 0$. These matrices on the edge are called *positive semidefinite*. Here are two examples (not invertible):

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \text{ are positive semidefinite.}$$

A has eigenvalues 5 and 0. Its upper left determinants are 1 and 0. Its rank is only 1. This matrix A factors into $R^{\mathrm{T}}R$ with **dependent columns** in R:

Dependent columns
Positive semidefinite

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} = R^{\mathrm{T}}R.$$

If 4 is increased by any small number, the matrix will become positive definite.

The cyclic B also has zero determinant (computed above when $b = -1$). It is singular. The eigenvector $\boldsymbol{x} = (1, 1, 1)$ has $B\boldsymbol{x} = \mathbf{0}$ and $\boldsymbol{x}^{\mathrm{T}}B\boldsymbol{x} = 0$. Vectors $\boldsymbol{x}$ in all other directions do give positive energy. This B can be written as $R^{\mathrm{T}}R$ in many ways, but R will always have *dependent* columns, with $(1, 1, 1)$ in its nullspace:

Second differences A from first differences $R^{\mathrm{T}}R$ Cyclic A from cyclic R

$$\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}.$$

Positive semidefinite matrices have all $\lambda \geq 0$ and all $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} \geq 0$. Those weak inequalities ($\geq$ instead of $>$) include positive definite matrices and the singular matrices at the edge.

First Application: The Ellipse $ax^2 + 2bxy + cy^2 = 1$

Think of a tilted ellipse $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} = 1$. Its center is $(0, 0)$, as in Figure 6.7a. Turn it to line up with the coordinate axes (X and Y axes). That is Figure 6.7b. These two pictures show the geometry behind the factorization $A = Q\Lambda Q^{-1} = Q\Lambda Q^{\mathrm{T}}$:

1. The tilted ellipse is associated with A. Its equation is $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} = 1$.
2. The lined-up ellipse is associated with Λ. Its equation is $\boldsymbol{X}^{\mathrm{T}}\Lambda\boldsymbol{X} = 1$.
3. The rotation matrix that lines up the ellipse is the eigenvector matrix Q.

Example 2 Find the axes of this tilted ellipse $5x^2 + 8xy + 5y^2 = 1$.

Solution Start with the positive definite matrix that matches this equation:

The equation is $\begin{bmatrix} x & y \end{bmatrix}\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = 1.$ The matrix is $A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}.$

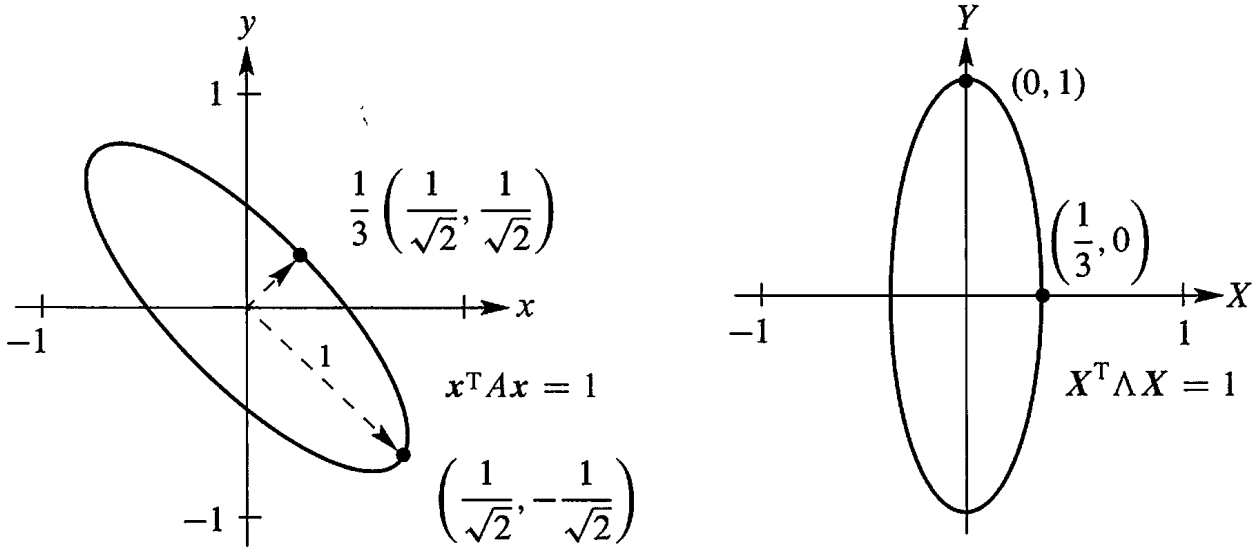


Figure 6.7: The tilted ellipse $5x^2 + 8xy + 5y^2 = 1$. Lined up it is $9X^2 + Y^2 = 1$.

The eigenvectors are $\begin{bmatrix}1\\1\end{bmatrix}$ and $\begin{bmatrix}-1\\1\end{bmatrix}$. Divide by $\sqrt{2}$ for unit vectors. Then $A = Q\Lambda Q^{\mathrm{T}}$:

$$\begin{matrix}\textbf{Eigenvectors in } Q\\ \textbf{Eigenvalues 9 and 1}\end{matrix} \quad \begin{bmatrix}5 & 4\\ 4 & 5\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}\mathbf{9} & \mathbf{0}\\ \mathbf{0} & \mathbf{1}\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}.$$

Now multiply by $\begin{bmatrix}x & y\end{bmatrix}$ on the left and $\begin{bmatrix}x\\ y\end{bmatrix}$ on the right to get back to $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x}$:

$$\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} = \textbf{sum of squares} \quad 5x^2 + 8xy + 5y^2 = 9\left(\frac{x+y}{\sqrt{2}}\right)^2 + 1\left(\frac{x-y}{\sqrt{2}}\right)^2. \tag{3}$$

The coefficients are not the pivots 5 and 9/5 from D, they are the eigenvalues 9 and 1 from Λ. Inside *these* squares are the eigenvectors $(1,1)/\sqrt{2}$ and $(1,-1)/\sqrt{2}$.

The axes of the tilted ellipse point along the eigenvectors. This explains why $A = Q\Lambda Q^{\mathrm{T}}$ is called the "principal axis theorem"—it displays the axes. Not only the axis directions (from the eigenvectors) but also the axis lengths (from the eigenvalues). To see it all, use capital letters for the new coordinates that line up the ellipse:

$$\textbf{Lined up} \quad \frac{x+y}{\sqrt{2}} = X \quad \text{and} \quad \frac{x-y}{\sqrt{2}} = Y \quad \text{and} \quad 9X^2 + Y^2 = 1.$$

The largest value of X^2 is $1/9$. The endpoint of the shorter axis has $X = 1/3$ and $Y = 0$. Notice: The *bigger* eigenvalue λ_1 gives the *shorter* axis, of half-length $1/\sqrt{\lambda_1} = 1/3$. The smaller eigenvalue $\lambda_2 = 1$ gives the greater length $1/\sqrt{\lambda_2} = 1$.

In the xy system, the axes are along the eigenvectors of A. In the XY system, the axes are along the eigenvectors of Λ—the coordinate axes. All comes from $A = Q\Lambda Q^{\mathrm{T}}$.

Suppose $A = Q\Lambda Q^{\mathrm{T}}$ is positive definite, so $\lambda_i > 0$. The graph of $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} = 1$ is an ellipse:

$$\begin{bmatrix}x & y\end{bmatrix}Q\Lambda Q^{\mathrm{T}}\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}X & Y\end{bmatrix}\Lambda\begin{bmatrix}X\\ Y\end{bmatrix} = \lambda_1 X^2 + \lambda_2 Y^2 = 1.$$

The axes point along eigenvectors. The half-lengths are $1/\sqrt{\lambda_1}$ and $1/\sqrt{\lambda_2}$.

$A = I$ gives the circle $x^2 + y^2 = 1$. If one eigenvalue is negative (exchange 4's and 5's in A), we don't have an ellipse. The sum of squares becomes a *difference of squares*: $9X^2 - Y^2 = 1$. This indefinite matrix gives a *hyperbola*. For a negative definite matrix like $A = -I$, with both λ's negative, the graph of $-x^2 - y^2 = 1$ has no points at all.

■ REVIEW OF THE KEY IDEAS ■

1. Positive definite matrices have positive eigenvalues and positive pivots.
2. A quick test is given by the upper left determinants: $a > 0$ and $ac - b^2 > 0$.

3. The graph of $x^T Ax$ is then a "bowl" going up from $x = 0$:

$$x^T Ax = ax^2 + 2bxy + cy^2 \text{ is positive except at } (x, y) = (0, 0).$$

4. $A = R^T R$ is automatically positive definite if R has independent columns.

5. The ellipse $x^T Ax = 1$ has its axes along the eigenvectors of A. Lengths $1/\sqrt{\lambda}$.

■ WORKED EXAMPLES ■

6.5 A The great factorizations of a symmetric matrix are $A = LDL^T$ from pivots and multipliers, and $A = Q\Lambda Q^T$ from eigenvalues and eigenvectors. Show that $x^T Ax > 0$ for all nonzero x exactly when the pivots and eigenvalues are positive. Try these n by n tests on pascal(6) and ones(6) and hilb(6) and other matrices in MATLAB's gallery.

Solution To prove $x^T Ax > 0$, put parentheses into $x^T LDL^T x$ and $x^T Q\Lambda Q^T x$:

$$x^T Ax = (L^T x)^T D(L^T x) \quad \text{and} \quad x^T Ax = (Q^T x)^T \Lambda (Q^T x).$$

If x is nonzero, then $y = L^T x$ and $z = Q^T x$ are nonzero (those matrices are invertible). So $x^T Ax = y^T Dy = z^T \Lambda z$ becomes a sum of squares and A is shown as positive definite:

Pivots	$x^T Ax$	$=$	$y^T Dy$	$=$	$d_1 y_1^2 + \cdots + d_n y_n^2$	$>$	0
Eigenvalues	$x^T Ax$	$=$	$z^T \Lambda z$	$=$	$\lambda_1 z_1^2 + \cdots + \lambda_n z_n^2$	$>$	0

MATLAB has a gallery of unusual matrices (type help gallery) and here are four:

pascal(6) is positive definite because all its pivots are 1 (Worked Example **2.6 A**).

ones(6) is positive *semidefinite* because its eigenvalues are $0, 0, 0, 0, 0, 6$.

H=hilb(6) is positive definite even though eig(H) shows two eigenvalues very near zero.

Hilbert matrix $x^T Hx = \int_0^1 (x_1 + x_2 s + \cdots + x_6 s^5)^2 \, ds > 0$, $H_{ij} = 1/(i + j + 1)$.

rand(6)+rand(6)' can be positive definite or not. Experiments gave only 2 in 20000.

$n = 20000$; $p = 0$; for $k = 1{:}n$, $A =$ rand(6); $p = p +$ all(eig($A + A'$) > 0); end, p / n

6.5 B When is the symmetric block matrix $M = \begin{bmatrix} A & B \\ B^T & C \end{bmatrix}$ **positive definite?**

Solution Multiply the first row of M by $B^T A^{-1}$ and subtract from the second row, to get a block of zeros. The *Schur complement* $S = C - B^T A^{-1} B$ appears in the corner:

$$\begin{bmatrix} I & 0 \\ -B^T A^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ B^T & C \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & C - B^T A^{-1} B \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & S \end{bmatrix} \tag{4}$$

Those two blocks A and S must be positive definite. Their pivots are the pivots of M.

6.5 C *Second application: Test for a minimum.* Does $F(x, y)$ have a minimum if $\partial F/\partial x = 0$ and $\partial F/\partial y = 0$ at the point $(x, y) = (0, 0)$?

Solution For $f(x)$, the test for a minimum comes from calculus: $df/dx = 0$ and $d^2f/dx^2 > 0$. Moving to two variables x and y produces a symmetric matrix H. It contains the four second derivatives of $F(x, y)$. Positive f'' changes to positive definite H:

Second derivative matrix
$$H = \begin{bmatrix} \partial^2 F/\partial x^2 & \partial^2 F/\partial x \partial y \\ \partial^2 F/\partial y \partial x & \partial^2 F/\partial y^2 \end{bmatrix}$$

$F(x, y)$ has a minimum if H is positive definite. Reason: H reveals the important terms $ax^2 + 2bxy + cy^2$ near $(x, y) = (0, 0)$. The second derivatives of F are $2a, 2b, 2b, 2c$!

6.5 D Find the eigenvalues of the $-1, 2, -1$ tridiagonal n by n matrix K (my favorite).

Solution The best way is to guess λ and $\boldsymbol{x}$. Then check $K\boldsymbol{x} = \lambda \boldsymbol{x}$. Guessing could not work for most matrices, but special cases are a big part of mathematics (pure and applied).

The key is hidden in a differential equation. The second difference matrix K is like a *second derivative*, and those eigenvalues are much easier to see:

Eigenvalues $\lambda_1, \lambda_2, \ldots$
Eigenfunctions $y_1, y_2, \ldots$
$$\frac{d^2y}{dx^2} = \lambda y(x) \quad \textbf{with} \quad \begin{matrix} y(0) = 0 \\ y(1) = 0 \end{matrix} \tag{5}$$

Try $y = \sin cx$. Its second derivative is $y'' = -c^2 \sin cx$. So the eigenvalue will be $\lambda = -c^2$, provided $y(x)$ satisfies the end point conditions $y(0) = 0 = y(1)$.

Certainly $\sin 0 = 0$ (this is where cosines are eliminated by $\cos 0 = 1$). At $x = 1$, we need $y(1) = \sin c = 0$. The number c must be $k\pi$, a multiple of π, and λ is $-c^2$:

Eigenvalues $\lambda = -k^2\pi^2$
Eigenfunctions $y = \sin k\pi x$
$$\frac{d^2}{dx^2} \sin k\pi x = -k^2\pi^2 \sin k\pi x. \tag{6}$$

Now we go back to the matrix K and guess its eigenvectors. They come from $\sin k\pi x$ at n points $x = h, 2h, \ldots, nh$, equally spaced between 0 and 1. The spacing Δx is $h = 1/(n + 1)$, so the $(n + 1)$st point comes out at $(n + 1)h = 1$. Multiply that sine vector $\boldsymbol{s}$ by K:

Eigenvector of K = sine vector $\boldsymbol{s}$
$$\begin{aligned} K\boldsymbol{s} &= \lambda \boldsymbol{s} = (2 - 2\cos k\pi h)\, \boldsymbol{s} \\ \boldsymbol{s} &= (\sin k\pi h, \ldots, \sin nk\pi h). \end{aligned} \tag{7}$$

I will leave that multiplication $K\boldsymbol{s} = \lambda \boldsymbol{s}$ as a challenge problem. Notice what is important:

1. All eigenvalues $2 - 2\cos k\pi h$ are positive and K is positive definite.

2. The **sine matrix** S has orthogonal columns = eigenvectors $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n$ of K.

Discrete Sine Transform
The j,k entry is $\sin jk\pi h$

$$S = \begin{bmatrix} \sin \pi h & & \sin k\pi h & \\ \vdots & \cdots & \vdots & \cdots \\ \sin n\pi h & & \sin nk\pi h & \end{bmatrix}$$

Those eigenvectors are orthogonal just like the eigenfunctions: $\int_0^1 \sin j\pi x \sin k\pi x \, dx = 0$.

Problem Set 6.5

Problems 1–13 are about tests for positive definiteness.

1 Suppose the 2 by 2 tests $a > 0$ and $ac - b^2 > 0$ are passed. Then $c > b^2/a$ is also positive.

(i) λ_1 and λ_2 have the *same sign* because their product $\lambda_1\lambda_2$ equals _____.

(i) That sign is positive because $\lambda_1 + \lambda_2$ equals _____.

Conclusion: The tests $a > 0, ac - b^2 > 0$ guarantee positive eigenvalues λ_1, λ_2.

2 Which of A_1, A_2, A_3, A_4 has two positive eigenvalues? Use the test, don't compute the λ's. Find an x so that $x^{\mathrm{T}}A_1x < 0$, so A_1 fails the test.

$$A_1 = \begin{bmatrix} 5 & 6 \\ 6 & 7 \end{bmatrix} \quad A_2 = \begin{bmatrix} -1 & -2 \\ -2 & -5 \end{bmatrix} \quad A_3 = \begin{bmatrix} 1 & 10 \\ 10 & 100 \end{bmatrix} \quad A_4 = \begin{bmatrix} 1 & 10 \\ 10 & 101 \end{bmatrix}.$$

3 For which numbers b and c are these matrices positive definite?

$$A = \begin{bmatrix} 1 & b \\ b & 9 \end{bmatrix} \qquad A = \begin{bmatrix} 2 & 4 \\ 4 & c \end{bmatrix} \qquad A = \begin{bmatrix} c & b \\ b & c \end{bmatrix}.$$

With the pivots in D and multiplier in L, factor each A into LDL^{T}.

4 What is the quadratic $f = ax^2 + 2bxy + cy^2$ for each of these matrices? Complete the square to write f as a sum of one or two squares $d_1(\ \ )^2 + d_2(\ \ )^2$.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 9 \end{bmatrix} \qquad \text{and} \qquad A = \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}.$$

5 Write $f(x, y) = x^2 + 4xy + 3y^2$ as a *difference* of squares and find a point (x, y) where f is negative. The minimum is not at $(0, 0)$ even though f has positive coefficients.

6 The function $f(x, y) = 2xy$ certainly has a saddle point and not a minimum at $(0, 0)$. What symmetric matrix A produces this f? What are its eigenvalues?

7 Test to see if R^TR is positive definite in each case:

$$R = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad R = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix} \quad \text{and} \quad R = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}.$$

8 The function $f(x, y) = 3(x + 2y)^2 + 4y^2$ is positive except at $(0, 0)$. What is the matrix in $f = [x \quad y]A[x \quad y]^T$? Check that the pivots of A are 3 and 4.

9 Find the 3 by 3 matrix A and its pivots, rank, eigenvalues, and determinant:

$$[x_1 \quad x_2 \quad x_3] \begin{bmatrix} & A & \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 4(x_1 - x_2 + 2x_3)^2.$$

10 Which 3 by 3 symmetric matrices A and B produce these quadratics?

$x^TAx = 2(x_1^2 + x_2^2 + x_3^2 - x_1x_2 - x_2x_3)$. Why is A positive definite?

$x^TBx = 2(x_1^2 + x_2^2 + x_3^2 - x_1x_2 - x_1x_3 - x_2x_3)$. Why is B semidefinite?

11 Compute the three upper left determinants of A to establish positive definiteness. Verify that their ratios give the second and third pivots.

Pivots = ratios of determinants $\qquad A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 3 & 8 \end{bmatrix}.$

12 For what numbers c and d are A and B positive definite? Test the 3 determinants:

$$A = \begin{bmatrix} c & 1 & 1 \\ 1 & c & 1 \\ 1 & 1 & c \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & d & 4 \\ 3 & 4 & 5 \end{bmatrix}.$$

13 Find a matrix with $a > 0$ and $c > 0$ and $a + c > 2b$ that has a negative eigenvalue.

Problems 14–20 are about applications of the tests.

14 *If A is positive definite then A^{-1} is positive definite.* Best proof: The eigenvalues of A^{-1} are positive because _____. *Second proof* (only for 2 by 2):

The entries of $A^{-1} = \dfrac{1}{ac - b^2}\begin{bmatrix} c & -b \\ -b & a \end{bmatrix}$ pass the determinant tests _____.

15 *If A and B are positive definite, their sum $A + B$ is positive definite.* Pivots and eigenvalues are not convenient for $A + B$. Better to prove $x^T(A + B)x > 0$. Or if $A = R^TR$ and $B = S^TS$, show that $A + B = [R \; S]^T \begin{bmatrix} R \\ S \end{bmatrix}$ with independent columns.

16 A positive definite matrix cannot have a zero (or even worse, a negative number) on its diagonal. Show that this matrix fails to have $x^{\mathrm{T}}Ax > 0$:

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 4 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{ is not positive when } (x_1, x_2, x_3) = (\ \ ,\ \ ,\ \ ).$$

17 A diagonal entry a_{jj} of a symmetric matrix cannot be smaller than all the λ's. If it were, then $A - a_{jj}I$ would have _____ eigenvalues and would be positive definite. But $A - a_{jj}I$ has a _____ on the main diagonal.

18 If $Ax = \lambda x$ then $x^{\mathrm{T}}Ax =$ _____. If $x^{\mathrm{T}}Ax > 0$, prove that $\lambda > 0$.

19 Reverse Problem 18 to show that *if all* $\lambda > 0$ *then* $x^{\mathrm{T}}Ax > 0$. We must do this for *every* nonzero x, not just the eigenvectors. So write x as a combination of the eigenvectors and *explain why all "cross terms" are* $x_i^{\mathrm{T}}x_j = 0$. Then $x^{\mathrm{T}}Ax$ is

$$(c_1x_1 + \cdots + c_nx_n)^{\mathrm{T}}(c_1\lambda_1x_1 + \cdots + c_n\lambda_nx_n) = c_1^2\lambda_1x_1^{\mathrm{T}}x_1 + \cdots + c_n^2\lambda_nx_n^{\mathrm{T}}x_n > 0.$$

20 Give a quick reason why each of these statements is true:

(a) Every positive definite matrix is invertible.

(b) The only positive definite projection matrix is $P = I$.

(c) A diagonal matrix with positive diagonal entries is positive definite.

(d) A symmetric matrix with a positive determinant might not be positive definite!

Problems 21–24 use the eigenvalues; Problems 25–27 are based on pivots.

21 For which s and t do A and B have all $\lambda > 0$ (therefore positive definite)?

$$A = \begin{bmatrix} s & -4 & -4 \\ -4 & s & -4 \\ -4 & -4 & s \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} t & 3 & 0 \\ 3 & t & 4 \\ 0 & 4 & t \end{bmatrix}.$$

22 From $A = Q\Lambda Q^{\mathrm{T}}$ compute the positive definite symmetric square root $Q\Lambda^{1/2}Q^{\mathrm{T}}$ of each matrix. Check that this square root gives $R^2 = A$:

$$A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}.$$

23 You may have seen the equation for an ellipse as $x^2/a^2 + y^2/b^2 = 1$. What are a and b when the equation is written $\lambda_1x^2 + \lambda_2y^2 = 1$? The ellipse $9x^2 + 4y^2 = 1$ has axes with half-lengths $a =$ _____ and $b =$ _____.

24 Draw the tilted ellipse $x^2 + xy + y^2 = 1$ and find the half-lengths of its axes from the eigenvalues of the corresponding matrix A.

25 With positive pivots in D, the factorization $A = LDL^T$ becomes $L\sqrt{D}\sqrt{D}L^T$. (Square roots of the pivots give $D = \sqrt{D}\sqrt{D}$.) Then $C = \sqrt{D}L^T$ yields the ***Cholesky factorization*** $A = C^TC$ which is "symmetrized LU":

$$\text{From } C = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \text{ find } A. \qquad \text{From } A = \begin{bmatrix} 4 & 8 \\ 8 & 25 \end{bmatrix} \text{ find } C = \textbf{chol}(A).$$

26 In the Cholesky factorization $A = C^TC$, with $C^T = L\sqrt{D}$, the square roots of the pivots are on the diagonal of C. Find C (upper triangular) for

$$A = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 8 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 7 \end{bmatrix}.$$

27 The symmetric factorization $A = LDL^T$ means that $x^TAx = x^TLDL^Tx$:

$$\begin{bmatrix} x & y \end{bmatrix}\begin{bmatrix} a & b \\ b & c \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix}\begin{bmatrix} 1 & 0 \\ b/a & 1 \end{bmatrix}\begin{bmatrix} a & 0 \\ 0 & (ac-b^2)/a \end{bmatrix}\begin{bmatrix} 1 & b/a \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}.$$

The left side is $ax^2 + 2bxy + cy^2$. The right side is $a\left(x + \frac{b}{a}y\right)^2 + _____y^2$. The second pivot completes the square! Test with $a = 2, b = 4, c = 10$.

28 Without multiplying $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, find

(a) the determinant of A (b) the eigenvalues of A
(c) the eigenvectors of A (d) a reason why A is symmetric positive definite.

29 For $F_1(x,y) = \frac{1}{4}x^4 + x^2y + y^2$ and $F_2(x,y) = x^3 + xy - x$ find the second derivative matrices H_1 and H_2:

$$\textbf{Test for minimum: } H = \begin{bmatrix} \partial^2F/\partial x^2 & \partial^2F/\partial x\partial y \\ \partial^2F/\partial y\partial x & \partial^2F/\partial y^2 \end{bmatrix} \text{ is positive definite}$$

H_1 is positive definite so F_1 is concave up (= convex). Find the minimum point of F_1 and the saddle point of F_2 (look only where first derivatives are zero).

30 The graph of $z = x^2 + y^2$ is a bowl opening upward. *The graph of $z = x^2 - y^2$ is a saddle.* The graph of $z = -x^2 - y^2$ is a bowl opening downward. What is a test on a, b, c for $z = ax^2 + 2bxy + cy^2$ to have a saddle point at $(0,0)$?

31 Which values of c give a bowl and which c give a saddle point for the graph of $z = 4x^2 + 12xy + cy^2$? Describe this graph at the borderline value of c.

Challenge Problems

32 A *group* of nonsingular matrices includes AB and A^{-1} if it includes A and B. "Products and inverses stay in the group." Which of these are groups (as in 2.7.37)?

Invent a "subgroup" of two of these groups (not I by itself $=$ the smallest group).

(a) Positive definite symmetric matrices A.

(b) Orthogonal matrices Q.

(c) All exponentials e^{tA} of a fixed matrix A.

(d) Matrices P with positive eigenvalues.

(e) Matrices D with determinant 1.

33 When A and B are symmetric positive definite, AB might not even be symmetric. But its eigenvalues are still positive. Start from $ABx = \lambda x$ and take dot products with Bx. Then prove $\lambda > 0$.

34 Write down the 5 by 5 sine matrix S from Worked Example **6.5 D**, containing the eigenvectors of K when $n = 5$ and $h = 1/6$. Multiply K times S to see the five positive eigenvalues.

Their sum should equal the trace 10. Their product should be $\det K = 6$.

35 Suppose C is positive definite (so $y^{\mathrm{T}}Cy > 0$ whenever $y \neq \mathbf{0}$) and A has independent columns (so $Ax \neq \mathbf{0}$ whenever $x \neq \mathbf{0}$). Apply the energy test to $x^{\mathrm{T}}A^{\mathrm{T}}CAx$ to show that $A^{\mathrm{T}}CA$ is positive definite: *the crucial matrix in engineering.*

15 Why doesn't the SVD for $A + I$ just use $\Sigma + I$?

Challenge Problems

16 (Search engine) Run a **random walk** $x(2), \ldots, x(n)$ starting from web site $x(1) = 1$. Count the visits to each site. At each step the code chooses the next website $x(k)$ with probabilities given by column $x(k-1)$ of A. At the end, $\boldsymbol{p}$ gives the fraction of time at each site from a histogram: count visits. ***The rankings are based on p.***

Please compare $\boldsymbol{p}$ to the steady state eigenvector of the Markov matrix A:

$$A = [0 \ .1 \ .2 \ .7; \ .05 \ 0 \ .15 \ .8; \ .15 \ .25 \ 0 \ .6; \ .1 \ .3 \ .6 \ 0]'$$

$n = 100$; $x = \text{zeros}(1, n)$; $x(1) = 1$;
for $k = 2 : n$ $x(k) = \text{min}(\text{find}(\text{rand}<\text{cumsum}(A(:, x(k-1)))))$; end
$\boldsymbol{p} = \text{hist}(x, 1 : 4)/n$

17 The $1, -1$ first difference matrix A has $A^{\mathrm{T}}A$ = second difference matrix. The singular vectors of A are *sine vectors* $\boldsymbol{v}$ and *cosine vectors* $\boldsymbol{u}$. Then $A\boldsymbol{v} = \sigma\boldsymbol{u}$ is the discrete form of $d/dx(\sin cx) = c(\cos cx)$. This is the best SVD I have seen.

SVD of A
$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \qquad A^{\mathrm{T}}A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Orthogonal sine matrix
$$V = \frac{1}{\sqrt{2}} \begin{bmatrix} \sin \pi/4 & \sin 2\pi/4 & \sin 3\pi/4 \\ \sin 2\pi/4 & \sin 4\pi/4 & \sin 6\pi/4 \\ \sin 3\pi/4 & \sin 6\pi/4 & \sin 9\pi/4 \end{bmatrix}$$

(a) Put numbers in V: The unit eigenvectors of $A^{\mathrm{T}}A$ are singular vectors of A. Show that the columns of V have $A^{\mathrm{T}}A\boldsymbol{v} = \lambda\boldsymbol{v}$ with $\lambda = 2 - \sqrt{2}, \ 2, \ 2 + \sqrt{2}$.

(b) Multiply AV and verify that its columns are orthogonal. They are $\sigma_1\boldsymbol{u}_1$ and $\sigma_2\boldsymbol{u}_2$ and $\sigma_3\boldsymbol{u}_3$. The first columns of the cosine matrix U are $\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3$.

(c) Since A is 4 by 3, we need a fourth orthogonal vector $\boldsymbol{u}_4$. It comes from the nullspace of A^{T}. What is $\boldsymbol{u}_4$?

The cosine vectors in U are eigenvectors of AA^{T}. The fourth cosine is $(1, 1, 1, 1)/2$.

$$AA^{\mathrm{T}} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \qquad U = \frac{1}{\sqrt{2}} \begin{bmatrix} \cos \pi/8 & \cos 2\pi/8 & \cos 3\pi/8 \\ \cos 3\pi/8 & \cos 6\pi/8 & \cos 9\pi/8 \\ \cos 5\pi/8 & \cos 10\pi/8 & \cos 15\pi/8 \\ \cos 7\pi/8 & \cos 14\pi/8 & \cos 21\pi/8 \end{bmatrix}$$

Those angles $\pi/8, 3\pi/8, 5\pi/8, 7\pi/8$ fit 4 points with spacing $\pi/4$ between 0 and π. The sine transform has three points $\pi/4, 2\pi/4, 3\pi/4$. The full cosine transform includes $\boldsymbol{u}_4$ from the "zero frequency" or *direct current* eigenvector $(1, 1, 1, 1)$.

The 8 by 8 cosine transform in 2D is the workhorse of **jpeg compression**. Linear algebra (circulant, Toeplitz, orthogonal matrices) is at the heart of signal processing.

Table of Eigenvalues and Eigenvectors

How are the properties of a matrix reflected in its eigenvalues and eigenvectors? This question is fundamental throughout Chapter 6. A table that organizes the key facts may be helpful. Here are the special properties of the eigenvalues λ_i and the eigenvectors $\boldsymbol{x}_i$.

Symmetric: $A^{\mathrm{T}} = A$	real λ's	orthogonal $\boldsymbol{x}_i^{\mathrm{T}}\boldsymbol{x}_j = 0$
Orthogonal: $Q^{\mathrm{T}} = Q^{-1}$	all $\lvert\lambda\rvert = 1$	orthogonal $\overline{\boldsymbol{x}}_i^{\mathrm{T}}\boldsymbol{x}_j = 0$
Skew-symmetric: $A^{\mathrm{T}} = -A$	imaginary λ's	orthogonal $\overline{\boldsymbol{x}}_i^{\mathrm{T}}\boldsymbol{x}_j = 0$
Complex Hermitian: $\overline{A}^{\mathrm{T}} = A$	real λ's	orthogonal $\overline{\boldsymbol{x}}_i^{\mathrm{T}}\boldsymbol{x}_j = 0$
Positive Definite: $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} > 0$	all $\lambda > 0$	orthogonal since $A^{\mathrm{T}} = A$
Markov: $m_{ij} > 0, \sum_{i=1}^{n} m_{ij} = 1$	$\lambda_{\max} = 1$	steady state $\boldsymbol{x} > 0$
Similar: $B = M^{-1}AM$	$\lambda(B) = \lambda(A)$	$\boldsymbol{x}(B) = M^{-1}\boldsymbol{x}(A)$
Projection: $P = P^2 = P^{\mathrm{T}}$	$\lambda = 1;\ 0$	column space; nullspace
Plane Rotation	$e^{i\theta}$ and $e^{-i\theta}$	$\boldsymbol{x} = (1, i)$ and $(1, -i)$
Reflection: $I - 2\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}$	$\lambda = -1;\ 1, .., 1$	$\boldsymbol{u}$; whole plane $\boldsymbol{u}^{\perp}$
Rank One: $\boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}$	$\lambda = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{u};\ 0, .., 0$	$\boldsymbol{u}$; whole plane $\boldsymbol{v}^{\perp}$
Inverse: A^{-1}	$1/\lambda(A)$	keep eigenvectors of A
Shift: $A + cI$	$\lambda(A) + c$	keep eigenvectors of A
Stable Powers: $A^n \to 0$	all $\lvert\lambda\rvert < 1$	any eigenvectors
Stable Exponential: $e^{At} \to 0$	all Re $\lambda < 0$	any eigenvectors
Cyclic Permutation: row 1 of I last	$\lambda_k = e^{2\pi ik/n}$	$\boldsymbol{x}_k = (1, \lambda_k, \ldots, \lambda_k^{n-1})$
Tridiagonal: $-1, 2, -1$ on diagonals	$\lambda_k = 2 - 2\cos\frac{k\pi}{n+1}$	$\boldsymbol{x}_k = \left(\sin\frac{k\pi}{n+1}, \sin\frac{2k\pi}{n+1}, \ldots\right)$
Diagonalizable: $A = S\Lambda S^{-1}$	diagonal of Λ	columns of S are independent
Symmetric: $A = Q\Lambda Q^{\mathrm{T}}$	diagonal of Λ (real)	columns of Q are orthonormal
Schur: $A = QTQ^{-1}$	diagonal of T	columns of Q *if* $A^{\mathrm{T}}A = AA^{\mathrm{T}}$
Jordan: $J = M^{-1}AM$	diagonal of J	each block gives $\boldsymbol{x} = (0, .., 1, .., 0)$
Rectangular: $A = U\Sigma V^{\mathrm{T}}$	rank(A) = rank(Σ)	eigenvectors of $A^{\mathrm{T}}A$, AA^{T} in V, U

8.6 Linear Algebra for Statistics and Probability

Statistics deals with data, often in large quantities. Since data tends to go into rectangular matrices, we expect to see A^TA. The least squares problem $A\widehat{x} \approx \boldsymbol{b}$ is ***linear regression.*** The best solution $\widehat{x}$ fits m observations by $n < m$ parameters. This is a fundamental application of linear algebra to statistics.

This section goes beyond $A^TA\widehat{x} = A^T\boldsymbol{b}$. These unweighted equations assume that the measurements $b_1, \ldots, b_m$ are equally reliable. When there is good reason to expect higher accuracy (lower variance) in some b_i, those equations should be weighted more heavily. *With what weights* $w_1, \cdots, w_m$? And if the b_i are not independent, a ***covariance matrix*** Σ gives the statistics of the errors. Here are key topics in this section:

1. Weighted least squares and $A^TCA\widehat{x} = A^TC\boldsymbol{b}$
2. Variances $\sigma_1^2, \ldots, \sigma_m^2$ and the covariance matrix $\boldsymbol{\Sigma}$
3. Important probability distributions: binomial, Poisson, and normal
4. Principal Component Analysis (PCA) to find combinations with greatest variance.

Weighted Least Squares

To include weights in the m equations $A\boldsymbol{x} = \boldsymbol{b}$, multiply each equation i by a weight w_i. Put those m weights into a diagonal matrix W. *We are replacing* $A\boldsymbol{x} = \boldsymbol{b}$ *by* $WA\boldsymbol{x} = W\boldsymbol{b}$. The equations are no more and no less solvable—we expect to use least squares.

The least squares equation $A^TA\widehat{x} = A^T\boldsymbol{b}$ changes to $(WA)^TWA\widehat{x} = (WA)^TW\boldsymbol{b}$. The matrix $C = W^TW$ is inside $(WA)^TWA$, in the middle of weighted least squares.

Weighted least squares $\quad C = W^TW$ *is in the n equations for* $\widehat{x}$ $\quad A^TCA\widehat{x} = A^TC\boldsymbol{b}$ (1)

When $n = 1$ and $A =$ column of 1's, $\widehat{x}$ changes from an average to a weighted average:

$$\textbf{Simplest case} \quad \widehat{x} = \frac{b_1 + \cdots + b_m}{m} \quad \text{changes to} \quad \widehat{x}_W = \frac{w_1^2 b_1 + \cdots + w_m^2 b_m}{w_1^2 + \cdots + w_m^2}. \tag{2}$$

This average $\widehat{x}_W$ gives greatest weight to the observations b_i that have the largest w_i. We always assume that errors have *zero mean.* (Subtract the mean if necessary, so there is no one-sided bias in the measurements.)

How should we choose the weights w_i? This depends on the reliability of b_i. If that observation has variance σ_i^2, then the root mean square error in b_i is σ_i. **When we divide the equations by** $\sigma_1, \ldots \sigma_m$ (left side together with right side), **all variances will equal 1.** So the weight is $w_i = 1/\sigma_i$ and the diagonal of $C = W^TW$ contains the numbers $1/\sigma_i^2$.

The statistically correct matrix is $C = \operatorname{diag}(1/\sigma_1^2, \ldots, 1/\sigma_m^2)$.

This is correct provided the errors e_i and e_j in different equations are statistically independent. If the errors are dependent, off-diagonal entries show up in the covariance matrix $\boldsymbol{\Sigma}$. The good choice is still $C = \boldsymbol{\Sigma}^{-1}$ as described in this section.

Mean and Variance

The two crucial numbers for a random variable are its **mean** m and its **variance** σ^2. The "expected value" E[e] is found from the probabilities $p_1, p_2, \ldots$ of the possible errors $e_1, e_2, \ldots$ (and the variance σ^2 is always measured around the mean).

For a discrete random variable, the error e_j has probability p_j (the p_j add to 1):

$$\textbf{Mean } m = \mathrm{E}[e] = \sum e_j p_j \quad \textbf{Variance } \sigma^2 = \mathrm{E}[(e-m)^2] = \sum (e_j - m)^2 p_j \tag{3}$$

Example 1 Flip a fair coin. The result is 1 (for heads) or 0 (for tails). Those events have equal probabilities $p_0 = p_1 = 1/2$. The mean is $m = 1/2$ and the variance is $\sigma^2 = 1/4$:

$$\text{Mean} = (0)\frac{1}{2} + (1)\frac{1}{2} \qquad \text{Variance} = \left(0 - \frac{1}{2}\right)^2 \frac{1}{2} + \left(1 - \frac{1}{2}\right)^2 \frac{1}{2} = \frac{1}{4}.$$

Example 2 (**Binomial**) Flip the fair coin N times and count heads. With 3 flips, we see $M = 0, 1, 2$, or 3 heads. The chances are $1/8, 3/8, 3/8, 1/8$. There are three ways to see $M = 2$ heads: HHT, HTH, and THH, and only HHH for $M = 3$ heads.

For all N, the number of ways to see M heads is the binomial coefficient "N *choose* M". Divide by the total number 2^N of all possible outcomes to get the probability for each M:

$$\begin{matrix}\textbf{M heads in}\\ \textbf{N coin flips}\end{matrix} \quad p_M = \frac{1}{2^N}\binom{N}{M} = \frac{1}{2^N}\frac{N!}{M!(N-M)!} \qquad \text{Check } \frac{1}{2^3}\frac{3!}{2!\,1!} = \frac{3}{8} \tag{4}$$

Gamblers know this instinctively. The probabilities p_M add to $\left(\frac{1}{2} + \frac{1}{2}\right)^N = 1$. The mean value of the number of heads is $m = N/2$. The variance around m turns out to be $\sigma^2 = N/4$. The standard deviation $\sigma = \sqrt{N}/2$ measures the expected spread around the mean.

Example 3 (**Poisson**) A very unfair coin (small $p << \frac{1}{2}$) is flipped very often (large N). **The product $\lambda = pN$ is kept fixed.** The high probability of tails is $1 - p$ each time. So the chance p_0 of no heads in N flips (tails every time) is $(1-p)^N = (1 - \lambda/N)^N$. For large N this approaches $e^{-\lambda}$. The probability p_j of j heads in N very unfair flips comes out neatly in terms of the crucial number $\lambda = pN$:

$$\textbf{Poisson probabilities} \quad p_j = \frac{\lambda^j}{j!}e^{-\lambda} \quad \textbf{Mean } m = \lambda \quad \textbf{Variance } \sigma^2 = \lambda \tag{5}$$

Poisson applies to counting infrequent events (low p) over a long time T. Then $\lambda = pT$.

A *continuous* random variable will have a probability *density* function $p(x)$ instead of $p_1, p_2, \ldots$. "An outcome between x and $x + dx$ has probability $p(x)\,dx$." The total probability is $\int p(x)\,dx = 1$, since some outcome must happen. Sums become integrals:

$$\textbf{Mean } m = \textbf{Expected value} = \int x p(x)\,dx \quad \textbf{Variance } \sigma^2 = \int (x-m)^2 p(x)\,dx. \tag{6}$$

The outstanding example of a probability density function $p(x)$ (called the pdf) is the ***normal distribution*** $\mathbf{N}(0, \sigma)$. This has mean zero by symmetry. Its variance is σ^2:

$$\textbf{Normal (Gaussian)} \qquad p(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-x^2/2\sigma^2} \quad \text{with} \quad \int_{-\infty}^{\infty} p(x)\,dx = 1. \tag{7}$$

The graph of $p(x)$ is the famous bell-shaped curve. The integral of $p(x)$ from $-\sigma$ to σ is the probability that a random sample is less than one standard deviation σ from the mean. This is near $2/3$. MATLAB's **randn** uses the normal distribution with $\sigma = 1$.

This normal $p(x)$ appears everywhere because of the ***Central Limit Theorem***: The average over many independent trials of another distribution (like binomial) will approach a normal distribution as $N \to \infty$. A shift produces $m = 0$ and rescaling produces $\sigma = 1$.

$$\textbf{Normalized headcount} \qquad x = \frac{M - \text{mean}}{\sigma} = \frac{M - N/2}{\sqrt{N}/2} \longrightarrow \text{Normal } \mathbf{N}(0, 1).$$

The Covariance Matrix

Now run m different experiments at once. They might be independent, or there might be some correlation between them. Each measurement $\boldsymbol{b}$ is now a *vector* with m components. Those components are the outputs b_i from the m experiments.

If we measure distances from the means m_i, each error $e_i = b_i - m_i$ has *mean zero*. If two errors e_i and e_j are *independent* (no relation between them), their product $e_i e_j$ also has mean zero. But if the measurements are by the same observer at nearly the same time, the errors e_i and e_j could tend to have the same sign or the same size. ***The errors in the m experiments could be correlated.*** The products $e_i e_j$ are weighted by p_{ij} (their probability): *covariance* $\sigma_{ij} = \sum\sum p_{ij} e_i e_j$. The sum of $e_i^2 p_{ii}$ is the variance σ_i^2:

$$\textbf{Covariance} \qquad \sigma_{ij} = \sigma_{ji} = \mathbf{E}[e_i e_j] = \textbf{expected value of } (e_i \textbf{ times } e_j). \tag{8}$$

This is the (i, j) and (j, i) entry of the ***covariance matrix*** $\boldsymbol{\Sigma}$. The (i, i) entry is $\sigma_{ii} = \sigma_i^2$.

Example 4 **(Multivariate normal)** For m random variables, the probability density function moves from $p(x)$ to $p(\boldsymbol{b}) = p(b_1, \ldots, b_m)$. The normal distribution with mean zero was controlled by one positive number σ^2. Now $p(\boldsymbol{b})$ is controlled by an m by m positive definite matrix $\boldsymbol{\Sigma}$. This is the covariance matrix and its determinant is $|\boldsymbol{\Sigma}|$:

$$p(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-x^2/2\sigma^2} \quad \text{becomes} \quad p(\boldsymbol{b}) = \frac{1}{(2\pi)^{m/2} |\boldsymbol{\Sigma}|^{1/2}} e^{-\boldsymbol{b}^{\mathrm{T}} \boldsymbol{\Sigma}^{-1} \boldsymbol{b}/2}$$

The integral of $p(\boldsymbol{b})$ over m-dimensional space is 1. The integral of $\boldsymbol{b}\boldsymbol{b}^{\mathrm{T}} p(\boldsymbol{b})$ is $\boldsymbol{\Sigma}$.

The good way to handle that exponent $-\boldsymbol{b}^{\mathrm{T}}\Sigma^{-1}\boldsymbol{b}/2$ is to use the eigenvalues and orthonormal eigenvectors of Σ (*linear algebra enters here*). When $\Sigma = Q\Lambda Q^{\mathrm{T}} = Q\Lambda Q^{-1}$, replacing $\boldsymbol{b}$ by $Q\boldsymbol{c}$ will split $p(\boldsymbol{b})$ into m one-dimensional normal distributions:

$$\exp\left(-\boldsymbol{b}^{\mathrm{T}}\Sigma^{-1}\boldsymbol{b}/2\right) = \exp\left(-\boldsymbol{c}^{\mathrm{T}}\Lambda^{-1}\boldsymbol{c}/2\right) = \left(e^{-c_1^2/2\lambda_1}\right)\cdots\left(e^{-c_m^2/2\lambda_m}\right).$$

The determinant has $|\Sigma|^{1/2} = |\Lambda|^{1/2} = (\lambda_1 \cdots \lambda_m)^{1/2}$. Each integral over $-\infty < c_i < \infty$ is back to one dimension, where $\lambda = \sigma^2$. Notice the wonderful fact that after any linear transformation (here $\boldsymbol{c} = Q^{-1}\boldsymbol{b}$), we still have a multivariate normal distribution.

We could even reach variances $= 1$ by including $\sqrt{\Lambda}$ in the change from $\boldsymbol{b}$ to $\boldsymbol{z}$:

Standard normal $\quad \boldsymbol{b} = \sqrt{\Lambda}Q\boldsymbol{z}$ changes $p(\boldsymbol{b})d\boldsymbol{b}$ to $p(\boldsymbol{z})d\boldsymbol{z} = \dfrac{e^{-\boldsymbol{z}^{\mathrm{T}}\boldsymbol{z}/2}}{(2\pi)^{m/2}}d\boldsymbol{z}$

This tells us the right weight matrix W to bring $A\boldsymbol{x} = \boldsymbol{b}$ back to ordinary least squares for $WA\boldsymbol{x} = W\boldsymbol{b}$. We want $W\boldsymbol{b}$ to become the standard normal $\boldsymbol{z}$. So W will be the inverse of $\sqrt{\Lambda}Q$. Better than that, $\boldsymbol{C = W^{\mathrm{T}}W}$ ***is the inverse of*** $\boldsymbol{Q\Lambda Q^{\mathrm{T}}}$ ***which is*** $\boldsymbol{\Sigma}$.

Summary For independent errors, Σ is the diagonal matrix $\mathbf{diag}(\sigma_1^2, \ldots, \sigma_m^2)$. This is the usual choice. The right weights w_i for the equations $A\boldsymbol{x} = \boldsymbol{b}$ are $1/\sigma_1, \ldots, 1/\sigma_m$ (this will equalize all variances to 1). The right matrix $C = W^{\mathrm{T}}W$ in the middle of the weighted least squares equations is exactly Σ^{-1}:

Weighted least squares $\qquad A^{\mathrm{T}}\Sigma^{-1}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\Sigma^{-1}\boldsymbol{b}$ $\qquad$ (9)

This choice of weighting returns $A\boldsymbol{x} = \boldsymbol{b}$ to a least squares problem $WA\boldsymbol{x} = W\boldsymbol{b}$ with equally reliable and independent errors. The usual equation $(WA)^{\mathrm{T}}WA\widehat{\boldsymbol{x}} = (WA)^{\mathrm{T}}W\boldsymbol{b}$ is the same as (9).

It was Gauss who found this *best linear unbiased estimate* $\widehat{\boldsymbol{x}}$. Unbiased because the mean of $\boldsymbol{x} - \widehat{\boldsymbol{x}}$ is zero, linear because of equation (9), best because the covariance of $\boldsymbol{x} - \widehat{\boldsymbol{x}}$ is as small as possible. That covariance (for error in $\widehat{\boldsymbol{x}}$, not error in $\boldsymbol{b}$!) is important:

Covariance of the best $\widehat{\boldsymbol{x}}$ $\qquad P = \mathrm{E}\left[(\boldsymbol{x} - \widehat{\boldsymbol{x}})(\boldsymbol{x} - \widehat{\boldsymbol{x}})^{\mathrm{T}}\right] = \left(A^{\mathrm{T}}\Sigma^{-1}A\right)^{-1}.$ $\qquad$ (10)

Example 5 Your pulse rate is measured ten times by independent doctors, all equally reliable. The mean error of each b_i is zero, and each variance is σ^2. Then $\Sigma = \sigma^2 I$. The ten equations $x = b_i$ produce the 10 by 1 matrix A of all ones. The best estimate $\widehat{x}$ is the average of the ten b_i. *The variance of that average value* $\widehat{x}$ *is the number* P:

$$P = (A^{\mathrm{T}}\Sigma^{-1}A)^{-1} = \sigma^2/10 \quad \text{so averaging reduces the variance.}$$

This matrix $P = (A^{\mathrm{T}}\Sigma^{-1}A)^{-1}$ tells how reliable is the result $\widehat{\boldsymbol{x}}$ of the experiment (Problem 6). P does not depend on the b's in the actual experiment! Those b's have probability distributions. Each experiment produces a sample value of $\widehat{\boldsymbol{x}}$ from a sample $\boldsymbol{b}$.

When a small Σ gives good reliability of the inputs $\boldsymbol{b}$, a small P gives good reliability of the outputs $\widehat{x}$. The key formula $P = (A^{\mathrm{T}}\Sigma^{-1}A)^{-1}$ connects those covariances.

Principal Component Analysis

These paragraphs are about finding useful information in a data matrix A. Start by measuring m properties (m features) of n samples. These could be grades in m courses for n students (a row for each course, a column for each student). From each row, subtract its average so the sample means are zero. We look for a ***combination of courses*** and/or ***combination of students*** for which the data provides the most information.

Information is "distance from randomness" and it is measured by **variance**. A large variance in course grades means greater information than a small variance.

The key matrix idea is the Singular Value Decomposition $A = U\Sigma V^{\mathrm{T}}$. We are back again to $A^{\mathrm{T}}A$ and AA^{T}, because their unit eigenvectors are the singular vectors $v_1, \ldots, v_n$ in V and $u_1, \ldots, u_m$ in U. The singular values in the diagonal matrix Σ (not the covariance) are in decreasing order and σ_1 is the most important. Weighting the m courses by the components of u_1 gives a "***master course***" or "***eigencourse***" with the most significant grades.

Example 6 Suppose the grades A, B, C, F are worth $\mathbf{4, 2, 0, -6}$ points. If each course and each student has one of each grade, then all means are zero. Here is the grade matrix A with $(1, 1, 1, 1)$ in its nullspace (rank 3). To keep integers, the SVD of A will be written as $2U$ times $\Sigma/4$ times $(2V)^{\mathrm{T}}$. So the σ's are $12, 8, 4$:

$$\begin{bmatrix} -6 & 2 & 0 & 4 \\ 0 & 4 & -6 & 2 \\ 4 & 0 & 2 & -6 \\ 2 & -6 & 4 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 1 & -1 \\ -1 & -1 & 1 \\ 1 & -1 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & & \\ & 2 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 & -1 \\ -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

Weighting the rows (the courses) by $u_1 = \frac{1}{2}(-1, -1, 1, 1)$ will give the *eigencourse*. Weighting the columns (the students) by $v_1 = \frac{1}{2}(1, -1, 1, -1)$ gives the *eigenstudent*. The fraction of the grade matrix that is "explained" by that one course and student is $\sigma_1^2/(\sigma_1^2 + \sigma_2^2 + \sigma_3^2) = 9/14$. The σ's in the SVD are the variances σ^2.

I guess this master course is what a Director of Admissions is looking for. If all grades in gym are the same, that row of A will be all zero—and gym is not part of the master course. Probably calculus is a part, but what about students who don't take calculus? The problem of ***missing data*** (holes in the matrix A) is extremely difficult for social sciences and the census and so much of the statistics of experiments.

Gene expression data Determining the functions of genes, and combinations of genes, is a central problem of genetics. Which genes combine to give which properties? Which genes malfunction to give which diseases?

We now have an incredibly fast way to find gene expression data in the lab. A gene microarray is often packed onto an Affymetrix chip, measuring tens of thousands of genes from one sample (one person). The understanding of genetic data (*bioinformatics*) has become a tremendous application of linear algebra.

Problem Set 8.6

1 Which line $Ct + D$ is the best fit to the three independent measurements $1, 2, 4$ at times $t = 0, 1, 2$ if the variances $\sigma_1^2, \sigma_2^2, \sigma_3^2$ are $1, 1, 2$? Use weights $w_i = 1/\sigma_i$.

2 In Problem 1, suppose that the third measurement is **totally unreliable**. The variance σ_3^2 becomes infinite. Then the best line will not use _____. Find the line that goes through the first two points and solves the first two equations in $Ax = b$ exactly.

3 In Problem 1, suppose that the third measurement is **totally reliable**. The variance σ_3^2 approaches zero. Now the best line will go through the third point exactly. Choose that line to minimize the sum of squares of the first two errors.

4 A single flip of a fair coin (0 or 1) has mean $m = 1/2$ and variance $\sigma^2 = 1/4$. This was Example 1. For the sum of two flips, the mean is $m = 1$. Compute the variance σ^2 around this mean, using the outcomes $0, 1, 2$ with their probabilities.

5 Instead of adding the flip results, make them two independent experiments. The outcome is $(0, 0), (1, 0), (0, 1)$ or $(1, 1)$. What is the covariance matrix Σ?

6 Change Example 1 so that the coin flip can be unfair. The probability is p for heads and $1 - p$ for tails. Find the mean m and the variance σ^2 of this distribution.

7 For two independent measurements $x = b_1$ and $x = b_2$, the best $\widehat{x}$ should be some weighted average $\widehat{x} = ab_1 + (1 - a)b_2$. When b_1 and b_2 have mean zero and variances σ_1^2 and σ_2^2, the variance of $\widehat{x}$ will be $P = a^2\sigma_1^2 + (1-a)^2\sigma_2^2$. ***Choose the number a that minimizes P***: $dP/da = 0$.

Show that this a gives the $\widehat{x}$ in equation (2) which the text claimed is best, using weights $w_1 = 1/\sigma_1$ and $w_2 = 1/\sigma_2$.

8 The least squares estimate correctly weighted by Σ^{-1} is $\widehat{x} = (A^{\mathrm{T}}\Sigma^{-1}A)^{-1}A^{\mathrm{T}}\Sigma^{-1}b$. ***Call that*** $\widehat{x} = Lb$. If b contains an error vector e, then $\widehat{x}$ contains the error Le.

The covariance matrix of those output errors Le is their expected value (average value) $P = \mathrm{E}\left[(Le)(Le)^{\mathrm{T}}\right] = L\mathrm{E}\left[ee^{\mathrm{T}}\right]L^{\mathrm{T}} = L\Sigma L^{\mathrm{T}}$. **Problem**: Do the multiplication $L\Sigma L^{\mathrm{T}}$ to show that P equals $(A^{\mathrm{T}}\Sigma^{-1}A)^{-1}$ as predicted in equation (10).

9 Change the grades to $3, 1, -1, -3$ for A, B, C, F. Show that the SVD of this grade matrix has the same u_1, u_2, v_1, v_2 (same eigencourses) as in Example 5, but now A has rank 2.

Grade matrix
$$A = \begin{bmatrix} 3 & -1 & 1 & -3 \\ -1 & 3 & -3 & 1 \\ -3 & 1 & -1 & 3 \\ 1 & -3 & 3 & -1 \end{bmatrix}$$

Notes One way to deal with missing entries in A is to complete the matrix to have minimum rank. And statistics makes major use of the pseudoinverse A^+ (which is exactly the left inverse $(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ from the normal equation when $A^{\mathrm{T}}A$ is invertible).

■ REVIEW OF THE KEY IDEAS ■

1. Computer graphics needs shift operations $T(v) = v + v_0$ as well as linear operations $T(v) = Av$.
2. A shift in $\mathbf{R}^n$ can be executed by a matrix of order $n + 1$, using homogeneous coordinates.
3. The extra component 1 in $[x\ y\ z\ 1]$ is preserved when all matrices have the numbers $0, 0, 0, 1$ as last column.

Problem Set 8.7

1 A typical point in $\mathbf{R}^3$ is $xi + yj + zk$. The coordinate vectors i, j, and k are $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$. The coordinates of the point are (x, y, z).

This point in computer graphics is $xi + yj + zk +$ **origin**. Its homogeneous coordinates are (, , ,). Other coordinates for the same point are (, , ,).

2 A linear transformation T is determined when we know $T(i), T(j), T(k)$. For an affine transformation we also need $T($_____$)$. The input point $(x, y, z, 1)$ is transformed to $xT(i) + yT(j) + zT(k) +$ _____.

3 Multiply the 4 by 4 matrix T for translation along $(1, 4, 3)$ and the matrix T_1 for translation along $(0, 2, 5)$. The product TT_1 is translation along _____.

4 Write down the 4 by 4 matrix S that scales by a constant c. Multiply ST and also TS, where T is translation by $(1, 4, 3)$. To blow up the picture around the center point $(1, 4, 3)$, would you use vST or vTS?

5 What scaling matrix S (in homogeneous coordinates, so 3 by 3) would produce a 1 by 1 square page from a standard 8.5 by 11 page?

6 What 4 by 4 matrix would move a corner of a cube to the origin and then multiply all lengths by 2? The corner of the cube is originally at $(1, 1, 2)$.

7 When the three matrices in equation 1 multiply the unit vector a, show that they give $(\cos\theta)a$ and $(1 - \cos\theta)a$ and $\mathbf{0}$. Addition gives $aQ = a$ and the rotation axis is not moved.

8 If b is perpendicular to a, multiply by the three matrices in 1 to get $(\cos\theta)b$ and $\mathbf{0}$ and a vector perpendicular to b. So Qb makes an angle θ with b. ***This is rotation.***

9 What is the 3 by 3 projection matrix $I - nn^{\mathrm{T}}$ onto the plane $\frac{2}{3}x + \frac{2}{3}y + \frac{1}{3}z = 0$? In homogeneous coordinates add $0, 0, 0, 1$ as an extra row and column in P.

10 With the same 4 by 4 matrix P, multiply $T_- P T_+$ to find the projection matrix onto the plane $\frac{2}{3}x + \frac{2}{3}y + \frac{1}{3}z = 1$. The translation T_- moves a point on that plane (choose one) to $(0, 0, 0, 1)$. The inverse matrix T_+ moves it back.

11 Project $(3, 3, 3)$ onto those planes. Use P in Problem 9 and $T_- P T_+$ in Problem 10.

12 If you project a square onto a plane, what shape do you get?

13 If you project a cube onto a plane, what is the outline of the projection? Make the projection plane perpendicular to a diagonal of the cube.

14 The 3 by 3 mirror matrix that reflects through the plane $\boldsymbol{n}^{\mathrm{T}}\boldsymbol{v} = 0$ is $M = I - 2\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}}$. Find the reflection of the point $(3, 3, 3)$ in the plane $\frac{2}{3}x + \frac{2}{3}y + \frac{1}{3}z = 0$.

15 Find the reflection of $(3, 3, 3)$ in the plane $\frac{2}{3}x + \frac{2}{3}y + \frac{1}{3}z = 1$. Take three steps $T_- M T_+$ using 4 by 4 matrices: translate by T_- so the plane goes through the origin, reflect the translated point $(3, 3, 3, 1)T_-$ in that plane, then translate back by T_+.

16 The vector between the origin $(0, 0, 0, 1)$ and the point $(x, y, z, 1)$ is the difference $v =$ _____. In homogeneous coordinates, vectors end in _____. So we add a _____ to a point, not a point to a point.

17 If you multiply only the *last* coordinate of each point to get (x, y, z, c), you rescale the whole space by the number _____. This is because the point (x, y, z, c) is the same as $(\ ,\ ,\ , 1)$.